

Supplementary Files

Integrative Machine Learning and Network Biology Analysis Reveals Molecular Links Between Post-Traumatic Stress Disorder and Depression

Supplementary Tables:

Table S1: Differentially expressed probes in GSE98793 (MDD; 4,607 probes, 3,744 genes) and GSE63878 (PTSD; 858 probes, 829 genes) selected with GEO2R (limma) at a nominal p-value below 0.08, after removal of probes without a gene symbol. Benjamini–Hochberg-adjusted p-values are shown for reference; selection was based on the nominal p-value. The complete lists are provided as a separate Excel file.

Item	Description
Selection	GEO2R (limma) on the submitter-processed series matrix; nominal P-value < 0.08
Data cleaning	Probes without a gene symbol removed
GSE98793 (MDD)	4607 probes
GSE63878 (PTSD)	858 probes
Note	Adjusted P-values (Benjamini-Hochberg) are shown for reference; selection did not use them

Dataset	Probe ID	Gene symbol	log2FC	P-value (nominal)	Adjusted P-value (BH)	Direction
GSE98793 (MDD)	206666_at	GZMK	-0.415	1.36e-06	0.0297	Down
GSE98793 (MDD)	220570_at	RETN	0.252	1.63e-06	0.0297	Up
GSE98793 (MDD)	204970_s_at	MAFG	0.256	3.22e-06	0.038	Up
GSE98793 (MDD)	226733_at	PFKFB2	0.364	3.48e-06	0.038	Up
GSE98793 (MDD)	231688_at	MMP8	0.749	7.32e-06	0.0572	Up
GSE98793 (MDD)	209954_x_at	SS18	-0.207	7.32e-06	0.0572	Down
GSE98793 (MDD)	227232_at	EVL	-0.265	1.017e-05	0.0594	Down
GSE98793 (MDD)	210426_x_at	RORA	-0.25	1.157e-05	0.0594	Down
GSE98793 (MDD)	210279_at	GPR18	-0.238	1.293e-05	0.0594	Down
GSE98793 (MDD)	205798_at	IL7R	-0.294	1.305e-05	0.0594	Down
GSE98793 (MDD)	210479_s_at	RORA	-0.278	1.362e-05	0.0594	Down
GSE98793 (MDD)	239086_at	IDNK	-0.213	1.504e-05	0.0594	Down
GSE98793 (MDD)	228285_at	TDRD9	0.492	1.521e-05	0.0594	Up

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GSE98793 (MDD)	218865_at	MARC1	0.348	2.203e-05	0.0803	Up
GSE98793 (MDD)	227957_at	GSN	0.218	3.568e-05	0.1219	Up
GSE98793 (MDD)	208744_x_at	HSPH1	-0.271	3.848e-05	0.1227	Down
GSE98793 (MDD)	202934_at	HK2	0.241	4.038e-05	0.1227	Up
GSE98793 (MDD)	223832_s_at	CAPNS2	0.297	5.03e-05	0.1407	Up
GSE98793 (MDD)	235189_at	ICE2	-0.266	5.148e-05	0.1407	Down
GSE98793 (MDD)	215769_at	YME1L1	-0.205	6.123e-05	0.1594	Down
GSE63878 (PTSD)	7971296	EPSTI1	-0.5827	2.75e-05	0.51	Down
GSE63878 (PTSD)	8176730	RPS4Y2	0.1238	6.95e-05	0.51	Up
GSE63878 (PTSD)	7903765	GSTM1	0.7899	9.25e-05	0.51	Up
GSE63878 (PTSD)	8101637	HSD17B13	0.1883	0.0002021	0.891	Up
GSE63878 (PTSD)	8103563	DDX60	-0.3795	0.0003082	1	Down
GSE63878 (PTSD)	8040080	RSAD2	-0.7517	0.0005108	1	Down
GSE63878 (PTSD)	7902541	IFI44L	-0.6129	0.0005209	1	Down
GSE63878 (PTSD)	8154962	DNAJB5	0.1594	0.000689	1	Up
GSE63878 (PTSD)	7989670	RBPM52	0.1863	0.0007556	1	Up
GSE63878 (PTSD)	8161964	FRMD3	-0.1843	0.0007645	1	Down
GSE63878 (PTSD)	8050102	CMPK2	-0.3625	0.0009505	1	Down
GSE63878 (PTSD)	8126531	CRIP3///CRIP3	0.1404	0.0010887	1	Up
GSE63878 (PTSD)	7921434	AIM2	-0.3668	0.0010914	1	Down
GSE63878 (PTSD)	7967117	OASL	-0.3542	0.0012746	1	Down
GSE63878 (PTSD)	7958884	OAS1	-0.3705	0.0015525	1	Down
GSE63878 (PTSD)	8004184	XAF1	-0.3217	0.0017366	1	Down
GSE63878 (PTSD)	7969096	CDADC1	-0.1163	0.0017945	1	Down

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GSE63878 (PTSD)	8029507	NECTIN2	0.2609	0.0017951	1	Up
GSE63878 (PTSD)	7908125	RGL1	-0.1274	0.0019837	1	Down
GSE63878 (PTSD)	8046124	DHRS9	-0.3213	0.0019892	1	Down

Table S2: (A) Shared genes identified at each nominal p-value and Fisher-score cut-off (0.05/0.08 × Top 500/800/1,000), with the log2 fold change, direction of change and Fisher-score rank of each gene in MDD (GSE98793) and PTSD (GSE63878), and whether the direction was concordant between the two disorders.

Gene	log2FC MDD	Direction MDD	Fisher rank MDD 0.05	Fisher rank MDD 0.08	log2FC PTSD	Direction PTSD	Fisher rank PTSD 0.05	Fisher rank PTSD 0.08	Concordant direction
AHI1	-0.161	Down	630	630	0.146	Up	274	274	no
BARD1	-0.176	Down	817	817	0.092	Up		543	no
BTN3A2	-0.253	Down	298	298	0.141	Up		734	no
CCDC57	0.122	Up	837	837	-0.063	Down	203	203	no
CYP2U1	-0.225	Down	352	352	0.076	Up		546	no
DDX60L	0.251	Up	884	884	0.222	Up	136	136	yes
DKKL1	0.137	Up	506	506	-0.126	Down	187	187	no
DOCK4	0.4	Up	688	688	0.174	Up	454	460	yes
EIF4B	-0.093	Down	340	340	-0.142	Down	273	273	yes
FCER1A	-0.388	Down	212	212	-0.163	Down		823	yes
GALNT11	-0.151	Down	423	423	0.098	Up	53	53	no
GFI1B	0.161	Up	533	533	-0.074	Down		532	no
GRB10	0.324	Up	50	50	0.077	Up		815	yes
HGF	0.172	Up	433	433	0.093	Up		665	yes
HTRA1	0.18	Up	847	847	-0.078	Down	302	302	no
IGF1R	0.184	Up	393	393	-0.147	Down		741	no
OLFM4	1.026	Up	53	53	0.352	Up	465	479	yes
PLXNA2	0.217	Up	339	339	-0.063	Down		632	no
PML	0.082	Up	245	245	0.213	Up	100	100	yes
PTMA	-0.082	Down	610	610	-0.1	Down	321	321	yes
SCAPER	-0.264	Down	653	653	0.124	Up	63	63	no
SLC22A14	0.142	Up	680	680	-0.067	Down		736	no
STAT2	0.137	Up	775	775	0.177	Up		794	yes
SYTL2	-0.347	Down	86	86	0.129	Up	409	412	no

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TDRD9	0.492	Up	13	13	0.112	Up	149	149	yes
TGIF1	-0.173	Down	882	882	-0.054	Down		400	yes
TIGIT	-0.229	Down	982	982	0.124	Up		668	no

Table S2: (B) S_pathway_stability. Overlap of significantly enriched KEGG and WikiPathways terms across the tested thresholds, summarizing which pathway-level findings were stable versus threshold-dependent.

Library	Comparison	Significant_A	Significant_B	Shared_significant	Jaccard	Example_shared_terms
KEGG_2021_Human	Shared genes 0.08 Top800 (19) vs Shared genes 0.05 Top800	4	0	0	0	
KEGG_2021_Human	Hub genes 0.08 Top800 vs Hub genes 0.05 Top800	69	10	4	0.05	Coronavirus disease; HIF-1 signaling pathway; PI3K-Akt signaling pathway; Proteoglycans in cancer
WikiPathways_2024_Human	Shared genes 0.08 Top800 (19) vs Shared genes 0.05 Top800	3	0	0	0	
WikiPathways_2024_Human	Hub genes 0.08 Top800 vs Hub genes 0.05 Top800	159	29	18	0.11	Acute Myeloid Leukemia WP5293; Brain Derived Neurotrophic Factor BDNF Signaling WP2380; EGFR Tyrosine Kinase Inhibitor

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						Resistance WP4806; Focal Adhesion PI3K Akt mTOR Signaling WP3932; Head And Neck Squamous Cell Carcinoma WP4674
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Table S2: (C) S_pathway_top_terms_by_threshold. KEGG and WikiPathways terms enriched in the shared gene set and in the hub gene set at each nominal p-value and Fisher-score cut-off, with adjusted P-values, enrichment ratios, and contributing genes for every threshold combination.

Library	Gene_set	Term	Overlap	FDR	Genes
KEGG_2021_Human	Shared genes 0.08 Top800 (19)	Rap1 signaling pathway	3/193	0.033463358 38005839	DOCK4; HGF; IGF1R
KEGG_2021_Human	Shared genes 0.08 Top800 (19)	Proteoglycans in cancer	3/187	0.0334633 5838005839	EIF4B; HGF; IGF1R
KEGG_2021_Human	Shared genes 0.08 Top800 (19)	Pathways in cancer	4/490	0.0343600333 829135	HGF; IGF1R; PML; STAT2
KEGG_2021_Human	Shared genes 0.08 Top800 (19)	Melanoma	Feb-68	0.03518904 40052346	HGF; IGF1R
KEGG_2021_Human	Shared genes 0.08 Top800 (19)	PI3K-Akt signaling pathway	3/322	0.05708107 607305544	EIF4B; HGF; IGF1R
KEGG_2021_Human	Shared genes 0.08 Top800 (19)	Influenza A	2/144	0.079225814 91266341	PML; STAT2
KEGG_2021_Human	Shared genes 0.08	Transcriptional misregulation in cancer	2/157	0.07922581 491266341	IGF1R; PML

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	Top800 (19)				
KEGG_2021_Human	Shared genes 0.08 Top800 (19)	Hepatocellular carcinoma	2/150	0.079225814 91266341	HGF; IGF1R
KEGG_2021_Human	Shared genes 0.08 Top800 (19)	mTOR signaling pathway	2/141	0.079225814 91266341	EIF4B; IGF1R
KEGG_2021_Human	Shared genes 0.08 Top800 (19)	Focal adhesion	2/187	0.099258018 35601751	HGF; IGF1R
KEGG_2021_Human	Shared genes 0.05 Top800	Acute myeloid leukemia	Jan-62	0.166108229 5557605	PML
KEGG_2021_Human	Shared genes 0.05 Top800	Influenza A	1/144	0.1661082295557605	PML
KEGG_2021_Human	Shared genes 0.05 Top800	Other types of O-glycan biosynthesis	Jan-43	0.166108229 5557605	GALNT1 1
KEGG_2021_Human	Shared genes 0.05 Top800	Mucin type O-glycan biosynthesis	Jan-32	0.166108229 5557605	GALNT1 1
KEGG_2021_Human	Shared genes 0.05 Top800	Transcriptional misregulation in cancer	1/157	0.166108229 5557605	PML
KEGG_2021_Human	Shared genes 0.05 Top800	Rap1 signaling pathway	1/193	0.166108229 5557605	DOCK4
KEGG_2021_Human	Shared genes 0.05 Top800	RNA transport	1/141	0.166108229 5557605	EIF4B
KEGG_2021_Human	Shared genes 0.05 Top800	Proteoglycans in cancer	1/187	0.166108229 5557605	EIF4B

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KEGG_2021_Human	Shared genes 0.05 Top800	Ubiquitin mediated proteolysis	1/126	0.166108229 5557605	PML
KEGG_2021_Human	Shared genes 0.05 Top800	mTOR signaling pathway	1/141	0.166108229 5557605	EIF4B
KEGG_2021_Human	Hub genes 0.08 Top800	Pathways in cancer	10/490	3.203012234 926083e-14	ERBB2; HDAC1; HGF; IGF1R; JAK1; JAK2; MDM2; PML; STAT2; STAT3
KEGG_2021_Human	Hub genes 0.08 Top800	PI3K-Akt signaling pathway	6/322	3.763390612 418494e-07	ERBB2; HGF; IGF1R; JAK1; JAK2; MDM2
KEGG_2021_Human	Hub genes 0.08 Top800	Epstein-Barr virus infection	5/177	7.929851391 962401e-07	HDAC1; JAK1; MDM2; STAT2; STAT3
KEGG_2021_Human	Hub genes 0.08 Top800	Proteoglycans in cancer	5/187	7.929851391 962401e-07	ERBB2; HGF; IGF1R; STAT3
KEGG_2021_Human	Hub genes 0.08 Top800	Necroptosis	4/116	7.182976714 567033e-06	JAK1; JAK2; STAT2; STAT3
KEGG_2021_Human	Hub genes 0.08 Top800	Signaling pathways regulating pluripotency of stem cells	4/131	9.753049051 066915e-06	IGF1R; JAK1; JAK2; STAT3
KEGG_2021_Human	Hub genes 0.08 Top800	Influenza A	4/144	1.006351585 5385738e-05	JAK1; JAK2; PML; STAT2
KEGG_2021_Human	Hub genes	Hepatitis B	4/147	1.006351585 5385738e-05	JAK1; JAK2; STAT2;

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	0.08 Top800				STAT3
KEGG_2021_Human	Hub genes 0.08 Top800	Viral carcinogenesis	4/152	1.006351585 5385738e-05	HDAC1; JAK1; MDM2; STAT3
KEGG_2021_Human	Hub genes 0.08 Top800	Transcriptional misregulation in cancer	4/157	1.006351585 5385738e-05	HDAC1; IGF1R; MDM2; PML
KEGG_2021_Human	Hub genes 0.05 Top800	Ribosome	Apr-99	2.507080405 4159233e-06	RPL15; RPS19; RPS28; RPS6
KEGG_2021_Human	Hub genes 0.05 Top800	Coronavirus disease	4/185	1.547518399 1164064e-05	RPL15; RPS19; RPS28; RPS6
KEGG_2021_Human	Hub genes 0.05 Top800	mTOR signaling pathway	3/141	0.000338991 14683590955	EIF4B; RPS6; RPS6KB2
KEGG_2021_Human	Hub genes 0.05 Top800	Proteoglycans in cancer	3/187	0.00059014 60575390648	EIF4B; RPS6; RPS6KB2
KEGG_2021_Human	Hub genes 0.05 Top800	PI3K-Akt signaling pathway	3/322	0.00235463 12655307973	EIF4B; RPS6; RPS6KB2
KEGG_2021_Human	Hub genes 0.05 Top800	HIF-1 signaling pathway	2/103	0.00552408 0012748282	RPS6; RPS6KB2
KEGG_2021_Human	Hub genes 0.05 Top800	AMPK signaling pathway	2/115	0.00575247 9637104188	EEF2; RPS6KB2
KEGG_2021_Human	Hub genes 0.05 Top800	Insulin signaling pathway	2/129	0.00575247963710418 8	RPS6; RPS6KB2
KEGG_2021_Human	Hub genes 0.05 Top800	Apelin signaling pathway	2/128	0.00575247 9637104188	RPS6; RPS6KB2
KEGG_2021_Human	Hub genes	Thermogenesis	2/189	0.01098344 6625727337	RPS6; RPS6KB2

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	0.05 Top800				
WikiPathways _2024_Human	Shared genes 0.08 Top800 (19)	NRP1 Triggered Signaling In Pancreatic Cancer WP5144	Feb-51	0.04150235 371569023	HGF; PLXNA2
WikiPathways _2024_Human	Shared genes 0.08 Top800 (19)	Interferon Type I Signaling WP585	Feb-51	0.041502353 71569023	EIF4B; STAT2
WikiPathways _2024_Human	Shared genes 0.08 Top800 (19)	Cancer Pathways WP5434	4/470	0.041502353 71569023	HGF; IGF1R; PML; STAT2
WikiPathways _2024_Human	Shared genes 0.08 Top800 (19)	Focal Adhesion PI3K Akt mTOR Signaling WP3932	3/287	0.058247664 090601135	EIF4B; HGF; IGF1R
WikiPathways _2024_Human	Shared genes 0.08 Top800 (19)	EGFR Tyrosine Kinase Inhibitor Resistance WP4806	Feb-81	0.058247664 090601135	HGF; IGF1R
WikiPathways _2024_Human	Shared genes 0.08 Top800 (19)	PI3K Akt Signaling WP4172	3/309	0.059686064 68099827	EIF4B; HGF; IGF1R
WikiPathways _2024_Human	Shared genes 0.08 Top800 (19)	Somatroph Axis GH Relationship To Dietary Restriction And Aging WP4186	6-Jan	0.072995385 12813764	IGF1R
WikiPathways _2024_Human	Shared genes 0.08 Top800 (19)	Mod Of PI3K Akt mTOR Signaling By Bioactive Sphingolipids WP5192	11-Jan	0.075334262 48314797	IGF1R
WikiPathways _2024_Human	Shared genes 0.08 Top800 (19)	IL10 Anti Inflammatory Signaling WP4495	11-Jan	0.075334262 48314797	STAT2

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WikiPathways _2024_Human	Shared genes 0.08 Top800 (19)	Ectoderm Differentiation WP2858	2/127	0.075334262 48314797	AHI1; PLXNA2
WikiPathways _2024_Human	Shared genes 0.05 Top800	Perturbations Host Cell Autophagy Induced By SARS CoV 2 Prots WP4936	Jan-42	0.087287565 39514622	EIF4B
WikiPathways _2024_Human	Shared genes 0.05 Top800	Deregulation Of Rab And Rab Effector Genes In Bladder Cancer WP2291	16-Jan	0.087287565 39514622	SYTL2
WikiPathways _2024_Human	Shared genes 0.05 Top800	Translation Inhibitors In Chronically Activated PDGFRA Cells WP4566	Jan-43	0.087287565 39514622	EIF4B
WikiPathways _2024_Human	Shared genes 0.05 Top800	DNA IR Damage And Cellular Response Via ATR WP4016	Jan-75	0.087287565 39514622	PML
WikiPathways _2024_Human	Shared genes 0.05 Top800	Wnt Beta Catenin Signaling In Leukemia WP3658	25-Jan	0.087287565 39514622	PML
WikiPathways _2024_Human	Shared genes 0.05 Top800	miRNA Regulation Of DNA Damage Response WP1530	Jan-63	0.087287565 39514622	PML
WikiPathways _2024_Human	Shared genes 0.05 Top800	Interferon Type I Signaling WP585	Jan-51	0.087287565 39514622	EIF4B
WikiPathways _2024_Human	Shared genes 0.05 Top800	P53 Transcriptional Gene Network WP4963	Jan-84	0.087287565 39514622	PML
WikiPathways _2024_Human	Shared genes 0.05 Top800	Acute Myeloid Leukemia WP5293	Jan-63	0.087287565 39514622	PML

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WikiPathways _2024_Human	Shared genes 0.05 Top800	Orexin Receptor Pathway WP5094	Jan-86	0.087287565 39514622	EIF4B
WikiPathways _2024_Human	Hub genes 0.08 Top800	Cancer Pathways WP5434	10/470	4.506606676 641131e-14	ERBB2; HDAC1; HGF; IGF1R; JAK1; JAK2; MDM2; PML; STAT2; STAT3
WikiPathways _2024_Human	Hub genes 0.08 Top800	EGFR Tyrosine Kinase Inhibitor Resistance WP4806	Jun-81	1.8612896036 586413e-10	ERBB2; HGF; IGF1R; JAK1; JAK2; STAT3
WikiPathways _2024_Human	Hub genes 0.08 Top800	IL6 Signaling WP364	Apr-43	4.514361324 831531e-07	HDAC1; JAK1; JAK2; STAT3
WikiPathways _2024_Human	Hub genes 0.08 Top800	Thyroid Stimulating Hormone TSH Signaling WP2032	Apr-53	8.01087961 240804e-07	IGF1R; JAK1; JAK2; STAT3
WikiPathways _2024_Human	Hub genes 0.08 Top800	IL10 Anti Inflammatory Signaling WP4495	11-Mar	8.822853156 388361e-07	JAK1; STAT2; STAT3
WikiPathways _2024_Human	Hub genes 0.08 Top800	ncRNAs Involved In STAT3 Signaling In Hepatocellular Carcinoma WP4337	13-Mar	1.273620646 6983013e-06	JAK1; JAK2; STAT3
WikiPathways _2024_Human	Hub genes 0.08 Top800	Leptin Signaling WP2034	Apr-69	1.345309666 988548e-06	ERBB2; JAK1; JAK2; STAT3
WikiPathways _2024_Human	Hub genes 0.08 Top800	Prolactin Signaling WP2037	Apr-72	1.399653986 4367853e-06	ERBB2; JAK1; JAK2; STAT3

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WikiPathways _2024_Human	Hub genes 0.08 Top800	Immune Response To Tuberculosis WP4197	22-Mar	4.559206129 04034e-06	JAK1; JAK2; STAT2
WikiPathways _2024_Human	Hub genes 0.08 Top800	Androgen Receptor Network In Prostate Cancer WP2263	4/103	4.7697625252 52114e-06	HGF; JAK1; MDM2; STAT3
WikiPathways _2024_Human	Hub genes 0.05 Top800	Cytoplasmic Ribosomal Proteins WP477	May-59	1.038245188 4498779e-09	RPL15; RPS19; RPS28; RPS6; RPS6KB2
WikiPathways _2024_Human	Hub genes 0.05 Top800	Translation Inhibitors In Chronically Activated PDGFRA Cells WP4566	Mar-43	1.82743330439 7776e-05	EIF4B; RPS6; RPS6KB2
WikiPathways _2024_Human	Hub genes 0.05 Top800	Follicle Stimulating Hormone FSH Signaling WP2035	22-Feb	0.00064532581 98857195	RPS6; RPS6KB2
WikiPathways _2024_Human	Hub genes 0.05 Top800	Perturbations Host Cell Autophagy Induced By SARS CoV 2 Prots WP4936	Feb-42	0.0010259768 339842233	EIF4B; RPS6
WikiPathways _2024_Human	Hub genes 0.05 Top800	Translation Factors WP107	Feb-42	0.0010259768 339842233	EEF2; EIF4B
WikiPathways _2024_Human	Hub genes 0.05 Top800	IL2 Signaling WP49	Feb-36	0.00102597683 39842233	RPS6; RPS6KB2
WikiPathways _2024_Human	Hub genes 0.05 Top800	IL5 Signaling WP127	Feb-36	0.00102597683 39842233	RPS6; RPS6KB2
WikiPathways _2024_Human	Hub genes 0.05 Top800	Interferon Type I Signaling WP585	Feb-51	0.00121504265 15335647	EIF4B; RPS6
WikiPathways _2024_Human	Hub genes	Focal Adhesion PI3K Akt	3/287	0.00121504265 15335647	EIF4B; RPS6;

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	0.05 Top800	mTOR Signaling WP3932			RPS6KB2
WikiPathways _2024_Human	Hub genes 0.05 Top800	PI3K Akt Signaling WP4172	3/309	0.001359092974 3520655	EIF4B; RPS6; RPS6KB2

Table S2: (D) S_rank_correlations. Spearman rank correlations of Fisher scores between feature-selection thresholds within each dataset (MDD and PTSD), and between the two datasets at each threshold, summarizing the stability and cross-disorder consistency of gene rankings.

Comparison	Genes _compared	Spearman _rho	Top_N	Top_N _overlap	Jaccard	Spearman_p
MDD: 0.05 vs 0.08	3115	0.903	500	479	0.92	
MDD: 0.05 vs 0.08	3115	0.903	800	760	0.9	
MDD: 0.05 vs 0.08	3115	0.903	1000	949	0.9	
PTSD: 0.05 vs 0.08	496	0.976	500	472	0.9	
PTSD: 0.05 vs 0.08	496	0.976	800	495	0.62	
PTSD: 0.05 vs 0.08	496	0.976	1000	496	0.58	
MDD vs PTSD at 0.05	55	-0.051				0.709541504 8399276
MDD vs PTSD at 0.08	140	-0.001				0.9895253439 835724

Table S3: (A) Bootstrap stability of Fisher-score selection across cut-offs from 25 to 3,000 genes (200 bootstrap samples), reported as the Nogueira stability index for MDD (GSE98793) and PTSD (GSE63878), with the knee point of each ranked Fisher-score curve and the primary cut-off of 800 genes marked.

Dataset	Genes	K	Nogueira_stability	Note
MDD (GSE98793)	4607	25	0.104	
MDD (GSE98793)	4607	50	0.112	
MDD (GSE98793)	4607	100	0.122	
MDD (GSE98793)	4607	200	0.129	
MDD (GSE98793)	4607	300	0.132	
MDD (GSE98793)	4607	400	0.134	
MDD (GSE98793)	4607	500	0.136	
MDD (GSE98793)	4607	579	0.136	knee
MDD (GSE98793)	4607	600	0.136	

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MDD (GSE98793)	4607	700	0.135	
MDD (GSE98793)	4607	800	0.135	primary
MDD (GSE98793)	4607	900	0.134	
MDD (GSE98793)	4607	1000	0.133	
MDD (GSE98793)	4607	1500	0.125	
MDD (GSE98793)	4607	2000	0.114	
MDD (GSE98793)	4607	3000	0.086	
PTSD (GSE63878)	858	25	0.103	
PTSD (GSE63878)	858	50	0.107	
PTSD (GSE63878)	858	80	0.112	knee
PTSD (GSE63878)	858	100	0.114	
PTSD (GSE63878)	858	200	0.115	
PTSD (GSE63878)	858	300	0.105	
PTSD (GSE63878)	858	400	0.093	
PTSD (GSE63878)	858	500	0.081	
PTSD (GSE63878)	858	600	0.069	
PTSD (GSE63878)	858	700	0.056	
PTSD (GSE63878)	858	800	0.035	primary

Table S3: (B) Bootstrap selection frequency of each of the 19 shared differentially expressed genes, reported as the proportion of 200 bootstrap resamples in which the gene was retained among the top-ranked features in both the MDD (GSE98793) and PTSD (GSE63878) datasets.

Gene	In primary 19	Selected in MDD Top-800 (% bootstraps)	Selected in PTSD Top-800 (% bootstraps)	Shared in both (% bootstraps)
TDRD9	yes	90	94.5	85
SYTL2	yes	66	96	63
OLFM4	yes	68	92	62.5
PML	yes	48	100	48
PLXNA2	yes	43.5	88	38
CYP2U1	yes	40.5	91.5	37
BTN3A2	yes	38	91.5	35.5
DKKL1	yes	35	98.5	35
HGF	yes	37	89	33.5
GALNT11	yes	33	100	33
EIF4B	yes	38.5	87.5	33
IGF1R	yes	41	80	32.5
AHI1	yes	30.5	97	30
DOCK4	yes	31.5	93.5	30
GFI1B	yes	34.5	88.5	28.5
SLC22A14	yes	29	89	28
PTMA	yes	27.5	93.5	26.5

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STAT2	yes	28.5	90	26.5
SCAPER	yes	24.5	100	24.5

Table S3: (C) Summary of the bootstrap stability of the 19-gene shared set, including the median and interquartile range of the Jaccard similarity between the shared sets obtained from each bootstrap resample and the original (full-dataset) shared set.

Bootstrap _samples	Shared_ genes_ observe d	Shared_ genes_ bootstrap _median	Shared_ genes_ bootstrap _IQR	Jaccard_ to_ observed _median	Jaccard _IQR	Genes_ shared_ in_50pct_of _bootstraps	Genes_ shared_ in_80pct_ of_ bootstrap s
200	19	21	18-24	0.22	0.17-0.26	3	1

Table S3: (D) Observed overlap between the top-ranked gene lists of the MDD (GSE98793) and PTSD (GSE63878) datasets compared with the overlap expected by chance, assessed by a hypergeometric test against the background of genes differentially expressed in both datasets.

K	Note	Observed _shared	In_ Universe _MDD	In_ Universe _PTSD	Universe	Expected	Hypergeo metric_p	Permuta tion_mean	Permut ation_p
80	knee PTSD	0	3	16	140	0.34	1	0.22	1
100		0	4	20	140	0.57	1	0.32	1
200		1	4	37	140	1.06	0.712	1.33	0.706
300		2	7	55	140	2.75	0.839	3.02	0.776
500		6	13	87	140	8.08	0.937	8.52	0.836
579	knee MDD	9	15	95	140	10.18	0.837	11.71	0.806
800	primary	19	21	131	140	19.65	0.866	22.52	0.786
1000		27	27	140	140	27	1	30.06	0.711

Table S4: (A) Cross-validated classification performance within each discovery dataset (PTSD, GSE63878; MDD, GSE98793). Logistic regression, linear SVM and random forest were evaluated under in-fold gene selection (A; 5-fold and 5-fold $\times$ 10), leaky selection before CV (B), nested selection in inner folds (C), shared genes re-derived in each fold (D), a fixed set of 19 shared genes (D-ref, leaky), and nominal-p screening with Fisher selection in each fold (E), together with a label-permutation test (P; 100 permutations). Values are mean $\pm$ SD across folds.

Dataset	Model	Scheme	Genes	AUC	Balanced Acc	Sensiti vity	Specifi city	F1	MCC
PTSD (GSE63878)	LogReg	A: in-fold selection, 5-fold	800	0.947 $\pm$ 0.039	0.894 $\pm$ 0.073	0.918 $\pm$ 0.117	0.871 $\pm$ 0.110	0.898 $\pm$ 0.076	0.803 $\pm$ 0.136

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PTSD (GSE63878)	LogReg	A: in-fold selection, 5-fold x 10	800	0.943 ± 0.053	0.862 ± 0.085	0.871 ± 0.120	0.853 ± 0.116	0.862 ± 0.088	0.734 ± 0.168
PTSD (GSE63878)	LogReg	B: leaky (selection before CV)	800	0.952 ± 0.045	0.916 ± 0.068	0.918 ± 0.117	0.913 ± 0.083	0.916 ± 0.075	0.843 ± 0.131
PTSD (GSE63878)	Linear SVM	A: in-fold selection, 5-fold	800	0.958 ± 0.035	0.906 ± 0.066	0.918 ± 0.117	0.893 ± 0.070	0.906 ± 0.074	0.821 ± 0.129
PTSD (GSE63878)	LinearS VM	A: in-fold selection, 5-fold x 10	800	0.953 ± 0.042	0.883 ± 0.078	0.886 ± 0.113	0.879 ± 0.108	0.881 ± 0.081	0.773 ± 0.154
PTSD (GSE63878)	Linear SVM	B: leaky (selection before CV)	800	0.967 ± 0.039	0.906 ± 0.066	0.918 ± 0.117	0.893 ± 0.070	0.906 ± 0.074	0.821 ± 0.129
PTSD (GSE63878)	Random Forest	A: in-fold selection, 5-fold	800	0.810 ± 0.088	0.738 ± 0.100	0.709 ± 0.073	0.767 ± 0.167	0.735 ± 0.083	0.481 ± 0.200
PTSD (GSE63878)	Random Forest	A: in-fold selection, 5-fold x 10	800	0.797 ± 0.112	0.711 ± 0.104	0.688 ± 0.155	0.735 ± 0.156	0.701 ± 0.113	0.434 ± 0.213
PTSD (GSE63878)	Random Forest	B: leaky (selection before CV)	800	0.822 ± 0.089	0.749 ± 0.146	0.733 ± 0.096	0.764 ± 0.236	0.756 ± 0.118	0.505 ± 0.297
PTSD (GSE63878)	LogReg	C: nested (genes chosen in inner folds)	800; 800; 800; 800; 800	0.947 ± 0.039					
PTSD (GSE63878)	LogReg	D: shared genes re-derived in each fold	21; 18; 20; 16; 21	0.799 ± 0.140					
PTSD (GSE63878)	LogReg	D-ref: fixed 19 shared genes (leaky)	19	0.810 ± 0.145	0.706 ± 0.098	0.660 ± 0.154	0.751 ± 0.121	0.687 ± 0.121	0.423 ± 0.202
PTSD (GSE63878)	LogReg	E: nominal- p screening and Fisher selection in each fold	18761 genes; nominal p<0.08 per fold [667, 761, 508, 1123, 569]	0.763 ± 0.075	0.676 ± 0.121	0.751 ± 0.076	0.600 ± 0.234	0.706 ± 0.089	0.356 ± 0.243

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PTSD (GSE63878)	LogReg	P: label permutation (100)	800	observed 0.947; null 0.488; p = 0.0099					
MDD (GSE98793)	LogReg	A: in-fold selection, 5-fold	800	0.732 ± 0.034	0.653 ± 0.044	0.836 ± 0.051	0.471 ± 0.080	0.795 ± 0.032	0.330 ± 0.092
MDD (GSE98793)	LogReg	A: in-fold selection, 5-fold x 10	800	0.728 ± 0.074	0.620 ± 0.074	0.826 ± 0.075	0.414 ± 0.134	0.779 ± 0.049	0.261 ± 0.159
MDD (GSE98793)	LogReg	B: leaky (selection before CV)	800	0.938 ± 0.013	0.860 ± 0.017	0.906 ± 0.030	0.813 ± 0.036	0.906 ± 0.015	0.721 ± 0.036
MDD (GSE98793)	Linear SVM	A: in-fold selection, 5-fold	800	0.728 ± 0.018	0.610 ± 0.028	0.829 ± 0.052	0.391 ± 0.085	0.776 ± 0.021	0.243 ± 0.049
MDD (GSE98793)	Linear SVM	A: in-fold selection, 5-fold x 10	800	0.720 ± 0.072	0.605 ± 0.068	0.833 ± 0.071	0.378 ± 0.124	0.776 ± 0.045	0.235 ± 0.150
MDD (GSE98793)	Linear SVM	B: leaky (selection before CV)	800	0.942 ± 0.007	0.845 ± 0.060	0.906 ± 0.018	0.783 ± 0.131	0.900 ± 0.023	0.695 ± 0.093
MDD (GSE98793)	Random Forest	A: in-fold selection, 5-fold	800	0.647 ± 0.027	0.539 ± 0.046	0.907 ± 0.039	0.171 ± 0.056	0.781 ± 0.023	0.117 ± 0.134
MDD (GSE98793)	Random Forest	A: in-fold selection, 5-fold x 10	800	0.663 ± 0.071	0.546 ± 0.054	0.886 ± 0.079	0.206 ± 0.097	0.775 ± 0.042	0.131 ± 0.152
MDD (GSE98793)	Random Forest	B: leaky (selection before CV)	800	0.816 ± 0.022	0.676 ± 0.033	0.946 ± 0.046	0.405 ± 0.050	0.843 ± 0.020	0.448 ± 0.102
MDD (GSE98793)	LogReg	C: nested (genes chosen in inner folds)	1000; 1000; 800; 1000; 1000	0.737 ± 0.050					
MDD (GSE98793)	LogReg	D: shared genes re-derived in each fold	19; 19; 24; 17; 22	0.593 ± 0.034					
MDD (GSE98793)	LogReg	D-ref: fixed 19 shared genes (leaky)	19	0.781 ± 0.078	0.724 ± 0.088	0.743 ± 0.068	0.704 ± 0.121	0.786 ± 0.062	0.430 ± 0.172
MDD (GSE98793)	LogReg	E: nominal-p screening	20848 genes; nominal	0.656 ± 0.036	0.591 ± 0.074	0.789 ± 0.038	0.394 ± 0.156	0.754 ± 0.032	0.186 ± 0.151

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		and Fisher selection in each fold	p<0.08 per fold [1705, 1963, 2301, 2761, 3040]						
MDD (GSE98793)	LogReg	P: label permutation (100)	800	observed 0.732; null 0.508; p = 0.0099					

Table S4: (B) Fold-level cross-validation results (AUC, balanced accuracy, sensitivity, specificity, F1, MCC) for scheme A (in-fold selection, 5-fold CV) for each classifier in both datasets.

Dataset	Model	Scheme	Fold	AUC	BalancedAcc	Sensitivity	Specificity	F1	MCC
PTSD	LogReg	A	1	0.88	0.8	0.7	0.9	0.7778	0.6124
PTSD	LogReg	A	2	0.9444	0.8333	1	0.6667	0.8696	0.7161
PTSD	LogReg	A	3	0.9667	0.9444	1	0.8889	0.9524	0.8989
PTSD	LogReg	A	4	0.9444	0.8944	0.8889	0.9	0.8889	0.7889
PTSD	LogReg	A	5	1	1	1	1	1	1
PTSD	Linear SVM	A	1	0.9	0.8	0.7	0.9	0.7778	0.6124
PTSD	Linear SVM	A	2	0.9556	0.8889	1	0.7778	0.9091	0.8051
PTSD	Linear SVM	A	3	0.9889	0.9444	1	0.8889	0.9524	0.8989
PTSD	Linear SVM	A	4	0.9444	0.8944	0.8889	0.9	0.8889	0.7889
PTSD	Linear SVM	A	5	1	1	1	1	1	1
PTSD	Random Forest	A	1	0.745	0.7	0.6	0.8	0.6667	0.4082
PTSD	Random Forest	A	2	0.6944	0.5722	0.7	0.4444	0.6364	0.1495
PTSD	Random Forest	A	3	0.9111	0.8444	0.8	0.8889	0.8421	0.6889
PTSD	Random Forest	A	4	0.7889	0.7333	0.6667	0.8	0.7059	0.4719
PTSD	Random Forest	A	5	0.9111	0.8389	0.7778	0.9	0.8235	0.6854
MDD	LogReg	A	1	0.7249	0.5769	0.7692	0.3846	0.7407	0.1612
MDD	LogReg	A	2	0.7367	0.6538	0.8462	0.4615	0.8	0.3322
MDD	LogReg	A	3	0.7815	0.6523	0.92	0.3846	0.8214	0.3728

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MDD	LogReg	A	4	0.6769	0.6692	0.8	0.5385	0.7843	0.3454
MDD	LogReg	A	5	0.7404	0.7147	0.8462	0.5833	0.8302	0.4402
MDD	Linear SVM	A	1	0.7337	0.5769	0.7692	0.3846	0.7407	0.1612
MDD	Linear SVM	A	2	0.7396	0.6346	0.8077	0.4615	0.7778	0.282
MDD	Linear SVM	A	3	0.7292	0.5754	0.92	0.2308	0.7931	0.2116
MDD	Linear SVM	A	4	0.6923	0.6308	0.8	0.4615	0.7692	0.2736
MDD	Linear SVM	A	5	0.7436	0.6314	0.8462	0.4167	0.8	0.2874
MDD	Random Forest	A	1	0.6672	0.5192	0.8846	0.1538	0.7667	0.0542
MDD	Random Forest	A	2	0.6553	0.5385	0.9231	0.1538	0.7869	0.1195
MDD	Random Forest	A	3	0.6815	0.5754	0.92	0.2308	0.7931	0.2116
MDD	Random Forest	A	4	0.6246	0.5954	0.96	0.2308	0.8136	0.2949
MDD	Random Forest	A	5	0.6074	0.4647	0.8462	0.0833	0.7458	-0.097

Table S5: (A) Differential expression of the 19 shared genes in long format (one row per gene and dataset): log2 fold change, nominal p-value, Benjamini–Hochberg-adjusted p-value, direction of change, Fisher score and Fisher-score rank.

Gene	Disorder	Dataset	log2FC	P_value	adj_P	Direction	Fisher_score	Fisher_rank	Genes_ranked
TDRD9	MDD	GSE98793	0.4916939 09648437	1.5041 8295 42361 5e-05	0.044 7988 6604 2736	Up	0.1017035 83107502	13	4607
OLFM4	MDD	GSE98793	1.025513 79782812	0.000 24964 20318 81252	0.202 2532 5989 3403	Up	0.0715859 901931311	53	4607
SYTL2	MDD	GSE98793	-0.282 881115	0.001 63695 98792 089	0.304 7083 8894 417	Down	0.0644177 267838584	86	4607
PML	MDD	GSE98793	0.05429236 03124997	0.2110 9858 6023 358	0.825 8874 8452 1201	Up	0.0501451 245418336	245	4607
BTN3A2	MDD	GSE98793	-0.25285	0.0103	0.482	Down	0.0477267	298	4607

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			3474	7389 26412 596	4901 5263 9169		424860599		
PLXNA2	MDD	GSE98793	0.07688859 60052083	0.0284 9322 39246 844	0.579 5586 6183 5187	Up	0.0463210 292347639	339	4607
EIF4B	MDD	GSE98793	-0.09343 4017	0.0018 6780 64128 5572	0.314 0324 8463 884	Down	0.0463167 872956826	340	4607
CYP2U1	MDD	GSE98793	-0.0929 43472	0.0236 9966 7947 1091	0.556 2747 3381 8775	Down	0.0460148 614115867	352	4607
IGF1R	MDD	GSE98793	0.1541271 49021875	0.0007 7500 51753 02617	0.244 3066 3766 4484	Up	0.0445124 728471667	393	4607
GALNT11	MDD	GSE98793	-0.15140 6653	0.0042 8467 34120 0865	0.376 9066 2993 0617	Down	0.04371209 35649609	423	4607
HGF	MDD	GSE98793	0.09736132 62218746	0.011 30851 24051 009	0.484 6892 8189 1917	Up	0.04339440 12366486	433	4607
DKKL1	MDD	GSE98793	0.1367805 27992187	0.005 39879 20476 1626	0.403 6272 2751 4571	Up	0.04154970 06567192	506	4607
GFI1B	MDD	GSE98793	0.1542629 47570312	0.035 0709 8325 39022	0.619 5176 1697 988	Up	0.04100978 82151768	533	4607
PTMA	MDD	GSE98793	-0.05732 5514	0.010 33356 35013 127	0.482 4901 5263 9169	Down	0.03904715 90922254	610	4607
AHI1	MDD	GSE98793	-0.07131 0415	0.0653 7136 37581 797	0.705 8241 3669 1019	Down	0.03875975 04964006	630	4607
SCAPER	MDD	GSE98793	-0.11381 8928	0.161 5838 0331 4479	0.791 5176 5307 807	Down	0.03830173 97115078	653	4607
SLC22A14	MDD	GSE98793	0.1422134 18140625	0.007 72982	0.432 8933	Up	0.03789498 00233012	680	4607

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				92291 0484	7856 9633				
DOCK4	MDD	GSE98793	0.123221 7686875	0.0628 0749 71082 228	0.701 9491 6899 526	Up	0.0377874 507087722	688	4607
STAT2	MDD	GSE98793	0.0692572 001848954	0.0203 7212 6065 1095	0.539 3957 7500 1486	Up	0.0363333 543436421	775	4607
TDRD9	PTSD	GSE63878	0.1124580 90333334	0.0151 5133 38618 435	0.999 9079 8861 3799	Up	0.0655375 82412832	149	858
OLFM4	PTSD	GSE63878	0.35159 3043	0.0412 5381 4477 8537	0.999 9079 8861 3799	Up	0.0433292 526486972	479	858
SYTL2	PTSD	GSE63878	0.1286220 77333334	0.0374 8874 2233 1435	0.999 9079 8861 3799	Up	0.0464005 292957898	412	858
PML	PTSD	GSE63878	0.21271497 2604167	0.0089 5061 74722 8717	0.999 9079 8861 3799	Up	0.0731560 749687013	100	858
BTN3A2	PTSD	GSE63878	0.1405671 41666667	0.062 6599 2296 94918	0.999 9079 8861 3799	Up	0.0365742 561303767	734	858
PLXNA2	PTSD	GSE63878	-0.06336 6838	0.065 7997 56412 3711	0.999 9079 8861 3799	Down	0.0390639 758694099	632	858
EIF4B	PTSD	GSE63878	-0.07335 1227	0.415 3062 5374 2068	0.999 9079 88613 799	Down	0.0537560 427120008	273	858
CYP2U1	PTSD	GSE63878	0.07584453 18333337	0.054 7665 40897 7158	0.999 9079 88613 799	Up	0.0414710 500195557	546	858
IGF1R	PTSD	GSE63878	-0.11751 6244	0.073 1884 96198 4662	0.999 9079 8861 3799	Down	0.0364239 256491172	741	858
GALNT11	PTSD	GSE63878	0.0975079 503541669	0.006 42052 62735	0.999 9079 8861	Up	0.0869649 891587304	53	858

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				0075	3799				
HGF	PTSD	GSE63878	0.0933666 919791674	0.060 7202 2387 60104	0.999 9079 8861 3799	Up	0.03825795 1198919	665	858
DKKL1	PTSD	GSE63878	-0.12612 1489	0.017 8454 7220 23961	0.999 9079 88613 799	Down	0.0613025 056526537	187	858
GFI1B	PTSD	GSE63878	-0.0735 81289	0.054 2078 12652 0436	0.999 9079 8861 3799	Down	0.0419158 275386644	532	858
PTMA	PTSD	GSE63878	-0.0668 17656	0.041 8335 09225 1867	0.999 9079 8861 3799	Down	0.0509385 995607623	321	858
AHI1	PTSD	GSE63878	0.1463765 79354167	0.025 1450 2157 30093	0.999 9079 8861 3799	Up	0.05375308 83082927	274	858
SCAPER	PTSD	GSE63878	0.1235698 41041667	0.006 3254 13727 76395	0.999 9079 8861 3799	Up	0.08378492 61296336	63	858
SLC22A14	PTSD	GSE63878	-0.06718 8073	0.072 68134 9256 4018	0.999 9079 8861 3799	Down	0.0364795 894972618	736	858
DOCK4	PTSD	GSE63878	0.1743952 80770834	0.0409 7945 3717 6084	0.999 9079 8861 3799	Up	0.04402854 52857901	460	858
STAT2	PTSD	GSE63878	0.1771395 73958334	0.066 98465 6350 7512	0.999 9079 8861 3799	Up	0.03508251 08405193	794	858

Table S5: (B) The same statistics shown side by side for MDD (GSE98793) and PTSD (GSE63878), one row per gene, with concordance of direction. Values were obtained with limma on gene-level expression (probes averaged per gene); because genes were originally selected at the probe level, a few genes have gene-level p-values above 0.08.

Gene	MDD log2FC	MDD P	MDD adj. P (BH)	MDD direc tion	MDD Fisher rank	PTSD log2FC	PTSD P	PTSD adj. P (BH)	PTSD direc tion	PTSD Fisher rank	Concor dance
DOCK4	0.123	0.06 28	0.702	Up	688/ 4607	0.174	0.041	1	Up	460/ 858	Concor dant

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EIF4B	-0.093	0.00187	0.314	Down	340/4607	-0.073	0.415	1	Down	273/858	Concordant
HGF	0.097	0.0113	0.485	Up	433/4607	0.093	0.0607	1	Up	665/858	Concordant
OLFM4	1.026	0.00025	0.202	Up	53/4607	0.352	0.0413	1	Up	479/858	Concordant
PML	0.054	0.211	0.826	Up	245/4607	0.213	0.00895	1	Up	100/858	Concordant
PTMA	-0.057	0.0103	0.482	Down	610/4607	-0.067	0.0418	1	Down	321/858	Concordant
STAT2	0.069	0.0204	0.539	Up	775/4607	0.177	0.067	1	Up	794/858	Concordant
TDRD9	0.492	1.5e-05	0.0448	Up	13/4607	0.112	0.0152	1	Up	149/858	Concordant
AHI1	-0.071	0.0654	0.706	Down	630/4607	0.146	0.0251	1	Up	274/858	Discordant
BTN3A2	-0.253	0.0104	0.482	Down	298/4607	0.141	0.0627	1	Up	734/858	Discordant
CYP2U1	-0.093	0.0237	0.556	Down	352/4607	0.076	0.0548	1	Up	546/858	Discordant
DKKL1	0.137	0.0054	0.404	Up	506/4607	-0.126	0.0178	1	Down	187/858	Discordant
GALNT11	-0.151	0.00428	0.377	Down	423/4607	0.098	0.00642	1	Up	53/858	Discordant
GFI1B	0.154	0.0351	0.62	Up	533/4607	-0.074	0.0542	1	Down	532/858	Discordant
IGF1R	0.154	0.000775	0.244	Up	393/4607	-0.118	0.0732	1	Down	741/858	Discordant
PLXNA2	0.077	0.0285	0.58	Up	339/4607	-0.063	0.0658	1	Down	632/858	Discordant
SCAPER	-0.114	0.162	0.792	Down	653/4607	0.124	0.00633	1	Up	63/858	Discordant
SLC22A14	0.142	0.00773	0.433	Up	680/4607	-0.067	0.0727	1	Down	736/858	Discordant
SYTL2	-0.283	0.00164	0.305	Down	86/4607	0.129	0.0375	1	Up	412/858	Discordant

Table S6: Topological properties of the ten hub genes in the PPI network, calculated with Cytoscape NetworkAnalyzer.

Genes Name	Degree	Betweenness Centrality	Closeness Centrality	Clustering coefficient	Topological coefficient
IGF1R	59	0.8747149122807021	0.5962732919254659	5.844535359438924E-4	0.023112480739599383
STAT2	21	0.3343201754385965	0.38247011952191234	0.0	0.11904761904761904
PML	13	0.20741228070175446	0.3595505617977528	0.0	0.11538461538461539
HGF	9	0.1412280701754386	0.39834024896265563	0.027777777777777776	0.11487758945386065
HDAC1	2	0.023662280701754378	0.2981366459627329	0.0	0.5161290322580645
JAK2	2	0.08594298245614039	0.4403669724770642	0.0	0.5131578947368421
JAK1	2	0.08594298245614039	0.4403669724770642	0.0	0.5131578947368421
MDM2	2	0.16129385964912282	0.41739130434782606	0.0	0.5

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ERBB2	2	0.0	0.3870967741935484	1.0	0.5151515151515151
STAT3	2	0.08594298245614039	0.4403669724770642	0.0	0.5131578947368421

Table S7: (A) Enrichment analysis settings: gene universe, gene lists, library versions, statistical test, term-size limits and multiple-testing procedure.

Item	Setting
Gene universe	16,911 genes measured on both platforms (GSE98793, GSE63878)
Gene lists	Shared DEGs (19): TDRD9, OLFM4, SYTL2, PML, BTN3A2, PLXNA2, EIF4B, CYP2U1, IGF1R, GALNT11, HGF, DKKL1, GFI1B, PTMA, AHI1, SCAPER, SLC22A14, DOCK4, STAT2; Hub genes (10): IGF1R, STAT2, PML, HGF, HDAC1, JAK2, JAK1, MDM2, ERBB2, STAT3
Genes not in universe	Shared DEGs (19): none; Hub genes (10): none
Main libraries (Enrichr)	GO_Biological_Process_2025, GO_Molecular_Function_2025, GO_Cellular_Component_2025, KEGG_2021_Human, WikiPathways_2024_Human
Supplementary libraries (Enrichr)	Reactome_Pathways_2024, BioCarta_2016, BioPlanet_2019, DSigDB
Library download date	10/3/2026
Test	One-sided hypergeometric test; terms with at least one input gene
Term size filter	5-2000 genes within the universe
Multiple testing	Benjamini-Hochberg FDR within each library and gene list; significance FDR < 0.05
Enrichment ratio	(k/n)/(K/N): k overlap, n input genes, K term size, N universe size

Table S7: (B) Number of terms in each library, number tested, number significant (FDR < 0.05) and lowest FDR for the shared DEGs and hub genes.

Gene_list	Library	Terms_in_library	Terms_tested	Significant_FDR_0_05	Min_FDR
Shared DEGs (19)	GO_Biological_Process_2025	5343	220	0	0.1048
Hub genes (10)	GO_Biological_Process_2025	5343	440	335	0
Shared DEGs (19)	GO_Molecular_Function_2025	1174	45	0	0.1001
Hub genes (10)	GO_Molecular_Function_2025	1174	51	41	0
Shared DEGs (19)	GO_Cellular_Component_2025	468	30	0	0.1235
Hub genes (10)	GO_Cellular_Component_2025	468	24	10	0.0061

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Shared DEGs (19)	KEGG_2021_Human	320	54	4	0.0335
Hub genes (10)	KEGG_2021_Human	320	84	69	0
Shared DEGs (19)	WikiPathways_2024_Human	829	76	3	0.0415
Hub genes (10)	WikiPathways_2024_Human	829	180	159	0
Shared DEGs (19)	Reactome_Pathways_2024	2105	129	0	0.122
Hub genes (10)	Reactome_Pathways_2024	2105	273	234	0
Shared DEGs (19)	BioCarta_2016	238	8	8	0.0266
Hub genes (10)	BioCarta_2016	238	47	47	0
Shared DEGs (19)	BioPlanet_2019	1510	114	0	0.0962
Hub genes (10)	BioPlanet_2019	1510	236	198	0
Shared DEGs (19)	DSigDB	4027	457	0	0.1865
Hub genes (10)	DSigDB	4027	1075	744	0

Table S7: © Top 15 terms per library for the 19 shared DEGs and 10 hub genes (GO BP, MF and CC; KEGG; WikiPathways; Reactome; BioCarta; BioPlanet; DSigDB), ranked by FDR, with overlap, p-value, FDR, enrichment ratio, odds ratio and contributing genes. Results for all tested terms are provided as a separate Excel file.

Gene list	Library	Term	Overlap (k/K)	P-value	FDR	Enrichment ratio	Odds ratio	Significant (FDR < 0.05)	Contributing genes
Shared DEGs (19)	GO BP 2025	Regulation of Apoptotic Process	4/613	0.004	0.105	5.8	7.1	No	AHI1, DKKL1, IGF1R, PTMA
Shared DEGs (19)	GO BP 2025	Cytochrome Metabolic Process	5-Jan	0.006	0.105	178	234.6	No	CYP2U1
Shared DEGs (19)	GO BP 2025	Negative Regulation of Vasoconstriction	5-Jan	0.006	0.105	178	234.6	No	DOCK4
Shared DEGs (19)	GO BP 2025	Regulation of Translational Initiation in Response to Stress	6-Jan	0.007	0.105	148.3	187.6	No	PML

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Shared DEGs (19)	GO BP 2025	Notch Receptor Processing	7-Jan	0.008	0.105	127.2	156.4	No	GALNT11
Shared DEGs (19)	GO BP 2025	Dendritic Spine Maintenance	7-Jan	0.008	0.105	127.2	156.4	No	IGF1R
Shared DEGs (19)	GO BP 2025	Formation of Translation Preinitiation Complex	7-Jan	0.008	0.105	127.2	156.4	No	EIF4B
Shared DEGs (19)	GO BP 2025	Protein O-linked Glycosylation via Threonine	7-Jan	0.008	0.105	127.2	156.4	No	GALNT11
Shared DEGs (19)	GO BP 2025	Positive Regulation of Vascular Associated Smooth Muscle Cell Migration	8-Jan	0.009	0.105	111.3	134	No	DOCK4
Shared DEGs (19)	GO BP 2025	Negative Regulation of Steroid Biosynthetic Process	9-Jan	0.01	0.105	98.9	117.2	No	DKKL1
Shared DEGs (19)	GO BP 2025	Organic Acid Metabolic Process	9-Jan	0.01	0.105	98.9	117.2	No	CYP2U1
Shared DEGs (19)	GO BP 2025	Peptidyl-Tyrosine Autophosphorylation	9-Jan	0.01	0.105	98.9	117.2	No	IGF1R
Shared DEGs (19)	GO BP 2025	Negative Regulation of Apoptotic Process	3/421	0.011	0.105	6.3	7.4	No	AHI1, IGF1R, PTMA
Shared DEGs (19)	GO BP 2025	Cellular Component Maintenance	10-Jan	0.011	0.105	89	104.2	No	IGF1R
Shared DEGs (19)	GO BP 2025	Gene Silencing by piRNA-directed DNA Methylation	10-Jan	0.011	0.105	89	104.2	No	TDRD9
Shared DEGs (19)	GO MF 2025	Riboflavin Transmembrane Transporter Activity	5-Jan	0.006	0.1	178	234.6	No	SLC22A14
Shared DEGs (19)	GO MF 2025	Neurexin Family Protein Binding	10-Jan	0.011	0.1	89	104.2	No	SYTL2
Shared DEGs (19)	GO MF 2025	Semaphorin Receptor Activity	10-Jan	0.011	0.1	89	104.2	No	PLXNA2
Shared DEGs (19)	GO MF 2025	SUMO Binding	13-Jan	0.015	0.1	68.5	78.1	No	PML
Shared DEGs (19)	GO MF 2025	Insulin-Like Growth Factor Binding	14-Jan	0.016	0.1	63.6	72.1	No	IGF1R

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Shared DEGs (19)	GO MF 2025	Insulin Receptor Binding	17-Jan	0.019	0.1	52.4	58.6	No	IGF1R
Shared DEGs (19)	GO MF 2025	Receptor Antagonist Activity	17-Jan	0.019	0.1	52.4	58.6	No	DKKL1
Shared DEGs (19)	GO MF 2025	Ribosomal Small Subunit Binding	18-Jan	0.02	0.1	49.4	55.1	No	EIF4B
Shared DEGs (19)	GO MF 2025	Notch Binding	20-Jan	0.022	0.1	44.5	49.3	No	GALNT11
Shared DEGs (19)	GO MF 2025	Steroid Hydroxylase Activity	20-Jan	0.022	0.1	44.5	49.3	No	CYP2U1
Shared DEGs (19)	GO MF 2025	SUMO Transferase Activity	23-Jan	0.026	0.104	38.7	42.6	No	PML
Shared DEGs (19)	GO MF 2025	Oxidoreduc tase, Paired Donors, W Incorp or Reduc of O2, Reduced Flav in or Flavoprotein as One Donor, and Incorp of O	26-Jan	0.029	0.108	34.2	37.5	No	CYP2U1
Shared DEGs (19)	GO MF 2025	Ubiquitin-Like Protein Ligase Binding	2/253	0.032	0.112	7	7.8	No	PML, STAT2
Shared DEGs (19)	GO MF 2025	DNA-binding Transcription Factor Binding	2/274	0.037	0.12	6.5	7.2	No	GFI1B, PTMA
Shared DEGs (19)	GO MF 2025	Single-Stranded RNA Binding	Jan-43	0.047	0.139	20.7	22.3	No	EIF4B
Shared DEGs (19)	GO CC 2025	Transferase Complex, Transferring Phosphorus-Containing Groups	7-Jan	0.008	0.123	127.2	156.4	No	IGF1R
Shared DEGs (19)	GO CC 2025	Primary Lysosome	11-Jan	0.012	0.123	80.9	93.8	No	OLFM4
Shared DEGs (19)	GO CC 2025	Protein Kinase Complex	21-Jan	0.023	0.123	42.4	46.9	No	IGF1R
Shared DEGs (19)	GO CC 2025	P Granule	23-Jan	0.026	0.123	38.7	42.6	No	TDRD9
Shared DEGs (19)	GO CC 2025	Pigment Granule	24-Jan	0.027	0.123	37.1	40.7	No	SYTL2
Shared DEGs (19)	GO CC 2025	Exocytic Vesicle	26-Jan	0.029	0.123	34.2	37.5	No	SYTL2
Shared DEGs (19)	GO CC 2025	Secretory Vesicle	28-Jan	0.031	0.123	31.8	34.7	No	SYTL2

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Shared DEGs (19)	GO CC 2025	Melanosome	31-Jan	0.034	0.123	28.7	31.2	No	SYTL2
Shared DEGs (19)	GO CC 2025	Nucleolus	3/685	0.04	0.123	3.9	4.5	No	DOCK4, IGF1R, PML
Shared DEGs (19)	GO CC 2025	Nuclear Lumen	3/696	0.041	0.123	3.8	4.4	No	DOCK4, IGF1R, PML
Shared DEGs (19)	GO CC 2025	Tertiary Granule Lumen	Jan-50	0.055	0.138	17.8	19.1	No	OLFM4
Shared DEGs (19)	GO CC 2025	Cilium	2/341	0.055	0.138	5.2	5.7	No	AHI1, IGF1R
Shared DEGs (19)	GO CC 2025	Acrosomal Vesicle	Jan-57	0.062	0.138	15.6	16.7	No	DKKL1
Shared DEGs (19)	GO CC 2025	Specific Granule Lumen	Jan-59	0.064	0.138	15.1	16.1	No	OLFM4
Shared DEGs (19)	GO CC 2025	Intracellular Membrane less Organelle	3/1128	0.129	0.252	2.4	2.6	No	DOCK4, IGF1R, PML
Shared DEGs (19)	KEGG 2021	Proteoglycans in cancer	3/187	0.001	0.033	14.3	17	Yes	EIF4B, HGF, IGF1R
Shared DEGs (19)	KEGG 2021	Rap1 signaling pathway	3/193	0.001	0.033	13.8	16.5	Yes	DOCK4, HGF, IGF1R
Shared DEGs (19)	KEGG 2021	Pathways in cancer	4/490	0.002	0.034	7.3	9	Yes	HGF, IGF1R, PML, STAT2
Shared DEGs (19)	KEGG 2021	Melanoma	Feb-68	0.003	0.035	26.2	30	Yes	HGF, IGF1R
Shared DEGs (19)	KEGG 2021	PI3K-Akt signaling pathway	3/322	0.005	0.057	8.3	9.7	No	EIF4B, HGF, IGF1R
Shared DEGs (19)	KEGG 2021	mTOR signaling pathway	2/141	0.011	0.079	12.6	14.2	No	EIF4B, IGF1R
Shared DEGs (19)	KEGG 2021	Influenza A	2/144	0.011	0.079	12.4	13.9	No	PML, STAT2
Shared DEGs (19)	KEGG 2021	Hepatocellular carcinoma	2/150	0.012	0.079	11.9	13.3	No	HGF, IGF1R
Shared DEGs (19)	KEGG 2021	Transcriptional misregulation in cancer	2/157	0.013	0.079	11.3	12.7	No	IGF1R, PML
Shared DEGs (19)	KEGG 2021	Focal adhesion	2/187	0.018	0.099	9.5	10.6	No	HGF, IGF1R
Shared DEGs (19)	KEGG 2021	Ras signaling pathway	2/208	0.022	0.11	8.6	9.5	No	HGF, IGF1R
Shared DEGs (19)	KEGG 2021	Endocytosis	2/222	0.025	0.114	8	8.9	No	IGF1R, PML
Shared DEGs (19)	KEGG 2021	Mucin type O-glycan	Jan-32	0.035	0.144	27.8	30.2	No	GALNT11

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		biosynthesis							
Shared DEGs (19)	KEGG 2021	MAPK signaling pathway	2/274	0.037	0.144	6.5	7.2	No	HGF, IGF1R
Shared DEGs (19)	KEGG 2021	Fatty acid degradation	Jan-41	0.045	0.148	21.7	23.4	No	CYP2U1
Shared DEGs (19)	WikiPath ways 2024	NRP1 Triggered Signaling In Pancreatic Cancer WP5144	Feb-51	0.001	0.042	34.9	40.4	Yes	HGF, PLXNA2
Shared DEGs (19)	WikiPath ways 2024	Interferon Type I Signaling WP585	Feb-51	0.001	0.042	34.9	40.4	Yes	EIF4B, STAT2
Shared DEGs (19)	WikiPath ways 2024	Cancer Pathways WP5434	4/470	0.002	0.042	7.6	9.4	Yes	HGF, IGF1R, PML, STAT2
Shared DEGs (19)	WikiPath ways 2024	EGFR Tyrosine Kinase Inhibitor Resistance WP4806	Feb-81	0.004	0.058	22	25	No	HGF, IGF1R
Shared DEGs (19)	WikiPath ways 2024	Focal Adhesion PI3K Akt mTOR Signaling WP3932	3/287	0.004	0.058	9.3	11	No	EIF4B, HGF, IGF1R
Shared DEGs (19)	WikiPath Ways 2024	PI3K Akt Signaling WP4172	3/309	0.005	0.06	8.6	10.2	No	EIF4B, HGF, IGF1R
Shared DEGs (19)	WikiPath ways 2024	Somatroph Axis GH Relationship To Dietary Restriction And Aging WP4186	6-Jan	0.007	0.073	148.3	187.6	No	IGF1R
Shared DEGs (19)	WikiPath ways 2024	Ectoderm Differentiation WP2858	2/127	0.009	0.075	14	15.8	No	AHI1, PLXNA2
Shared DEGs (19)	WikiPath ways 2024	Type III Interferon Signaling WP2113	8-Jan	0.009	0.075	111.3	134	No	STAT2
Shared DEGs (19)	WikiPath ways 2024	Pleural Mesothelioma WP5087	3/406	0.01	0.075	6.6	7.7	No	EIF4B, HGF, IGF1R
Shared DEGs (19)	WikiPath Ways 2024	Mod Of PI3K Akt mTOR Signaling By Bioactive Sphingolipids WP5192	11-Jan	0.012	0.075	80.9	93.8	No	IGF1R
Shared DEGs (19)	WikiPath ways 2024	IL10 Anti Inflammatory Signaling WP4495	11-Jan	0.012	0.075	80.9	93.8	No	STAT2

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Shared DEGs (19)	WikiPath ways 2024	Ciliopathies WP4803	2/155	0.013	0.075	11.5	12.9	No	AHI1, GALNT11
Shared DEGs (19)	WikiPath ways 2024	Focal Adhesion WP306	2/183	0.018	0.09	9.7	10.9	No	HGF, IGF1R
Shared DEGs (19)	WikiPath ways 2024	Deregulation Of Rab And Rab Effector Genes In Bladder Cancer WP2291	16-Jan	0.018	0.09	55.6	62.5	No	SYTL2
Shared DEGs (19)	Reactome 2024	MET Activates PTPN11	5-Jan	0.006	0.122	178	234.6	No	HGF
Shared DEGs (19)	Reactome 2024	Respiratory Syncytial Virus Infection Pathway	2/101	0.006	0.122	17.6	20	No	IGF1R, STAT2
Shared DEGs (19)	Reactome 2024	MET Receptor Activation	6-Jan	0.007	0.122	148.3	187.6	No	HGF
Shared DEGs (19)	Reactome 2024	MET Activates PI3K AKT Signaling	6-Jan	0.007	0.122	148.3	187.6	No	HGF
Shared DEGs (19)	Reactome 2024	MET Receptor Recycling	8-Jan	0.009	0.122	111.3	134	No	HGF
Shared DEGs (19)	Reactome 2024	SHC-related Events Triggered by IGF1R	8-Jan	0.009	0.122	111.3	134	No	IGF1R
Shared DEGs (19)	Reactome 2024	Regulation of PTEN Localization	9-Jan	0.01	0.122	98.9	117.2	No	PML
Shared DEGs (19)	Reactome 2024	Synthesis of (16-20)-Hydroxyei cosatetra enoic Acids (HETE)	9-Jan	0.01	0.122	98.9	117.2	No	CYP2U1
Shared DEGs (19)	Reactome 2024	Butyrophil in (BTN) Family Interactions	10-Jan	0.011	0.122	89	104.2	No	BTN3A2
Shared DEGs (19)	Reactome 2024	Z-decay Degrada tion of Maternal mRNAs by Zygotically Expressed Factors	10-Jan	0.011	0.122	89	104.2	No	EIF4B
Shared DEGs (19)	Reactome 2024	MET Activates RAP1 and RAC1	11-Jan	0.012	0.122	80.9	93.8	No	HGF
Shared DEGs (19)	Reactome 2024	MET Activates RAS Signaling	11-Jan	0.012	0.122	80.9	93.8	No	HGF
Shared DEGs (19)	Reactome 2024	Miscella neous Substrates	11-Jan	0.012	0.122	80.9	93.8	No	CYP2U1

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Shared DEGs (19)	Reactome 2024	SEMA3A-Plexin Repulsion Signaling by Inhibiting Integrin Adhesion	14-Jan	0.016	0.125	63.6	72.1	No	PLXNA2
Shared DEGs (19)	Reactome 2024	CRMPs in Sema3A Signaling	16-Jan	0.018	0.125	55.6	62.5	No	PLXNA2
Shared DEGs (19)	BioCarta 2016	IFN alpha signaling pathway Homo sapiens h ifnaPath way	7-Jan	0.008	0.027	127.2	156.4	Yes	STAT2
Shared DEGs (19)	BioCarta 2016	Regulation of transcriptional activity by PML Homo sapiens h pmlPath way	11-Jan	0.012	0.027	80.9	93.8	Yes	PML
Shared DEGs (19)	BioCarta 2016	Multiple antiapoptotic pathways from IGF-1R signaling lead to BAD phosphorylation Homo sapiens h igf1rPath way	12-Jan	0.013	0.027	74.2	85.3	Yes	IGF1R
Shared DEGs (19)	BioCarta 2016	The IGF-1 Receptor and Longevity Homo sapiens h longevity Pathway	16-Jan	0.018	0.027	55.6	62.5	Yes	IGF1R
Shared DEGs (19)	BioCarta 2016	IGF-1 Signaling Pathway Homo sapiens h igf1 Pathway	18-Jan	0.02	0.027	49.4	55.1	Yes	IGF1R
Shared DEGs (19)	BioCarta 2016	mTOR Signaling Pathway Homo sapiens h mTOR Pathway	22-Jan	0.024	0.027	40.5	44.6	Yes	EIF4B
Shared DEGs (19)	BioCarta 2016	Regulation of BAD phosphorylation Homo sapiens h	23-Jan	0.026	0.027	38.7	42.6	Yes	IGF1R

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		badPath way							
Shared DEGs (19)	BioCarta 2016	Skeletal muscle hypertrophy is regulated via AKT/mTOR pathway Homo sapiens h igf1mtor pathway	24-Jan	0.027	0.027	37.1	40.7	Yes	IGF1R
Shared DEGs (19)	BioPlanet 2019	BDNF signaling pathway	3/245	0.002	0.096	10.9	12.9	No	DOCK4, EIF4B, HGF
Shared DEGs (19)	BioPlanet 2019	Immune system signaling by interferons, interleukins, prolactin, and growth hormones	3/246	0.002	0.096	10.9	12.8	No	HGF, PML, STAT2
Shared DEGs (19)	BioPlanet 2019	Melanoma	Feb-67	0.003	0.096	26.6	30.5	No	HGF, IGF1R
Shared DEGs (19)	BioPlanet 2019	Pathways in cancer	3/309	0.005	0.1	8.6	10.2	No	HGF, IGF1R, PML
Shared DEGs (19)	BioPlanet 2019	Cytochrome P450 metabolism of miscellaneous substrates	5-Jan	0.006	0.1	178	234.6	No	CYP2U1
Shared DEGs (19)	BioPlanet 2019	mTOR signaling pathway	2/103	0.006	0.1	17.3	19.6	No	EIF4B, PML
Shared DEGs (19)	BioPlanet 2019	S6K1 signaling	7-Jan	0.008	0.1	127.2	156.4	No	EIF4B
Shared DEGs (19)	BioPlanet 2019	Interferon-alpha signaling pathway	8-Jan	0.009	0.1	111.3	134	No	STAT2
Shared DEGs (19)	BioPlanet 2019	p53 signaling pathway	2/130	0.009	0.1	13.7	15.4	No	HGF, PML
Shared DEGs (19)	BioPlanet 2019	cAMP receptor, G-protein-independent pathways inferred from amoeba model	9-Jan	0.01	0.1	98.9	117.2	No	STAT2
Shared DEGs (19)	BioPlanet 2019	Interferon signaling	2/138	0.01	0.1	12.9	14.5	No	PML, STAT2
Shared DEGs (19)	BioPlanet 2019	Mammalian target of rapamycin	10-Jan	0.011	0.1	89	104.2	No	EIF4B

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		complex 1 (mTORC1)-mediated signaling							
Shared DEGs (19)	BioPlanet 2019	Other semaphorin interactions	14-Jan	0.016	0.1	63.6	72.1	No	PLXNA2
Shared DEGs (19)	BioPlanet 2019	Sema3A-plexin repulsion signaling by inhibiting integrin adhesion	14-Jan	0.016	0.1	63.6	72.1	No	PLXNA2
Shared DEGs (19)	BioPlanet 2019	Sema3A/PAK-dependent axon repulsion	15-Jan	0.017	0.1	59.3	67	No	PLXNA2
Shared DEGs (19)	DSigDB	bucladesine HL60 DOWN	Feb-43	0.001	0.187	41.4	48.4	No	HGF, SCAPER
Shared DEGs (19)	DSigDB	Mitoxantrone CTD 00006345	3/230	0.002	0.187	11.6	13.8	No	IGF1R, PML, SYTL2
Shared DEGs (19)	DSigDB	(-)-Epigallocatechin gallate CTD 00002033	Jul-15	0.003	0.187	3.3	4.6	No	BTN3A2, EIF4B, HGF, IGF1R, PLXNA2, SCAPER, STAT2
Shared DEGs (19)	DSigDB	Doxycycline HL60 UP	3/303	0.004	0.187	8.8	10.4	No	AHI1, CYP2U1, PML
Shared DEGs (19)	DSigDB	Sulfaguanidine PC3 UP	4/622	0.005	0.187	5.7	7	No	AHI1, CYP2U1, DKKL1, PML
Shared DEGs (19)	DSigDB	Kaempferol CTD 00000297	Feb-95	0.005	0.187	18.7	21.3	No	HGF, STAT2
Shared DEGs (19)	DSigDB	rofecoxib CTD 00003631	Feb-95	0.005	0.187	18.7	21.3	No	HGF, PML
Shared DEGs (19)	DSigDB	15(S)-HETE-d8 CTD 00001221	5-Jan	0.006	0.187	178	234.6	No	HGF
Shared DEGs (19)	DSigDB	Metronidazole PC3 UP	5/1077	0.006	0.187	4.1	5.3	No	BTN3A2, IGF1R, PML, SCAPER, STAT2
Shared DEGs (19)	DSigDB	doxorubicin CTD 00005874	4/699	0.007	0.187	5.1	6.2	No	DOCK4, GF11B, HGF, STAT2
Shared DEGs (19)	DSigDB	paclitaxel PC3 UP	2/113	0.007	0.187	15.8	17.8	No	BTN3A2, EIF4B
Shared DEGs (19)	DSigDB	scopolamine PC3 UP	3/361	0.007	0.187	7.4	8.7	No	CYP2U1, DKKL1, PML
Shared DEGs (19)	DSigDB	primaquine PC3 UP	3/363	0.007	0.187	7.4	8.6	No	CYP2U1, DKKL1, PML

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Shared DEGs (19)	DSigDB	lycorine HL60 DOWN	4/732	0.008	0.187	4.9	5.9	No	BTN3A2, GALNT11, HGF, IGF1R
Shared DEGs (19)	DSigDB	Sanguinarine HL60 DOWN	3/387	0.009	0.187	6.9	8.1	No	BTN3A2, HGF, IGF1R
Hub genes (10)	GO BP 2025	Cell Surface Receptor Signaling Pathway via JAK-STAT	Apr-44	8.27e-09	2.08e-06	153.7	281	Yes	JAK1, JAK2, STAT2, STAT3
Hub genes (10)	GO BP 2025	Cell Surface Receptor Signaling Pathway via STAT	Apr-52	1.65e-08	2.08e-06	130.1	234.1	Yes	JAK1, JAK2, STAT2, STAT3
Hub genes (10)	GO BP 2025	Growth Hormone Receptor Signaling Pathway via JAK-STAT	10-Mar	1.78e-08	2.08e-06	507.3	1034.3	Yes	JAK1, JAK2, STAT3
Hub genes (10)	GO BP 2025	Regulation of Cell Population Proliferation	7/686	1.89e-08	2.08e-06	17.3	55.7	Yes	HDAC1, IGF1R, JAK2, MDM2, PML, STAT2, STAT3
Hub genes (10)	GO BP 2025	Interleukin-6-Mediated Signaling Pathway	12-Mar	3.27e-08	2.87e-06	422.8	804.4	Yes	JAK1, JAK2, STAT3
Hub genes (10)	GO BP 2025	Growth Hormone Receptor Signaling Pathway	19-Mar	1.44e-07	1.05e-05	267	452.3	Yes	JAK1, JAK2, STAT3
Hub genes (10)	GO BP 2025	Cellular Response to Interleukin-6	21-Mar	1.97e-07	1.24e-05	241.6	402	Yes	JAK1, JAK2, STAT3
Hub genes (10)	GO BP 2025	Phosphatidylinositol 3-Kinase/Protein Kinase B Signaling Transduction	Mar-37	1.14e-06	6.3e-05	137.1	212.6	Yes	ERBB2, IGF1R, STAT3
Hub genes (10)	GO BP 2025	Interferon-Mediated Signaling Pathway	Mar-46	2.23e-06	0.000106	110.3	168	Yes	JAK1, JAK2, STAT2
Hub genes (10)	GO BP 2025	Positive Regulation of Cell Population Proliferation	5/432	2.41e-06	0.000106	19.6	38.6	Yes	ERBB2, HDAC1, IGF1R, JAK2, MDM2

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Hub genes (10)	GO BP 2025	Interleukin-23-Mediated Signaling Pathway	5-Feb	3.14e-06	0.000126	676.4	1408.2	Yes	JAK2, STAT3
Hub genes (10)	GO BP 2025	Peptidyl-Tyrosine Phosphorylation	Mar-53	3.43e-06	0.000126	95.7	144.4	Yes	ERBB2, IGF1R, JAK2
Hub genes (10)	GO BP 2025	Cellular Response to Interleukin-2	7-Feb	6.6e-06	0.000181	483.2	844.8	Yes	JAK1, STAT3
Hub genes (10)	GO BP 2025	Cellular Response to Interleukin-9	7-Feb	6.6e-06	0.000181	483.2	844.8	Yes	JAK1, STAT3
Hub genes (10)	GO BP 2025	Interleukin-2-Mediated Signaling Pathway	7-Feb	6.6e-06	0.000181	483.2	844.8	Yes	JAK1, STAT3
Hub genes (10)	GO MF 2025	Protein Tyrosine Kinase Activity	Apr-93	1.75e-07	8.95e-06	72.7	125.9	Yes	ERBB2, IGF1R, JAK1, JAK2
Hub genes (10)	GO MF 2025	Growth Hormone Receptor Binding	8-Feb	8.8e-06	0.000163	422.8	704	Yes	JAK1, JAK2
Hub genes (10)	GO MF 2025	Ubiquitin-Like Protein Ligase Binding	4/253	9.57e-06	0.000163	26.7	44.6	Yes	JAK1, MDM2, PML, STAT2
Hub genes (10)	GO MF 2025	Hormone Receptor Binding	17-Feb	4.26e-05	0.000543	199	281.4	Yes	JAK1, JAK2
Hub genes (10)	GO MF 2025	SUMO Transferase Activity	23-Feb	7.91e-05	0.000807	147.1	201	Yes	MDM2, PML
Hub genes (10)	GO MF 2025	Non-Membrane Spanning Protein Tyrosine Kinase Activity	Feb-35	0.000185	0.002	96.6	127.8	Yes	JAK1, JAK2
Hub genes (10)	GO MF 2025	Ubiquitin Protein Ligase Binding	3/238	0.000307	0.002	21.3	30.4	Yes	JAK1, MDM2, PML
Hub genes (10)	GO MF 2025	Transmembrane Receptor Protein Tyrosine Kinase Activity	Feb-45	0.000307	0.002	75.2	98	Yes	ERBB2, IGF1R
Hub genes (10)	GO MF 2025	Cytokine Receptor Binding	Feb-98	0.001	0.008	34.5	43.8	Yes	JAK1, JAK2
Hub genes (10)	GO MF 2025	Phosphatase Binding	2/106	0.002	0.008	31.9	40.4	Yes	JAK1, STAT3
Hub genes (10)	GO MF 2025	Protein Phosphatase	2/108	0.002	0.008	31.3	39.6	Yes	JAK1, STAT3

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		Binding							
Hub genes (10)	GO MF 2025	5S rRNA Binding	5-Jan	0.003	0.011	338.2	469.4	Yes	MDM2
Hub genes (10)	GO MF 2025	lncRNA Binding	5-Jan	0.003	0.011	338.2	469.4	Yes	STAT3
Hub genes (10)	GO MF 2025	Primary miRNA Binding	5-Jan	0.003	0.011	338.2	469.4	Yes	STAT3
Hub genes (10)	GO MF 2025	Histone H3 Kinase Activity	9-Jan	0.005	0.017	187.9	234.6	Yes	JAK2
Hub genes (10)	GO CC 2025	Extrinsic Component of Cytoplasmic Side of Plasma Membrane	Feb-41	0.000255	0.006	82.5	108.1	Yes	JAK1, JAK2
Hub genes (10)	GO CC 2025	Cytoplasmic Side of Plasma Membrane	2/110	0.002	0.022	30.7	38.9	Yes	JAK1, JAK2
Hub genes (10)	GO CC 2025	Transferase Complex, Transferring Phosphorus-Containing Groups	7-Jan	0.004	0.032	241.6	312.9	Yes	IGF1R
Hub genes (10)	GO CC 2025	Nucleolus	3/685	0.006	0.032	7.4	10.2	Yes	IGF1R, MDM2, PML
Hub genes (10)	GO CC 2025	Nuclear Lumen	3/696	0.007	0.032	7.3	10	Yes	IGF1R, MDM2, PML
Hub genes (10)	GO CC 2025	Protein Kinase Complex	21-Jan	0.012	0.046	80.5	93.8	Yes	IGF1R
Hub genes (10)	GO CC 2025	Endosome Lumen	23-Jan	0.014	0.046	73.5	85.2	Yes	JAK2
Hub genes (10)	GO CC 2025	Focal Adhesion	2/348	0.017	0.047	9.7	12	Yes	JAK1, JAK2
Hub genes (10)	GO CC 2025	Cell-Substrate Junction	2/356	0.018	0.047	9.5	11.7	Yes	JAK1, JAK2
Hub genes (10)	GO CC 2025	Cytoplasmic Vesicle Membrane	2/374	0.02	0.047	9	11.1	Yes	ERBB2, MDM2
Hub genes (10)	GO CC 2025	Intracellular Membrane less Organelle	3/1128	0.025	0.054	4.5	6	No	IGF1R, MDM2, PML
Hub genes (10)	GO CC 2025	Caveola	Jan-55	0.032	0.064	30.7	34.7	No	JAK2
Hub genes (10)	GO CC 2025	Plasma Membrane Raft	Jan-74	0.043	0.079	22.9	25.6	No	JAK2
Hub genes (10)	GO CC 2025	Bounding Membrane of Organelle	2/789	0.076	0.131	4.3	5.1	No	ERBB2, MDM2
Hub genes (10)	GO CC 2025	Endocytic Vesicle Membrane	1/145	0.083	0.132	11.7	12.9	No	MDM2

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Hub genes (10)	KEGG 2021	Pathways in cancer	10/490	3.81e-16	3.2e-14	34.5	717.7	Yes	ERBB2, HDAC1, HGF, IGF1R, JAK1, JAK2, MDM2, PML, STAT2, STAT3
Hub genes (10)	KEGG 2021	PI3K-Akt signaling pathway	6/322	8.96e-09	3.76e-07	31.5	78.7	Yes	ERBB2, HGF, IGF1R, JAK1, JAK2, MDM2
Hub genes (10)	KEGG 2021	Epstein-Barr virus infection	5/177	2.87e-08	7.93e-07	47.8	97.3	Yes	HDAC1, JAK1, MDM2, STAT2, STAT3
Hub genes (10)	KEGG 2021	Proteoglycans in cancer	5/187	3.78e-08	7.93e-07	45.2	91.9	Yes	ERBB2, HGF, IGF1R, MDM2, STAT3
Hub genes (10)	KEGG 2021	Necroptosis	4/116	4.28e-07	7.18e-06	58.3	99.9	Yes	JAK1, JAK2, STAT2, STAT3
Hub genes (10)	KEGG 2021	Signaling pathways regulating pluripotency of stem cells	4/131	6.97e-07	9.75e-06	51.6	88.1	Yes	IGF1R, JAK1, JAK2, STAT3
Hub genes (10)	KEGG 2021	Influenza A	4/144	1.02e-06	1.01e-05	47	79.8	Yes	JAK1, JAK2, PML, STAT2
Hub genes (10)	KEGG 2021	JAK-STAT signaling pathway	4/146	1.08e-06	1.01e-05	46.3	78.7	Yes	JAK1, JAK2, STAT2, STAT3
Hub genes (10)	KEGG 2021	Hepatitis B	4/147	1.11e-06	1.01e-05	46	78.1	Yes	JAK1, JAK2, STAT2, STAT3
Hub genes (10)	KEGG 2021	Viral Carcinogenesis	4/152	1.26e-06	1.01e-05	44.5	75.5	Yes	HDAC1, JAK1, MDM2, STAT3
Hub genes (10)	KEGG 2021	MicroRNAs in cancer	4/155	1.37e-06	1.01e-05	43.6	74	Yes	ERBB2, HDAC1, MDM2, STAT3
Hub genes (10)	KEGG 2021	Transcriptional misregulation in cancer	4/157	1.44e-06	1.01e-05	43.1	73	Yes	HDAC1, IGF1R, MDM2, PML
Hub genes (10)	KEGG 2021	Kaposi sarcoma-associated herpesvirus infection	4/173	2.12e-06	1.37e-05	39.1	66	Yes	JAK1, JAK2, STAT2, STAT3
Hub genes (10)	KEGG 2021	Non-small cell lung cancer	Mar-66	6.68e-06	4.01e-05	76.9	114.5	Yes	ERBB2, HGF, STAT3

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Hub genes (10)	KEGG 2021	Melanoma	Mar-68	7.31e-06	4.1e-05	74.6	111	Yes	HGF, IGF1R, MDM2
Hub genes (10)	WikiPath ways 2024	Cancer Pathways WP5434	10/470	2.5e-16	4.51e-14	36	749.8	Yes	ERBB2, HDAC1, HGF, IGF1R, JAK1, JAK2, MDM2, PML, STAT2, STAT3
Hub genes (10)	WikiPath ways 2024	EGFR Tyrosine Kinase Inhibitor Resistance WP4806	Jun-81	2.07e-12	1.86e-10	125.3	336.5	Yes	ERBB2, HGF, IGF1R, JAK1, JAK2, STAT3
Hub genes (10)	WikiPath ways 2024	IL6 Signaling WP364	Apr-43	7.52e-09	4.51e-07	157.3	288.2	Yes	HDAC1, JAK1, JAK2, STAT3
Hub genes (10)	WikiPath ways 2024	Thyroid Stimulating Hormone TSH Signaling WP2032	Apr-53	1.78e-08	8.01e-07	127.6	229.3	Yes	IGF1R, JAK1, JAK2, STAT3
Hub genes (10)	WikiPath ways 2024	IL10 Anti Inflammatory Signaling WP4495	11-Mar	2.45e-08	8.82e-07	461.2	905	Yes	JAK1, STAT2, STAT3
Hub genes (10)	WikiPath ways 2024	ncRNAs Involved In STAT3 Signaling In Hepatocellular Carcinoma WP4337	13-Mar	4.25e-08	1.27e-06	390.3	723.9	Yes	JAK1, JAK2, STAT3
Hub genes (10)	WikiPath ways 2024	Leptin Signaling WP2034	Apr-69	5.23e-08	1.35e-06	98	172.7	Yes	ERBB2, JAK1, JAK2, STAT3
Hub genes (10)	WikiPath ways 2024	Prolactin Signaling WP2037	Apr-72	6.22e-08	1.4e-06	94	165	Yes	ERBB2, JAK1, JAK2, STAT3
Hub genes (10)	WikiPath ways 2024	Immune Response To Tuberculosis WP4197	22-Mar	2.28e-07	4.56e-06	230.6	380.8	Yes	JAK1, JAK2, STAT2
Hub genes (10)	WikiPath ways 2024	Androgen Receptor Network In Prostate Cancer WP2263	4/103	2.65e-07	4.77e-06	65.7	113.1	Yes	HGF, JAK1, MDM2, STAT3
Hub genes (10)	WikiPath ways 2024	Focal Adhesion PI3K Akt mTOR Signaling WP3932	5/287	3.2e-07	5.23e-06	29.5	58.9	Yes	HGF, IGF1R, JAK1, JAK2, MDM2

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Hub genes (10)	WikiPath ways 2024	Interferon Mediated Signaling WP4558	27-Mar	4.32e-07	5.93e-06	187.9	301.4	Yes	JAK1, JAK2, STAT2
Hub genes (10)	WikiPath ways 2024	PDGFR Beta Pathway WP3972	27-Mar	4.32e-07	5.93e-06	187.9	301.4	Yes	JAK1, JAK2, STAT3
Hub genes (10)	WikiPath ways 2024	PI3K Akt Signaling WP4172	5/309	4.61e-07	5.93e-06	27.4	54.6	Yes	HGF, IGF1R, JAK1, JAK2, MDM2
Hub genes (10)	WikiPath ways 2024	IL17 Signaling WP2112	Mar-32	7.32e-07	8.78e-06	158.5	249.3	Yes	JAK1, JAK2, STAT3
Hub genes (10)	Reactome 2024	Disease	10/1591	5.29e-11	1.45e-08	10.6	203.4	Yes	ERBB2, HDAC1, HGF, IGF1R, JAK1, JAK2, MDM2, PML, STAT2, STAT3
Hub genes (10)	Reactome 2024	Interleukin -20 Family Signaling	24-Apr	6.51e-10	8.89e-08	281.8	562.7	Yes	JAK1, JAK2, STAT2, STAT3
Hub genes (10)	Reactome 2024	Interleukin -27 Signaling	11-Mar	2.45e-08	1.67e-06	461.2	905	Yes	JAK1, JAK2, STAT3
Hub genes (10)	Reactome 2024	Interleukin -6 Signaling	11-Mar	2.45e-08	1.67e-06	461.2	905	Yes	JAK1, JAK2, STAT3
Hub genes (10)	Reactome 2024	Interleukin -35 Signalling	12-Mar	3.27e-08	1.78e-06	422.8	804.4	Yes	JAK1, JAK2, STAT3
Hub genes (10)	Reactome 2024	Diseases of Signal Transduction by Growth Factor Receptors and Second Messengers	6/416	4.13e-08	1.88e-06	24.4	60.3	Yes	ERBB2, HDAC1, HGF, JAK2, MDM2, STAT3
Hub genes (10)	Reactome 2024	Interferon Signaling	5/224	9.31e-08	3.63e-06	37.7	76.2	Yes	JAK1, JAK2, PML, STAT2, STAT3
Hub genes (10)	Reactome 2024	Signaling by Receptor Tyrosine Kinases	6/503	1.28e-07	4.35e-06	20.2	49.5	Yes	ERBB2, HDAC1, HGF, IGF1R, JAK2, STAT3
Hub genes (10)	Reactome 2024	PIP3 Activates AKT Signaling	5/250	1.61e-07	4.79e-06	33.8	68	Yes	ERBB2, HDAC1, HGF, MDM2, PML
Hub genes (10)	Reactome 2024	Potential Therapeutics	Apr-93	1.75e-07	4.79e-06	72.7	125.9	Yes	HDAC1, JAK1, JAK2,

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		for SARS							STAT2
Hub genes (10)	Reactome 2024	Interleukin -6 Family Signaling	22-Mar	2.28e-07	5.11e-06	230.6	380.8	Yes	JAK1, JAK2, STAT3
Hub genes (10)	Reactome 2024	Interleukin -4 and Interleukin -13 Signaling	4/101	2.45e-07	5.11e-06	67	115.5	Yes	HGF, JAK1, JAK2, STAT3
Hub genes (10)	Reactome 2024	Inactivation of CSF3 (G-CSF) Signaling	23-Mar	2.62e-07	5.11e-06	220.6	361.7	Yes	JAK1, JAK2, STAT3
Hub genes (10)	Reactome 2024	Interleukin -7 Signaling	23-Mar	2.62e-07	5.11e-06	220.6	361.7	Yes	HGF, JAK1, STAT3
Hub genes (10)	Reactome 2024	Intracellular Signaling by Second Messengers	5/290	3.37e-07	6.13e-06	29.2	58.3	Yes	ERBB2, HDAC1, HGF, MDM2, PML
Hub genes (10)	BioCarta 2016	Role of ERBB2 in Signal Transduction and Oncology Homo sapiens h her2 Pathway	27-Apr	1.07e-09	5.05e-08	250.5	489.2	Yes	ERBB2, JAK1, JAK2, STAT3
Hub genes (10)	BioCarta 2016	Stat3 Signaling Pathway Homo sapiens h stat3 Pathway	8-Mar	8.33e-09	1.96e-07	634.2	1448.2	Yes	JAK1, JAK2, STAT3
Hub genes (10)	BioCarta 2016	IL 6 signaling pathway Homo sapiens h il6Pathway	13-Mar	4.25e-08	6.65e-07	390.3	723.9	Yes	JAK1, JAK2, STAT3
Hub genes (10)	BioCarta 2016	IFN gamma signaling pathway Homo sapiens h ifngPathway	5-Feb	3.14e-06	3.69e-05	676.4	1408.2	Yes	JAK1, JAK2
Hub genes (10)	BioCarta 2016	IFN alpha signaling pathway Homo sapiens h ifnaPathway	7-Feb	6.6e-06	6.2e-05	483.2	844.8	Yes	JAK1, STAT2
Hub genes (10)	BioCarta 2016	IL22 Soluble Receptor Signaling Pathway Homo	11-Feb	1.73e-05	0.000135	307.5	469.2	Yes	JAK1, STAT3

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		sapiens h il22bppath way							
Hub genes (10)	BioCarta 2016	Sumoylation by RanBP2 Regulates Transcrip tional Repression Homo sapiens h ranbp2 Pathway	13-Feb	2.45e-05	0.000164	260.2	383.9	Yes	HDAC1, MDM2
Hub genes (10)	BioCarta 2016	EGF Signaling Pathway Homo sapiens h egfPathway	20-Feb	5.95e-05	0.000349	169.1	234.5	Yes	JAK1, STAT3
Hub genes (10)	BioCarta 2016	TPO Signaling Pathway Homo sapiens h TPOPath way	22-Feb	7.22e-05	0.000377	153.7	211	Yes	JAK2, STAT3
Hub genes (10)	BioCarta 2016	PDGF Signaling Pathway Homo sapiens h pdgfPath way	24-Feb	8.63e-05	0.000405	140.9	191.8	Yes	JAK1, STAT3
Hub genes (10)	BioCarta 2016	Bioactive Peptide Induced Signaling Pathway Homo sapiens h biopeptid es Pathway	Feb-33	0.000165	0.000703	102.5	136	Yes	JAK2, STAT3
Hub genes (10)	BioCarta 2016	NO2-depen dent IL 12 Pathway in NK cells Homo sapiens h no2il12 Pathway	9-Jan	0.005	0.016	187.9	234.6	Yes	JAK2
Hub genes (10)	BioCarta 2016	Overview of telomerase protein component gene hTert Transcrip tional Regulation Homo sapiens h tertpath way	10-Jan	0.006	0.016	169.1	208.5	Yes	HDAC1
Hub genes (10)	BioCarta 2016	EPO Signaling Pathway	11-Jan	0.006	0.016	153.7	187.7	Yes	JAK2

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		Homo sapiens h epoPath way							
Hub genes (10)	BioCarta 2016	Regulation Oftranscrip tional activity by PML Homo sapiens h pmlPath way	11-Jan	0.006	0.016	153.7	187.7	Yes	PML
Hub genes (10)	BioPlanet 2019	Pathways in cancer	8/309	4.95e-13	1.17e-10	43.8	220.6	Yes	ERBB2, HDAC1, HGF, IGF1R, JAK1, MDM2, PML, STAT3
Hub genes (10)	BioPlanet 2019	Interleukin -27-medi ated signaling events	26-Apr	9.16e-10	1.08e-07	260.2	511.5	Yes	JAK1, JAK2, STAT2, STAT3
Hub genes (10)	BioPlanet 2019	Immune system signaling by interferons, interleukins, prolactin, and growth hormones	6/246	1.78e-09	1.4e-07	41.2	104.1	Yes	HGF, JAK1, JAK2, PML, STAT2, STAT3
Hub genes (10)	BioPlanet 2019	Leptin signaling pathway	Apr-59	2.76e-08	1.4e-06	114.7	204.2	Yes	ERBB2, JAK1, JAK2, STAT3
Hub genes (10)	BioPlanet 2019	Thyroid- stimulating hormone signaling pathway	Apr-60	2.96e-08	1.4e-06	112.7	200.5	Yes	IGF1R, JAK1, JAK2, STAT3
Hub genes (10)	BioPlanet 2019	Interleukin -22 soluble receptor signaling pathway	13-Mar	4.25e-08	1.67e-06	390.3	723.9	Yes	JAK1, JAK2, STAT3
Hub genes (10)	BioPlanet 2019	Interleukin -6 signaling pathway	Apr-69	5.23e-08	1.76e-06	98	172.7	Yes	HDAC1, JAK1, JAK2, STAT3
Hub genes (10)	BioPlanet 2019	Prolactin activation of MAPK signaling	Apr-73	6.58e-08	1.93e-06	92.7	162.6	Yes	ERBB2, JAK1, JAK2, STAT3
Hub genes (10)	BioPlanet 2019	Interferon- gamma signaling pathway	Apr-77	8.17e-08	1.93e-06	87.8	153.7	Yes	JAK1, JAK2, PML, STAT3
Hub genes (10)	BioPlanet 2019	Immune system	7/848	8.18e-08	1.93e-06	14	44.6	Yes	HGF, JAK1, JAK2, MDM2, PML, STAT2, STAT3

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Hub genes (10)	BioPlanet 2019	STAT3 pathway	19-Mar	1.44e-07	3.08e-06	267	452.3	Yes	JAK1, JAK2, STAT3
Hub genes (10)	BioPlanet 2019	Signaling by interleukins	4/106	2.97e-07	5.85e-06	63.8	109.8	Yes	HGF, JAK1, JAK2, STAT3
Hub genes (10)	BioPlanet 2019	EGF/EGFR signaling pathway	4/138	8.58e-07	1.45e-05	49	83.4	Yes	ERBB2, JAK1, JAK2, STAT3
Hub genes (10)	BioPlanet 2019	Interferon signaling	4/138	8.58e-07	1.45e-05	49	83.4	Yes	JAK1, JAK2, PML, STAT2
Hub genes (10)	BioPlanet 2019	Interleukin -7 signaling pathway	Mar-35	9.65e-07	1.46e-05	145	225.9	Yes	HGF, JAK1, STAT3
Hub genes (10)	DSigDB	curcumin CTD 00000663	7/489	1.8e-09	1.94e-06	24.2	79.5	Yes	ERBB2, HDAC1, HGF, IGF1R, JAK2, MDM2, STAT3
Hub genes (10)	DSigDB	Arsenenous acid CTD 00000922	8/1167	2e-08	1.07e-05	11.6	54.3	Yes	ERBB2, HDAC1, HGF, JAK1, JAK2, MDM2, PML, STAT3
Hub genes (10)	DSigDB	Capsaicin CTD 00005570	5/190	4.09e-08	1.47e-05	44.5	90.4	Yes	ERBB2, IGF1R, JAK1, MDM2, STAT3
Hub genes (10)	DSigDB	NVP-TAE684 CTD 00004657	16-Mar	8.3e-08	1.54e-05	317.1	556.7	Yes	ERBB2, HGF, STAT3
Hub genes (10)	DSigDB	resveratrol CTD 00002483	8/1416	9.16e-08	1.54e-05	9.6	44	Yes	ERBB2, HDAC1, HGF, IGF1R, JAK1, MDM2, PML, STAT3
Hub genes (10)	DSigDB	Vorinostat BOSS	Apr-80	9.54e-08	1.54e-05	84.6	147.6	Yes	ERBB2, HDAC1, MDM2, STAT3
Hub genes (10)	DSigDB	172889-27-9 BOSS	Apr-81	1e-07	1.54e-05	83.5	145.7	Yes	ERBB2, JAK2, MDM2, STAT3
Hub genes (10)	DSigDB	Erlotinib BOSS	Apr-89	1.47e-07	1.97e-05	76	131.9	Yes	ERBB2, IGF1R, JAK2, STAT3
Hub genes (10)	DSigDB	Podophyllotoxin TTD 00010285	24-Mar	2.99e-07	3.58e-05	211.4	344.5	Yes	JAK2, MDM2, STAT3
Hub genes (10)	DSigDB	Podophyllotoxin CTD 00006577	27-Mar	4.32e-07	4.65e-05	187.9	301.4	Yes	JAK2, MDM2, STAT3

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Hub genes (10)	DSigDB	Cladribine CTD 00007175	29-Mar	5.4e-07	5.01e-05	174.9	278.2	Yes	JAK2, MDM2, STAT3
Hub genes (10)	DSigDB	Chromi m(VI) CTD 00002830	4/124	5.59e-07	5.01e-05	54.6	93.2	Yes	ERBB2, HDAC1, IGF1R, STAT3
Hub genes (10)	DSigDB	GNF-Pf-78 TTD 00008278	31-Mar	6.64e-07	5.49e-05	163.7	258.3	Yes	JAK2, MDM2, STAT3
Hub genes (10)	DSigDB	Doxoru bicin CTD 00005874	6/699	8.89e-07	6.83e-05	14.5	35.1	Yes	ERBB2, HDAC1, HGF, MDM2, STAT2, STAT3
Hub genes (10)	DSigDB	Sorafenib tosylate BOSS	4/146	1.08e-06	7.71e-05	46.3	78.7	Yes	ERBB2, JAK2, MDM2, STAT3

Table S8: (A) Parameters of the WGCNA analysis; the module assignment of all 5,017 analysed genes is provided as a separate Excel file.

Parameter	Setting
Reference network	GSE98793 (MDD; GPL570; 192 samples: 128 MDD, 64 controls)
Test network	GSE63878 (PTSD; GPL6244; 96 samples: 48 PTSD, 48 controls)
Input data	Full gene-level matrices without DEG or label-based filtering; probes averaged per gene symbol
Genes measured on both platforms	16,911
Genes analysed	5,017 (5,000 most variable in the reference + hub genes and shared DEGs)
Not used for network construction	Top-800 Fisher-score genes, combined feature sets and the 19 shared genes
Network type	Signed; Pearson correlation; signed TOM
Soft-thresholding power	12 (lowest power with signed $R^2 \geq 0.85$; $R^2 = 0.862$)
Module detection	blockwiseModules, single block; minModuleSize = 30; deepSplit = 2; mergeCutHeight = 0.25
Modules detected	6 (53–2450 genes); 510 unassigned (grey)
Module preservation	modulePreservation; reference GSE98793, test GSE63878; 200 permutations
Additional analyses	Reverse preservation (GSE63878 → GSE98793, 200 permutations); gene-set size 3,000 and 8,000 (100 permutations)
Random seed	12345

Table S8: (B) Module assignment, module membership (kME) in GSE98793 and GSE63878, gene significance for MDD and PTSD, and direction of change of the 10 hub genes and 19 shared DEGs.

Gene	Gene set	Module (GSE98793)	kME GSE98793	kME GSE63878	GS MDD (p)	GS PTSD (p)	Same direction in MDD
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							and PTSD
IGF1R	hub + shared DEG	grey	-0.53	-0.45	0.24 (0.001)	-0.18 (0.077)	no
STAT2	hub + shared DEG	brown	0.46	-0.66	0.17 (0.018)	0.18 (0.073)	yes
PML	hub + shared DEG	turquoise	0.57	0.78	0.09 (0.211)	0.26 (0.010)	yes
HGF	hub + shared DEG	grey	-0.22	-0.23	0.18 (0.011)	0.19 (0.061)	yes
HDAC1	hub (network)	turquoise	0.88	0.88	-0.08 (0.250)	-0.00 (0.962)	yes
JAK2	hub (network)	blue	0.64	0.76	0.09 (0.236)	0.12 (0.229)	yes
JAK1	hub (network)	brown	0.83	-0.75	-0.14 (0.060)	-0.01 (0.959)	yes
MDM2	hub (network)	brown	0.39	-0.72	0.07 (0.348)	-0.12 (0.236)	no
ERBB2	hub (network)	grey	-0.02	-0.45	0.03 (0.692)	-0.05 (0.629)	no
STAT3	hub (network)	turquoise	-0.06	0.75	-0.01 (0.896)	0.13 (0.220)	no
TDRD9	shared DEG	grey	-0.34	0.01	0.30 (0.000)	0.25 (0.015)	yes
OLFM4	shared DEG	grey	-0.28	-0.26	0.26 (0.000)	0.20 (0.046)	yes
SYTL2	shared DEG	turquoise	0.37	0.38	-0.22 (0.002)	0.21 (0.039)	no
BTN3A2	shared DEG	turquoise	0.52	0.7	-0.18 (0.011)	0.19 (0.067)	no
PLXNA2	shared DEG	turquoise	0.32	0.02	0.16 (0.027)	-0.19 (0.058)	no
EIF4B	shared DEG	turquoise	0.3	0.88	-0.23 (0.002)	-0.08 (0.424)	yes
CYP2U1	shared DEG	blue	0.43	0.44	-0.16 (0.023)	0.20 (0.051)	no
GALNT1 1	shared DEG	brown	0.32	-0.47	-0.20 (0.004)	0.28 (0.005)	no
DKKL1	shared DEG	grey	-0.21	0.56	0.20 (0.005)	-0.24 (0.018)	no
GFI1B	shared DEG	turquoise	0.31	0.16	0.15 (0.036)	-0.20 (0.050)	no
PTMA	shared DEG	brown	0.6	-0.44	-0.19 (0.008)	-0.22 (0.035)	yes
AHI1	shared DEG	grey	0.41	-0.15	-0.13 (0.064)	0.23 (0.027)	no
SCAPER	shared DEG	turquoise	0.65	0.17	-0.10 (0.165)	0.28 (0.006)	no

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SLC22A14	shared DEG	brown	0.34	0.65	0.19 (0.008)	-0.19 (0.067)	no
DOCK4	shared DEG	brown	0.39	-0.31	0.13 (0.065)	0.21 (0.045)	yes

Table S8: (C) Robustness of the WGCNA results: main analysis, analyses with 3,000 and 8,000 genes, and reverse-direction analysis (GSE63878 as reference).

Analysis	Reference	Test	Genes_Common	Genes_Analysed	Power	Signif_R2	Modules	Smallest_Module	Largest_module	Grey_Genes	Permutations	Strong	Module_Erate	Not_preserved	Gold_Zsummary	Trait-associated modules in reference (FDR < 0.05)	Trait-associated modules in test (FDR < 0.05)
Main	GSE98793	GSE63878	16911	5017	12	0.862	6	53	2450	510	200	2	3	1	12.18	0	0
Reverse	GSE63878	GSE98793	16911	5015	20	0.363	5	37	3196	92	200	4	1	0	27.06	1	0
Sensitivity_top3000	GSE98793	GSE63878	16911	3018	13	0.854	5	36	1623	295	100	2	2	1	9.76	0	0
Sensitivity_top8000	GSE98793	GSE63878	16911	8012	6	0.878	6	101	3199	1101	100	2	4	0	18.53	0	0

Table S8: (D) Preservation of GSE63878 co-expression modules in GSE98793 (exploratory; the GSE63878 network did not reach scale-free topology).

Module	Module size (genes)	Zsummary	Zdensity	Zconnectivity	Median rank	Preservation	r with PTSD (FDR)	r with MDD (FDR)
blue	1532	32.94	45.95	19.94	4	Strong	-0.02 (0.97)	0.03 (0.78)

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turquoise	3196	26.37	40.96	11.79	6	Strong	0.03 (0.97)	-0.01 (0.94)
yellow	60	12.65	18.5	6.8	2	Strong	0.30 (0.02)	0.09 (0.34)
brown	98	10.76	16.48	5.04	4	Strong	-0.00 (0.97)	-0.19 (0.05)
green	37	8.97	16.21	1.72	1	Moderate	-0.09 (0.97)	0.16 (0.09)
gold (random genes) [†]	1000	27.06	15.82	38.31		Reference		

Table S9: (A) All pre-specified validation analyses of the hub-gene classifiers: discovery and validation cohorts, sample size (cases/controls), genes available, discovery out-of-fold AUC, validation AUC with standard error, DeLong and bootstrap 95% confidence intervals, permutation p-values (raw and Holm-adjusted), decision threshold, confusion-matrix counts, sensitivity, specificity, accuracy and balanced accuracy. Internal analyses are within-cohort nested cross-validation and are not external validation.

Analyses	Panel	Phenotype	Discovery	Validation	n (case / control)	Genes used	Genes	Disc. OOF AUC	AUC	SE	DeLong 95% CI	Bootstrap 95% CI	p (DeLong vs 0.5)	p (perm)	p (perm, Holm)	Threshold	TP	FN	TN	FP	Sensitivity	Specificity	Acc.	Bal. acc.
Primary frozen	10 hub genes	MD	GSE 98793	GSE 32280	24 (16/8)	10	IGF1R, STAT2, PML, HGF, HDAC1, JAK2, JAK1, MDM2, ERBB2, STAT3	0.6943	0.5234	0.1352	0.2585–0.7884	0.2734–0.7656	0.8624	0.4479	0.7331	0.5507	6	10	4	4	0.3750	0.5000	0.4167	0.4375
Primary frozen	10 hub genes	MD	GSE 98793	GSE 39653	45 (21/24)	8	PML, HGF, HDAC1, JAK2, JAK1, MDM2, ERBB2, STAT3	0.6191	0.6250	0.0864	0.4556–0.7944	0.4504–0.7877	0.1482	0.0790	0.3159	0.5120	14	7	16	8	0.6667	0.6667	0.6667	0.6667
Primary frozen	10 hub genes	MD	GSE 98793	GSE 54566	28 (14/14)	10	IGF1R, STAT2, PML, HGF, HDAC1, JAK2, JAK1, MDM2, ERBB2, STAT3	0.6943	0.5408	0.1144	0.3166–0.7650	0.3265–0.7501	0.7213	0.3665	0.7331	0.5507	6	8	8	6	0.4286	0.5714	0.5000	0.5000

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Primary frozen	10 hub genes	Com orbid PTS D+M DD	GSE 6387 8	GSE 6766 3	184 (112/72)	7	STAT2, PML, HDAC 1, JAK2, JAK1, ERBB 2, STAT3	0.6063	0.5365	0.0449	0.448 4–0.624 5	0.4482–0.6277	0.4 171	0.211 6	0.634 7	0.4723	64	48	40	32	0.57 14	0.55 56	0.5652	0.5635
Seed-gene panel (frozen)	4 seed genes	MD D	GSE 9879 3	GSE 3228 0	24 (16/8)	4	IGF1R, STAT2, HGF, PML	0.6783	0.5625	0.1370	0.294 0–0.831 0	0.2969–0.8047	0.6 482	0.326 5	1.000 0	0.5969	5	11	5	3	0.31 25	0.62 50	0.4167	0.4688
Seed-gene panel (frozen)	4 seed genes	MD D	GSE 9879 3	GSE 3965 3	45 (21/24)	2	HGF, PML	0.5735	0.5119	0.0900	0.335 5–0.688 3	0.3253–0.6825	0.8 948	0.448 9	1.000 0	0.5347	9	12	16	8	0.42 86	0.66 67	0.5556	0.5476
Seed-gene panel (frozen)	4 seed genes	MD D	GSE 9879 3	GSE 5456 6	28 (14/14)	4	IGF1R, STAT2, HGF, PML	0.6783	0.4133	0.1156	0.186 7–0.639 8	0.1990–0.6276	0.4 531	0.771 8	1.000 0	0.5969	3	11	11	3	0.21 43	0.78 57	0.5000	0.5000
Seed-gene panel (frozen)	4 seed genes	Com orbid PTS D+M DD	GSE 6387 8	GSE 6766 3	184 (112/72)	2	STAT2, PML	0.6276	0.4938	0.0445	0.406 6–0.581 0	0.4049–0.5852	0.8 892	0.564 3	1.000 0	0.5080	52	60	42	30	0.46 43	0.58 33	0.5109	0.5238
Sensitivity: single 10-gene model	10 hub genes	Com orbid PTS D+M DD	GSE 6387 8	GSE 6766 3	184	7	–	–	0.4126	0.0439	0.326 6–0.498 6	0.3302–0.5001	0.0 463	0.977 0	1.000 0	0.5663	41	71	33	39	0.36 61	0.45 83	0.4022	0.4122
Sensitivity: single 10-gene model	10 hub genes	MD D	GSE 9879 3	GSE 3228 0	24	10	–	–	0.5234	0.1352	0.258 5–0.788 4	0.2578–0.7656	0.8 624	0.433 3	1.000 0	0.5507	6	10	4	4	0.37 50	0.50 00	0.4167	0.4375
Sensitivity: single 10-gene model	10 hub genes	MD D	GSE 9879 3	GSE 3965 3	45	8	–	–	0.6032	0.0871	0.432 5–0.773 9	0.4425–0.7679	0.2 361	0.126 8	0.507 1	0.5507	11	10	14	10	0.52 38	0.58 33	0.5556	0.5536
Sensitivity: single 10-gene model	10 hub genes	MD D	GSE 9879 3	GSE 5456 6	28	10	–	–	0.5408	0.1144	0.316 6–0.765 0	0.3214–0.7551	0.7 213	0.366 5	1.000 0	0.5507	6	8	8	6	0.42 86	0.57 14	0.5000	0.5000
Secondary: leave-one-cohort-out	10 hub genes	MD D	GSE 3228 0+G SE39 653+ GSE 5456 6	GSE 9879 3	192	8	–	–	0.5286	0.0448	0.440 7–0.616 4	0.4348–0.6146	0.5 238	0.257 9	0.815 8	0.4257	92	36	23	41	0.71 88	0.35 94	0.5990	0.5391
Secondary: leave-one-cohort-out	10 hub genes	MD D	GSE 9879 3+G SE39 653+ GSE 5456 6	GSE 3228 0	24	8	–	–	0.6875	0.1165	0.459 2–0.915 8	0.4375–0.8750	0.1 075	0.071 8	0.430 7	0.4306	13	3	3	5	0.81 25	0.37 50	0.6667	0.5938

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Internal : within-cohort nested CV (not external)	10 hub genes	Com orbid PTS D+M DD	GSE 6766 3	GSE 6766 3	184	7	–	–	0.5694	0.0437	0.483 8–0.655 1	0.4808–0.6538	0.1 119	–	–	–	–	–	–	–	–	–	–
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Table S9: (B) Random-effects (DerSimonian–Laird) pooled AUCs across all external cohorts and across the MDD cohorts, with 95% confidence intervals, between-study variance and I².

Analysis	Pool	k	Pooled_AUC	CI_low	CI_high	tau2	I2_percent	p_vs_0_5
Primary frozen	All validation cohorts	4	0.5515	0.4805	0.6226	0	0	0.1553
Primary frozen	MDD cohorts	3	0.5797	0.4593	0.7002	0	0	0.1943
Seed-gene panel (frozen)	All validation cohorts	4	0.4936	0.4224	0.5649	0	0	0.8609
Seed-gene panel (frozen)	MDD cohorts	3	0.4933	0.3697	0.6169	0	0	0.9151
Sensitivity: single 10-gene model	All validation cohorts	4	0.4929	0.39	0.5957	0.0039	34.3788	0.8922
Sensitivity: single 10-gene model	MDD cohorts	3	0.5685	0.4476	0.6893	0	0	0.2668

Table S9: © Characteristics of the discovery (GSE98793, GSE63878) and validation (GSE32280, GSE39653, GSE54566, GSE67663) cohorts: platform, tissue, phenotype, case and control definitions, samples analysed and excluded (subsyndromal symptomatic depression, bipolar disorder), log2 transformation and availability of the ten hub genes.

Characteristic	GSE98793	GSE63878	GSE32280	GSE39653	GSE54566	GSE67663

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Role	Discovery	Discovery	Validation	Validation	Validation	Validation
Phenotype	MDD	PTSD	MDD	MDD	MDD	Comorbid PTSD+MDD
Rows in expression matrix	192	96	37	53	28	184
GEO samples	192	96	37	53	28	184
Status	used	used	used	used	used	used
Samples used (n)	192	96	24	45	28	184
Cases	128	48	16	21	14	112
Controls	64	48	8	24	14	72
log2 transform applied	False	False	False	False	True	False
Hub genes available	10	10	10	8	10	7
Hub genes missing	none	none	none	IGF1R, STAT2	none	IGF1R, HGF, MDM2
Probes: IGF1R	5	2	5	0	5	0
Probes: STAT2	3	1	3	0	3	1
Probes: PML	10	1	10	7	10	1
Probes: HGF	5	1	5	2	5	0
Probes: HDAC1	1	1	1	1	1	1
Probes: JAK2	3	1	3	1	3	1
Probes: JAK1	5	1	5	1	5	1
Probes: MDM2	10	1	10	2	10	0
Probes: ERBB2	3	1	3	3	3	1
Probes: STAT3	4	1	4	3	4	3
Excluded: bipolar	—	—	0	8	0	0
Excluded: schizophrenia-spectrum	—	—	13	0	0	0

Table S9: (D) Coefficients, intercepts and decision thresholds of the frozen logistic-regression models fitted in the discovery cohorts.

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Panel	Disco very	Valid ation	Interc ept	Thre shold _You den_ disco very	IGF1R	STAT2	PML	HGF	HDAC1	JAK2	JAK1	MDM2	ERBB2	STAT3
10 hub genes	GSE9 8793	GSE3 2280	0.17 19	0.55 07	0.8164	0.407	0.19 91	0.23 09	-0.3929	0.03 75	-0.64 97	-0.0421	0.0866	-0.3164
10 hub genes	GSE9 8793	GSE3 9653	0.09 13	0.5 12			0.28 66	0.26 99	-0.727	-0.05 42	-0.68 41	0.1302	0.048	0.0343
10 hub genes	GSE9 8793	GSE5 4566	0.17 19	0.55 07	0.8164	0.407	0.19 91	0.23 09	-0.3929	0.03 75	-0.64 97	-0.0421	0.0866	-0.3164
10 hub genes	GSE6 3878	GSE6 7663	0.02 19	0.47 23		-0.0935	1.15 33		-0.3533	-0.15 78	-0.08 71		-0.2339	-0.0378
4 seed genes	GSE9 8793	GSE3 2280	0.1	0.59 69	0.5494	0.2824	0.20 33	0.38						
4 seed genes	GSE9 8793	GSE3 9653	0.03 47	0.53 47			0.16 79	0.40 97						
4 seed genes	GSE9 8793	GSE5 4566	0.1	0.59 69	0.5494	0.2824	0.20 33	0.38						
4 seed genes	GSE6 3878	GSE6 7663	0.00 37	0.50 8		-0.2447	0.75 06							
10 hub genes (single model)	GSE6 3878	all	0.09 83	0.56 63	-0.9642	-0.1138	1.07 11	0.5 05	-0.5314	-0.11 59	0.77 72	-0.8344	-0.2393	0.556
10 hub genes (single model)	GSE9 8793	all	0.17 19	0.55 07	0.8164	0.407	0.19 91	0.23 09	-0.3929	0.03 75	-0.64 97	-0.0421	0.0866	-0.3164

Table S10: Composition of the protein–protein interaction network. (A) Mapping of the 19 shared DEGs to STRING seed nodes. (B) All 97 network nodes, classified as seed genes (shared DEGs) or first-order STRING interactors, with hub rank, degree, betweenness centrality, closeness centrality and stress.

(A) Mapping of the 19 shared DEGs to STRING seed nodes

Shared DEG	Mapped as STRING seed	Degree in network	Top-10 hub
IGF1R	Yes	59	Yes (rank 1)
STAT2	Yes	21	Yes (rank 2)
PML	Yes	13	Yes (rank 3)
HGF	Yes	9	Yes (rank 4)
AHI1	No	—	No
BTN3A2	No	—	No
CYP2U1	No	—	No
DKKL1	No	—	No
DOCK4	No	—	No

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EIF4B	No	–	No
GALNT1 1	No	–	No
GFI1B	No	–	No
OLFM4	No	–	No
PLXNA2	No	–	No
PTMA	No	–	No
SCAPER	No	–	No
SLC22A 14	No	–	No
SYTL2	No	–	No
TDRD9	No	–	No

(B) All 97 network nodes

Gene	Node type	Top-10 hub rank	Degree	Betweenness	Closeness	Stress
IGF1R	Seed (shared DEG)	1	59	0.875	0.596	13066
STAT2	Seed (shared DEG)	2	21	0.334	0.382	8220
PML	Seed (shared DEG)	3	13	0.207	0.36	2596
HGF	Seed (shared DEG)	4	9	0.141	0.398	1848
ERBB2	First-order interactor (STRING)	5	2	0	0.387	0
HDAC1	First-order interactor (STRING)	5	2	0.024	0.298	474
JAK1	First-order interactor (STRING)	5	2	0.086	0.44	2428
JAK2	First-order interactor (STRING)	5	2	0.086	0.44	2428
MDM2	First-order interactor (STRING)	5	2	0.161	0.417	1704

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STAT1	First-order interactor (STRING)	Tied (not listed)	2	0.024	0.298	474
STAT3	First-order interactor (STRING)	5	2	0.086	0.44	2428
APPL1	First-order interactor (STRING)	—	1	0	0.375	0
BDNF	First-order interactor (STRING)	—	1	0	0.375	0
CAV1	First-order interactor (STRING)	—	1	0	0.375	0
CD44	First-order interactor (STRING)	—	1	0	0.286	0
CISH	First-order interactor (STRING)	—	1	0	0.277	0
CREBBP	First-order interactor (STRING)	—	1	0	0.277	0
CRKL	First-order interactor (STRING)	—	1	0	0.375	0
CTNNA1	First-order interactor (STRING)	—	1	0	0.375	0
DAXX	First-order interactor (STRING)	—	1	0	0.265	0
EGF	First-order interactor (STRING)	—	1	0	0.375	0
EGFR	First-order interactor (STRING)	—	1	0	0.375	0
EP300	First-order interactor (STRING)	—	1	0	0.277	0
ERBB3	First-order interactor (STRING)	—	1	0	0.375	0
ESR1	First-order interactor (STRING)	—	1	0	0.375	0
FN1	First-order interactor (STRING)	—	1	0	0.286	0

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GNAI2	First-order interactor (STRING)	—	1	0	0.375	0
GNAI3	First-order interactor (STRING)	—	1	0	0.375	0
GNAS	First-order interactor (STRING)	—	1	0	0.375	0
GNAZ	First-order interactor (STRING)	—	1	0	0.375	0
GRB10	First-order interactor (STRING)	—	1	0	0.375	0
GRB2	First-order interactor (STRING)	—	1	0	0.375	0
HRAS	First-order interactor (STRING)	—	1	0	0.375	0
IFNAR1	First-order interactor (STRING)	—	1	0	0.277	0
IFNAR2	First-order interactor (STRING)	—	1	0	0.277	0
IGF1	First-order interactor (STRING)	—	1	0	0.375	0
IGF2	First-order interactor (STRING)	—	1	0	0.375	0
IGFBP3	First-order interactor (STRING)	—	1	0	0.375	0
INS	First-order interactor (STRING)	—	1	0	0.375	0

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INS-IGF2	First-order interactor (STRING)	—	1	0	0.375	0
INSR	First-order interactor (STRING)	—	1	0	0.375	0
INSRR	First-order interactor (STRING)	—	1	0	0.375	0
IRF9	First-order interactor (STRING)	—	1	0	0.277	0
IRS1	First-order interactor (STRING)	—	1	0	0.375	0
IRS2	First-order interactor (STRING)	—	1	0	0.375	0
IRS4	First-order interactor (STRING)	—	1	0	0.375	0
ISG15	First-order interactor (STRING)	—	1	0	0.277	0
ITGB3	First-order interactor (STRING)	—	1	0	0.375	0
JAK3	First-order interactor (STRING)	—	1	0	0.277	0
KPNA1	First-order interactor (STRING)	—	1	0	0.277	0
KRAS	First-order interactor (STRING)	—	1	0	0.375	0
MET	First-order interactor (STRING)	—	1	0	0.286	0
MST1R	First-order interactor (STRING)	—	1	0	0.286	0
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NRAS	First-order interactor (STRING)	—	1	0	0.375	0
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SHC1	First-order interactor (STRING)	—	1	0	0.375	0
SHC2	First-order interactor (STRING)	—	1	0	0.375	0
SHC3	First-order interactor (STRING)	—	1	0	0.375	0
SHC4	First-order interactor (STRING)	—	1	0	0.375	0
SOCS1	First-order interactor (STRING)	—	1	0	0.277	0
SOCS3	First-order interactor (STRING)	—	1	0	0.277	0
SP100	First-order interactor (STRING)	—	1	0	0.265	0
SPI1	First-order interactor (STRING)	—	1	0	0.265	0
SPINT1	First-order interactor (STRING)	—	1	0	0.286	0
SRC	First-order interactor (STRING)	—	1	0	0.375	0
STAT6	First-order interactor (STRING)	—	1	0	0.277	0
SUMO1	First-order interactor (STRING)	—	1	0	0.265	0
SUMO2	First-order interactor (STRING)	—	1	0	0.265	0
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TYK2	First-order interactor (STRING)	—	1	0	0.277	0
UBE2I	First-order interactor (STRING)	—	1	0	0.265	0
USP18	First-order interactor (STRING)	—	1	0	0.277	0

Table S11: Protein targets, PDB structures, and molecular docking grid parameters used in the virtual screening analysis.

Gene	PDB ID	Protein structure	Center X	Center Y	Center Z	Grid size (X × Y × Z, Å)	Exhaustiveness	Energy range (kcal/mol)
ERBB2	3N85	HER2 extracellular region + antibody Fab complex	90.202	5.735	−27.187	40 × 40 × 40	8	4
HDAC1	6Z2K	Tetrameric HDAC1/MI DEAS/DNT TIP1 MiDAC complex	216.003	215.994	215.974	40 × 40 × 40	8	4
HGF	1NK1	NK1 fragment of human HGF/SF	8.252	31.175	12.288	40 × 40 × 40	8	4
IGF1R	7XLC	Apo-IGF1R, conformation II	137.277	137.501	126.431	40 × 40 × 40	8	4
JAK1	5KHW	JAK1 + ADP	44.867	0.795	19.890	40 × 40 × 40	8	4
JAK2	7F7W	JAK2-JH2	8.445	−22.689	20.535	40 × 40 × 40	8	4
MDM2	6Y4Q	MDM2 + stapled peptide	0.990	−17.952	−18.440	40 × 40 × 40	8	4

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PML	8J25	PML B-box2 mutant	-3.63 9	-44.3 97	-3.115	40 × 40 × 40	8	4
STAT2	7ZN7	RCMV E27–DDB1 complex with STAT2 CCD	138.16 2	129.32 9	132.117	40 × 40 × 40	8	4
STAT3	6TLC	Unphosphorylated STAT3 + monobody	0.231	32.470	33.516	40 × 40 × 40	8	4

Note: Molecular docking was performed using AutoDock Vina. The grid box dimensions were 40 × 40 × 40 Å for all protein targets. We specified the grid-center coordinates individually for each target structure. We set the exhaustiveness and energy-range parameters to 8 and 4 kcal/mol, respectively.

Table S12: Molecular docking analysis of drugs with target genes

Gene	Drug Compounds	Binding Energy (kcal/mol)
IGF1R	3-acetylursolic acid	-7.1
	Luteolin	-6.8
	Quercetin	-6.7
	Apigenin	-6.5
STAT2	3-acetylursolic acid	-8.6
	Luteolin	-7.7
	Quercetin	-7.7
	Apigenin	-7.5
PML	3-acetylursolic acid	-7.5
	Luteolin	-7.2
	Quercetin	-7.4
	Apigenin	-7.1
HGF	3-acetylursolic acid	-8.2
	Luteolin	-8.4
	Quercetin	-8.2
	Apigenin	-7.8
HDAC1	3-acetylursolic acid	-6.2
	Luteolin	-6.2
	Quercetin	-6.1
	Apigenin	-5.9
JAK2	3-acetylursolic acid	-10.4
	Luteolin	-8.4
	Quercetin	-8.3
	Apigenin	-8.5
JAK1	3-acetylursolic acid	-8.6
	Luteolin	-8.2
	Quercetin	-8.2
	Apigenin	-7.9

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MDM2	3-acetylursolic acid	-7.8
	Luteolin	-7.0
	Quercetin	-7.1
	Apigenin	-7.1
ERBB2	3-acetylursolic acid	-8.4
	Luteolin	-7.4
	Quercetin	-7.3
	Apigenin	-6.9
STAT3	3-acetylursolic acid	-8.6
	Luteolin	-7.8
	Quercetin	-7.6
	Apigenin	-7.5

Supplementary Figures:

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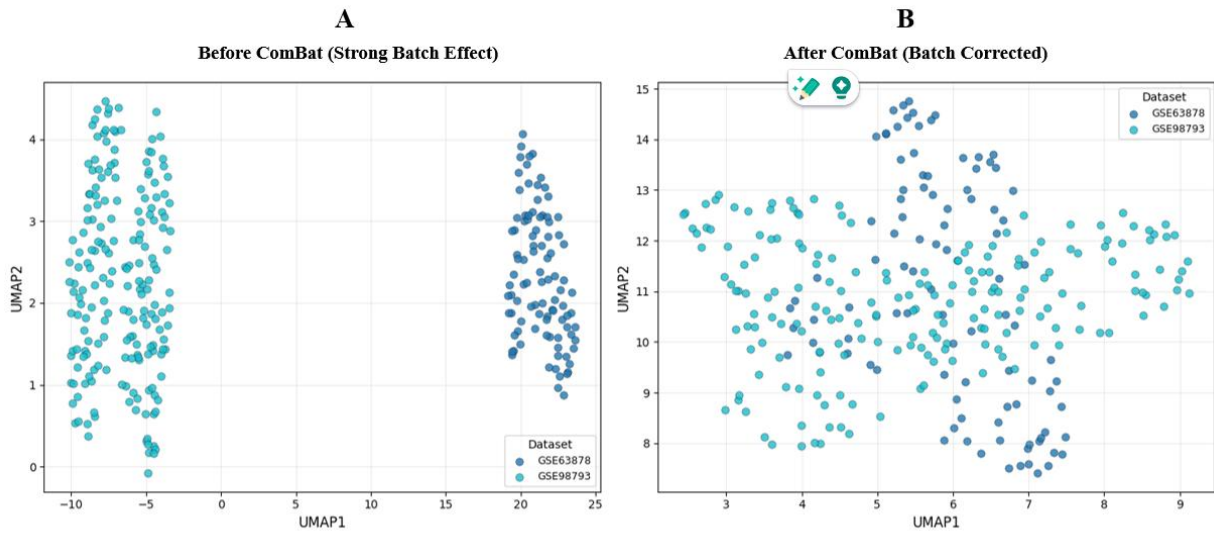


Fig. S1: UMAP of the selected feature matrices before (A) and after (B) ComBat batch correction. Samples separate by dataset before correction, reflecting platform-associated differences, and this separation is

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reduced after correction. The plot serves as a visual quality-control check of platform effects.

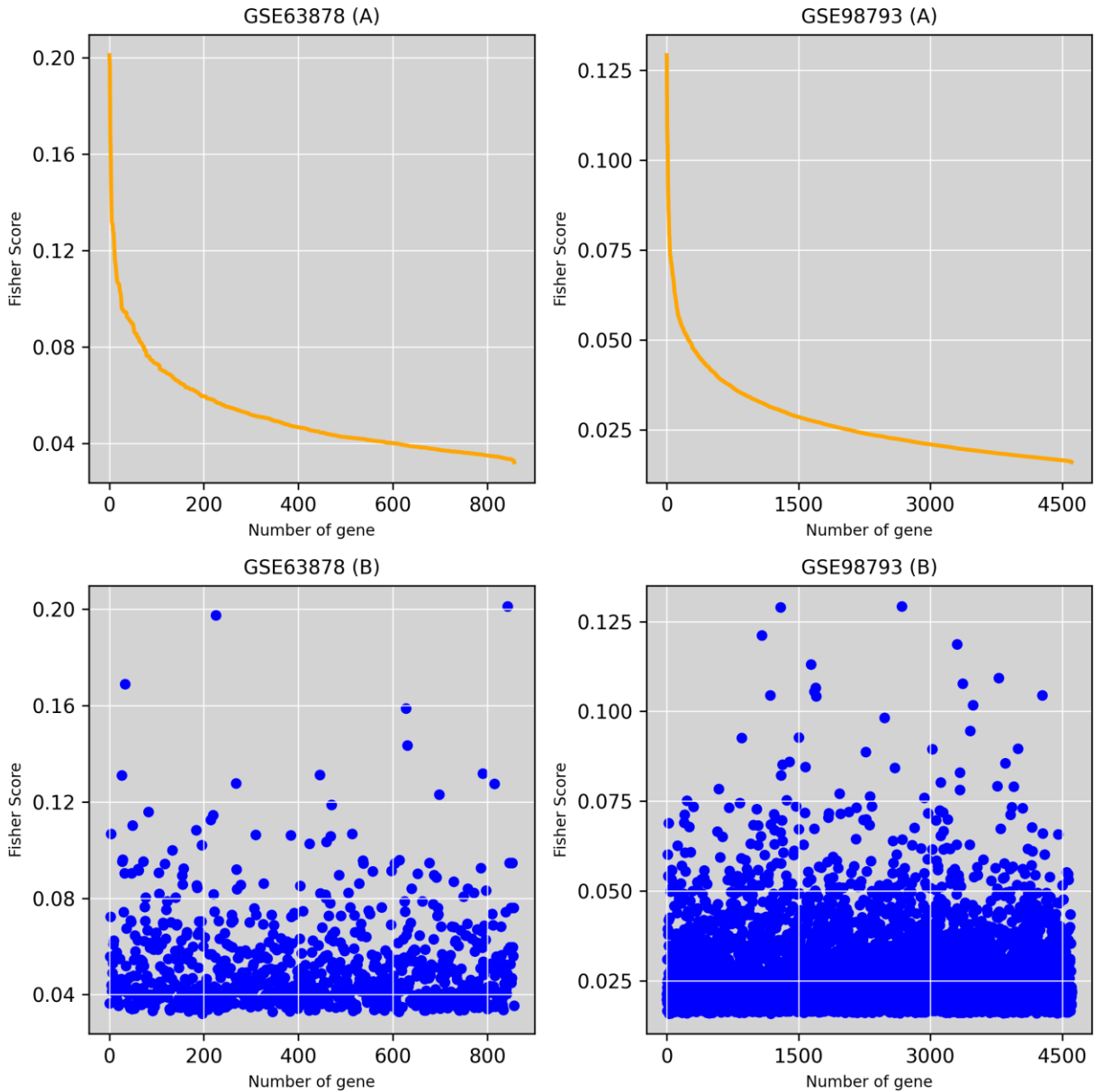


Fig. S2: Distribution of Fisher scores in the PTSD dataset (GSE63878) and the depression dataset (GSE98793). (A) Fisher scores ranked in descending order, showing a sharp decline among top-ranked genes followed by a plateau. (B) Fisher scores in the original gene order. The top 800 genes were retained

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in each dataset; the cut-off is assessed in Supplementary Table S3.

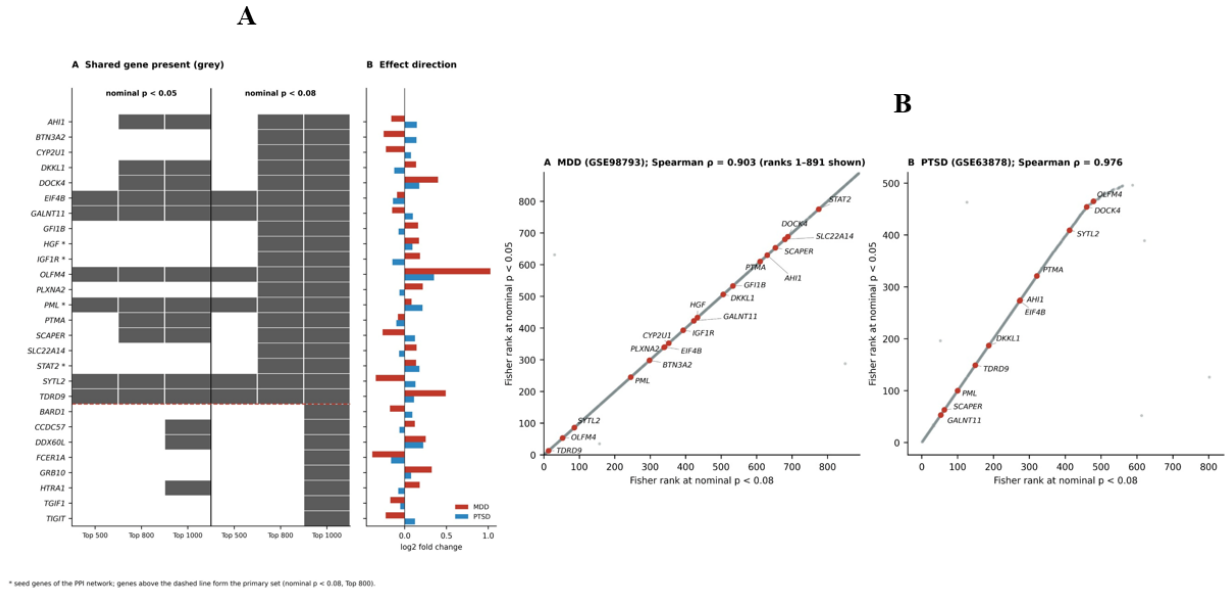


Fig. S3: (A) Presence of each shared gene across the six threshold and cut-off combinations, with log2 fold changes in MDD and PTSD. (B) Agreement of Fisher-score ranks between nominal $p < 0.05$ and < 0.08 .

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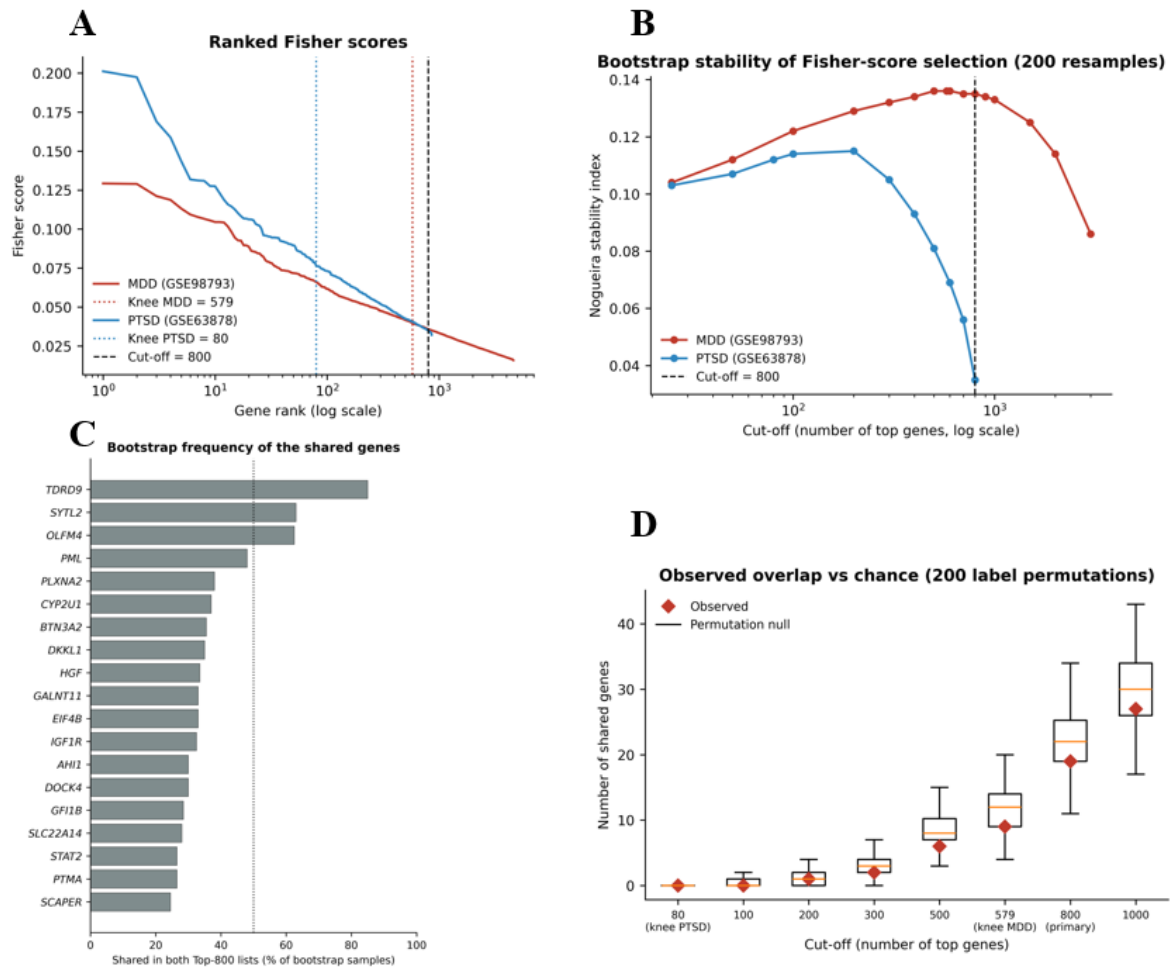


Fig. S4: (A) Ranked Fisher-score curves with knee points and the 800-gene cut-off. (B) Bootstrap stability across cut-offs. (C) Bootstrap frequency of each shared gene. (D) Observed overlap versus the label-permutation null.

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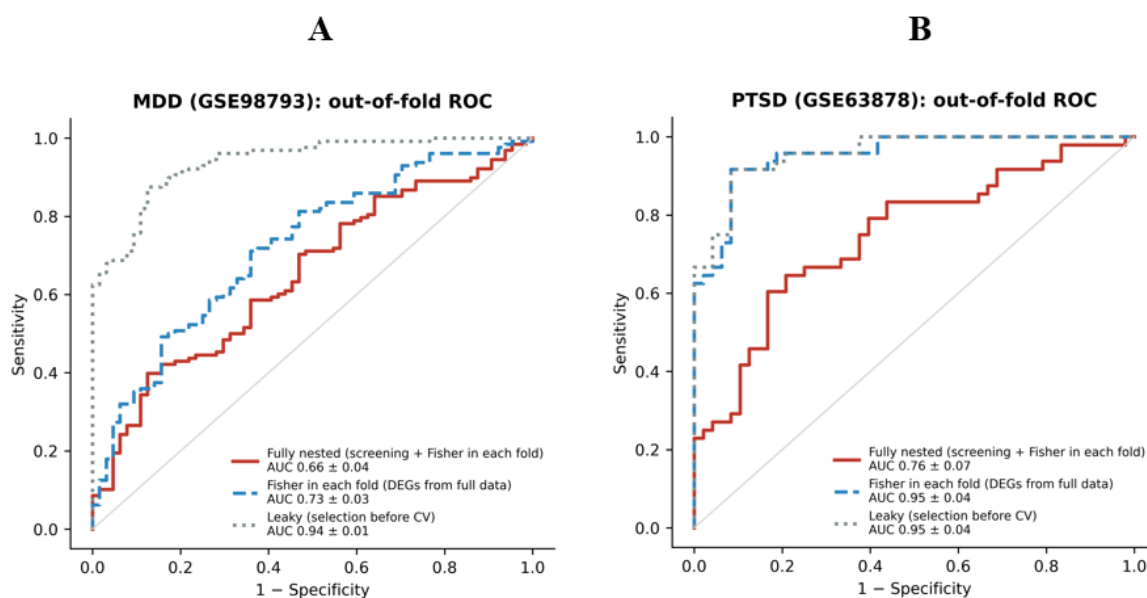


Fig. S5: Out-of-fold ROC curves for logistic regression in (A) MDD and (B) PTSD under three schemes: nominal-p screening and Fisher selection within each training fold (fully nested; solid), Fisher selection within folds applied to DEGs pre-selected on the complete dataset (dashed), and both steps performed before cross-validation (dotted). Legends give the mean $\pm$ SD of the fold-level AUCs.

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Direction of the 19 shared genes in MDD and PTSD

* BH-adjusted $P < 0.08$ (limma)

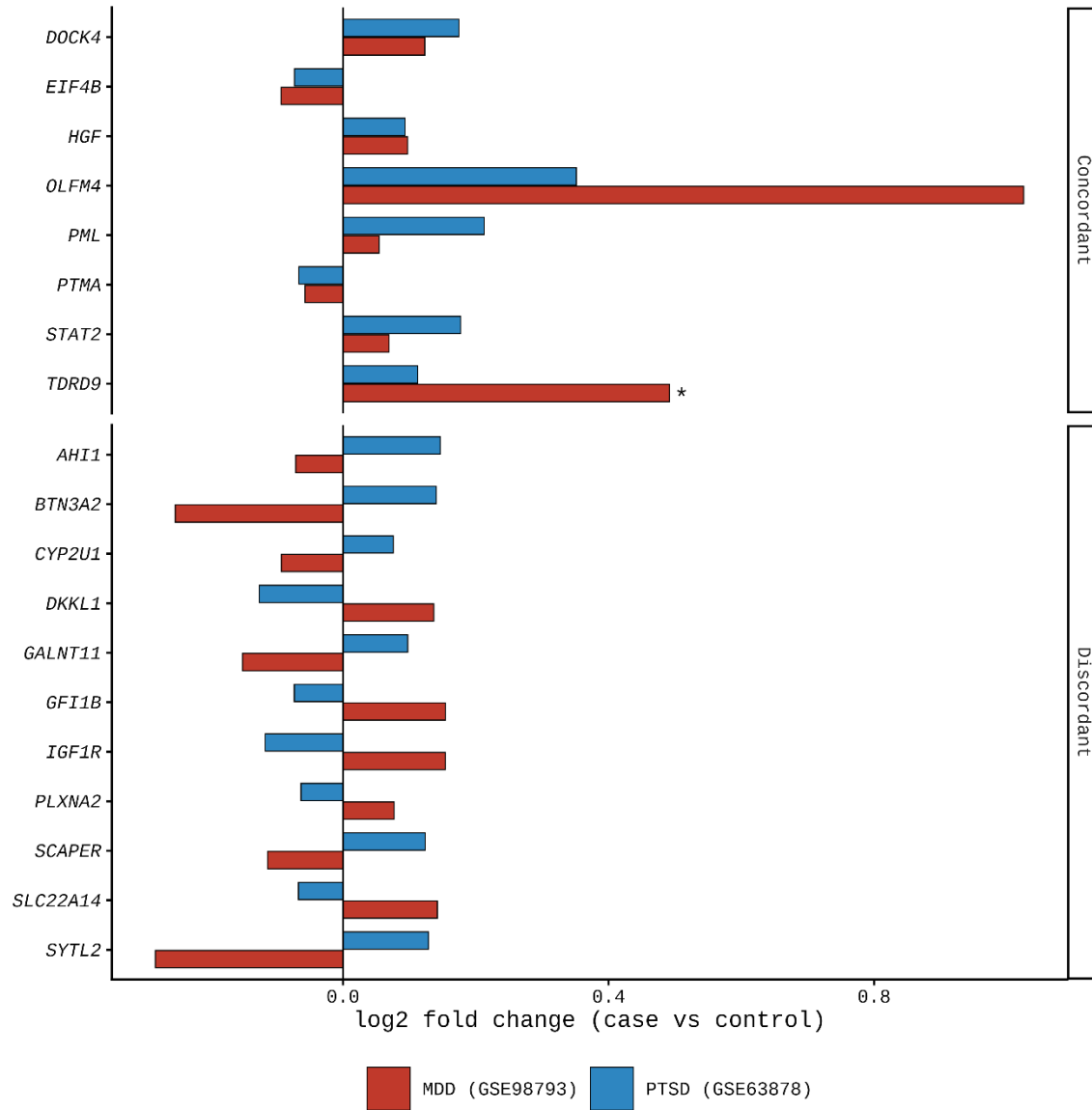


Fig. S6: log2 fold changes of the shared genes in MDD and PTSD, grouped into concordant and discordant genes; asterisks mark BH-adjusted $p < 0.08$.

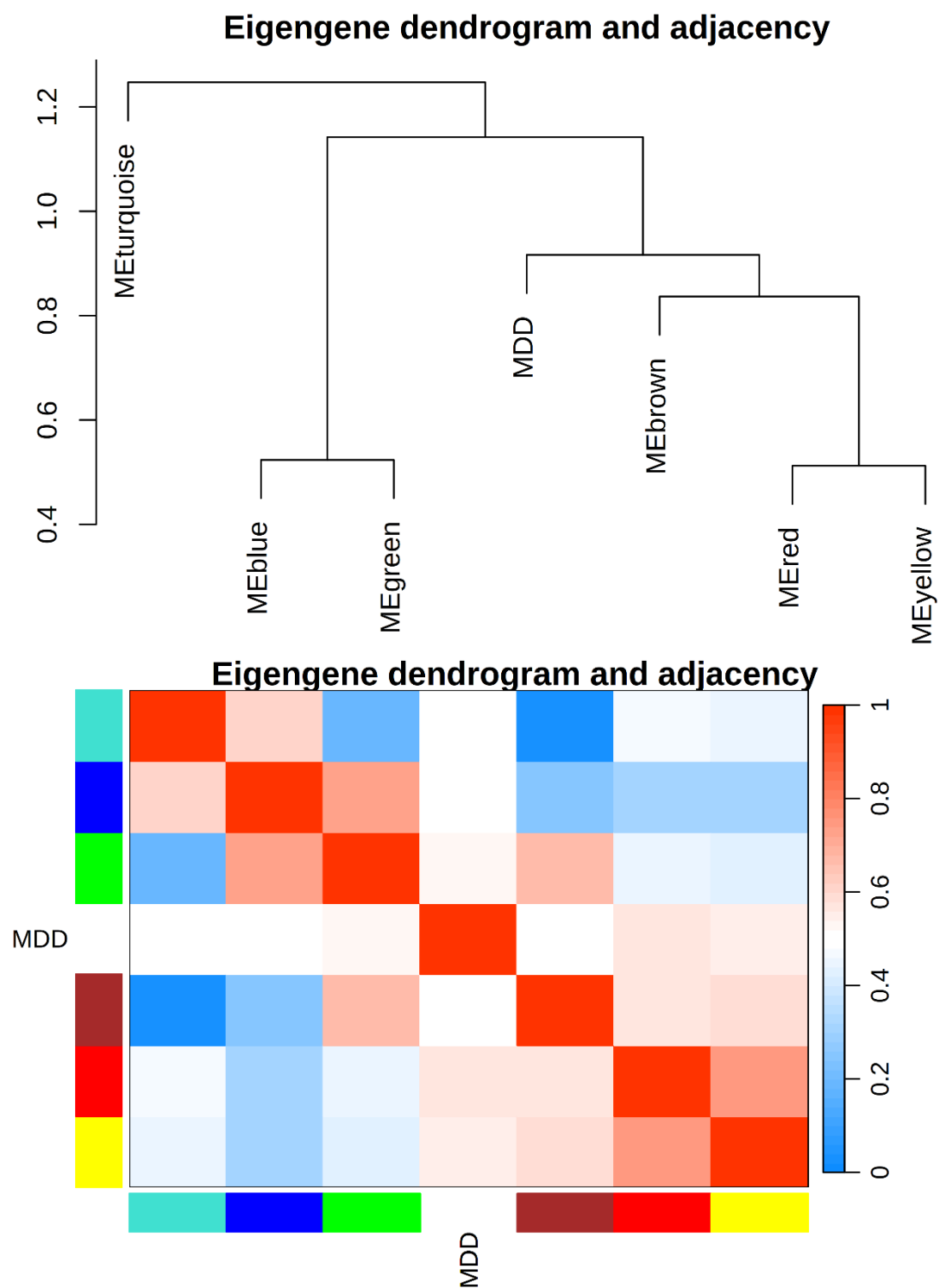


Fig. S7: Module eigengene dendrogram and adjacency heatmap for GSE98793, including the MDD trait, showing how the co-expression modules relate to one another and to disease status. No module eigengene

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was significantly associated with MDD after FDR correction.

Network heatmap plot (TOM), 400 randomly selected genes

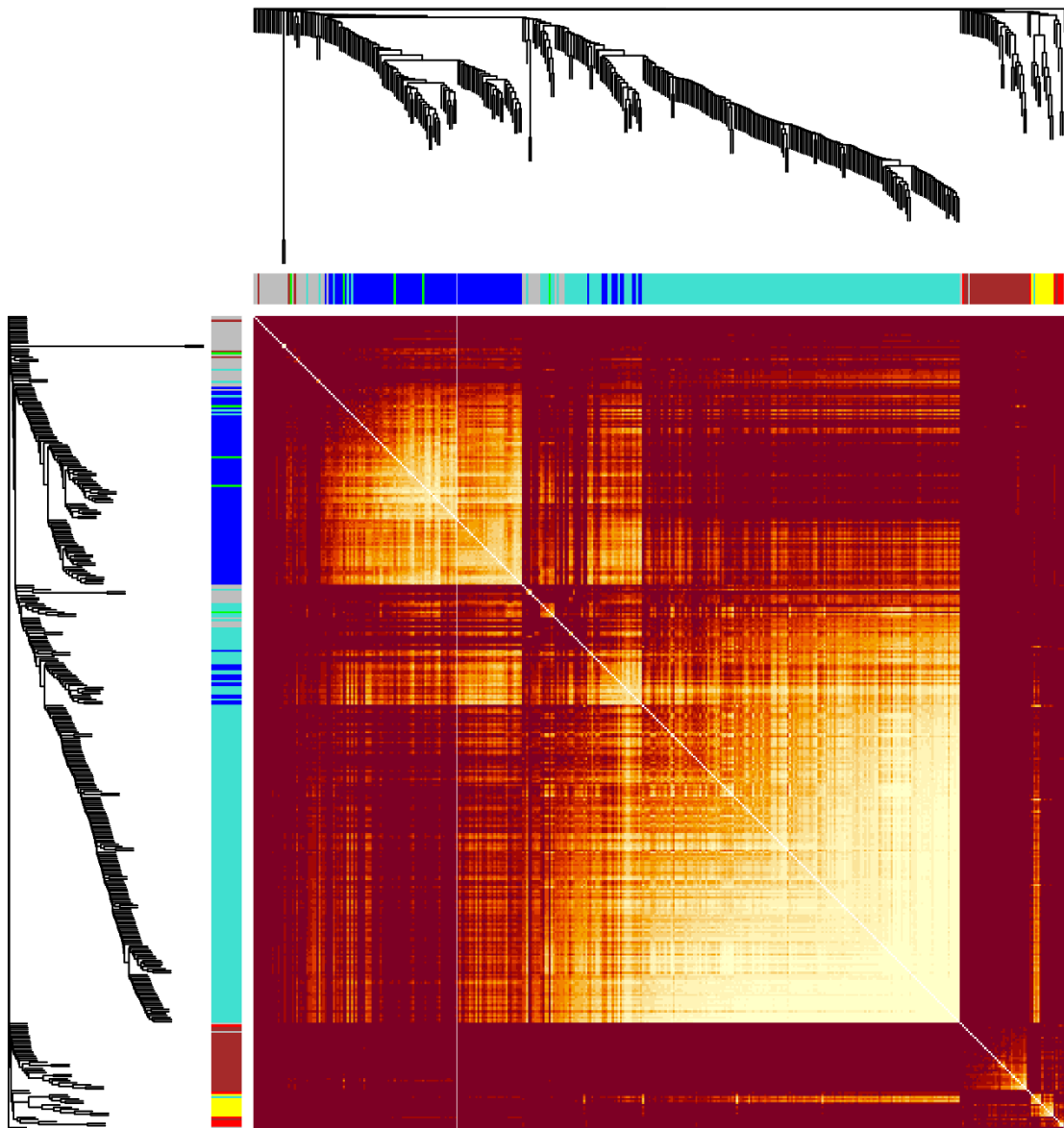


Fig. S8: TOM heatmap of 400 randomly selected genes, with the gene dendrogram and module colors. Clear blocks for the blue and turquoise modules indicate well-defined modules.

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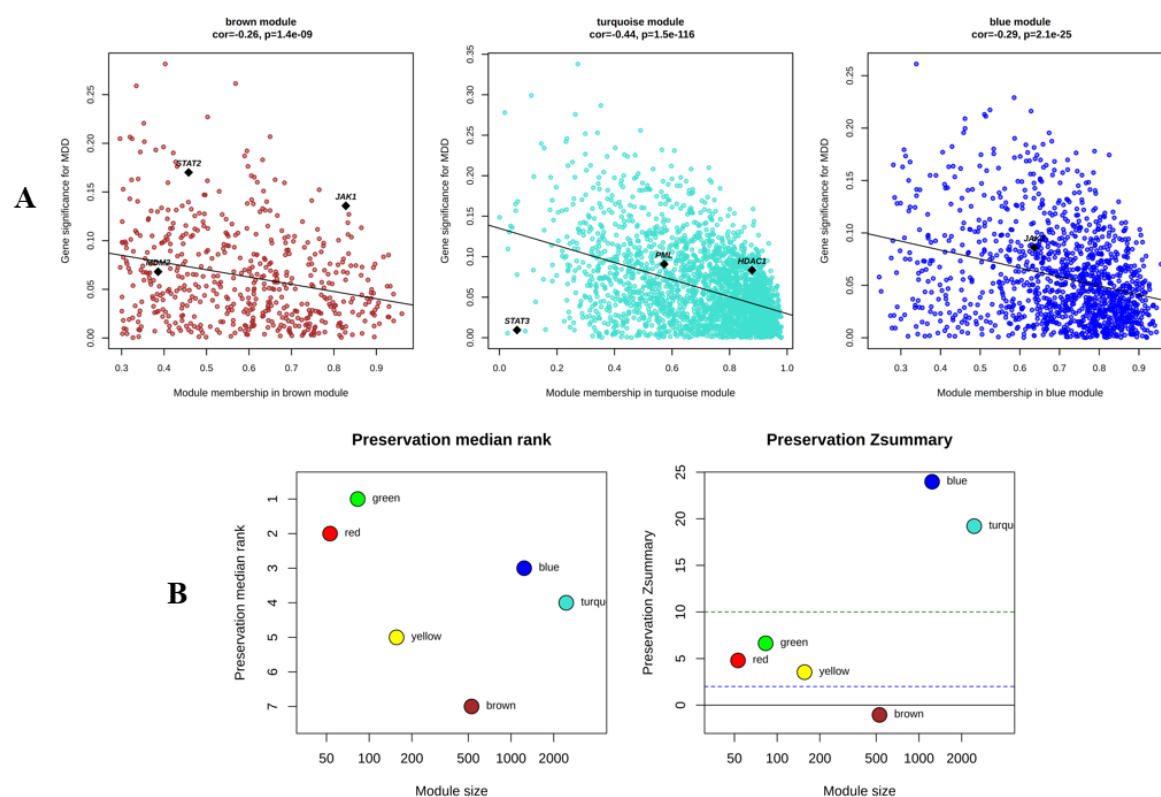


Fig. S9: (A) Module membership versus gene significance for MDD in the brown, turquoise, and blue modules, with hub genes marked (STAT2 and JAK1 in brown; HDAC1, PML, and STAT3 in turquoise; JAK2 in blue). (B) Preservation median rank and Zsummary against module size for the GSE98793 modules tested in GSE63878. The blue and turquoise modules were strongly preserved (Zsummary > 10);

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the brown module showed no preservation.

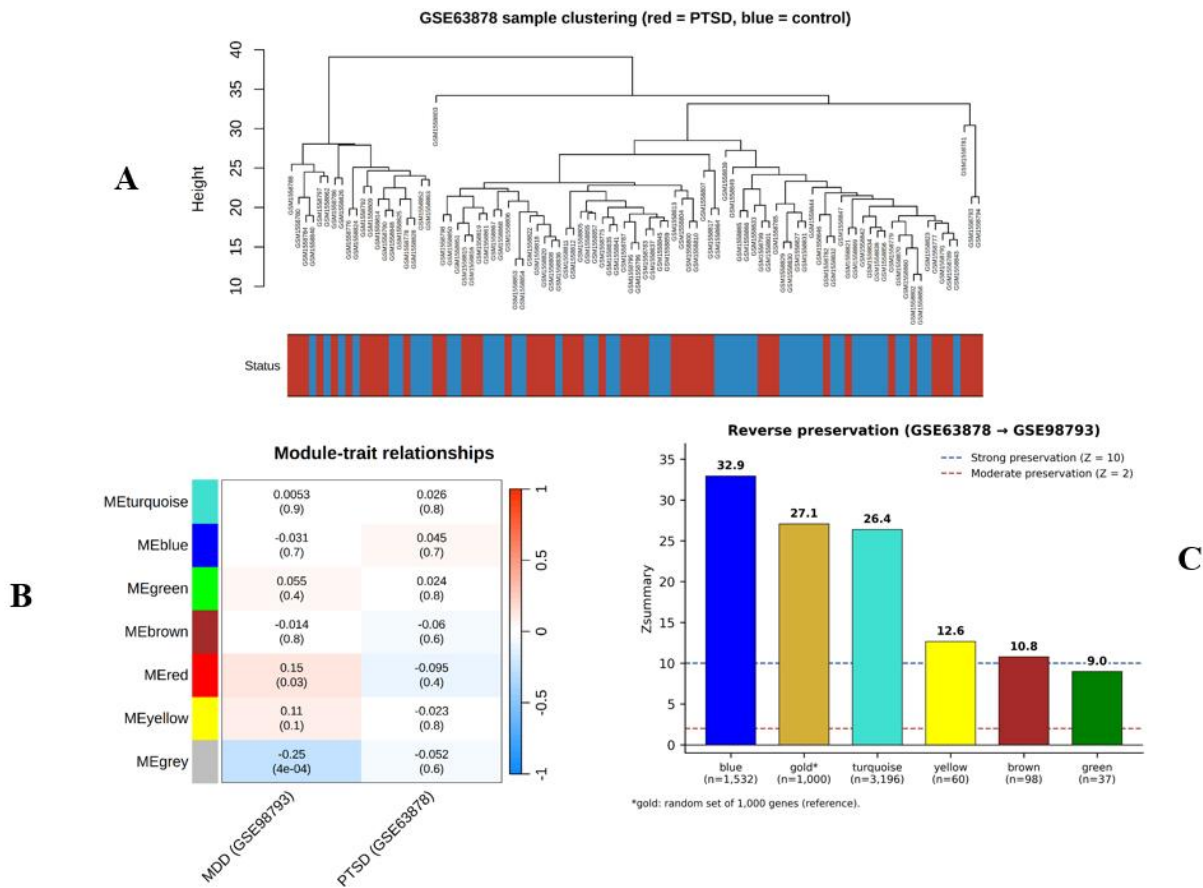


Fig. S10: (A) Sample clustering of GSE63878 (red, PTSD; blue, control). (B) Module–trait relationships for MDD and PTSD, shown with nominal p-values; none remained significant after FDR correction. (C) Exploratory reverse preservation analysis (GSE63878 → GSE98793). Zsummary was above 10 for the blue (32.9), turquoise (26.4), yellow (12.7), and brown (10.8) modules and 9.0 for the green module. The random

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gold reference (Zsummary = 27.1) is shown as a benchmark.

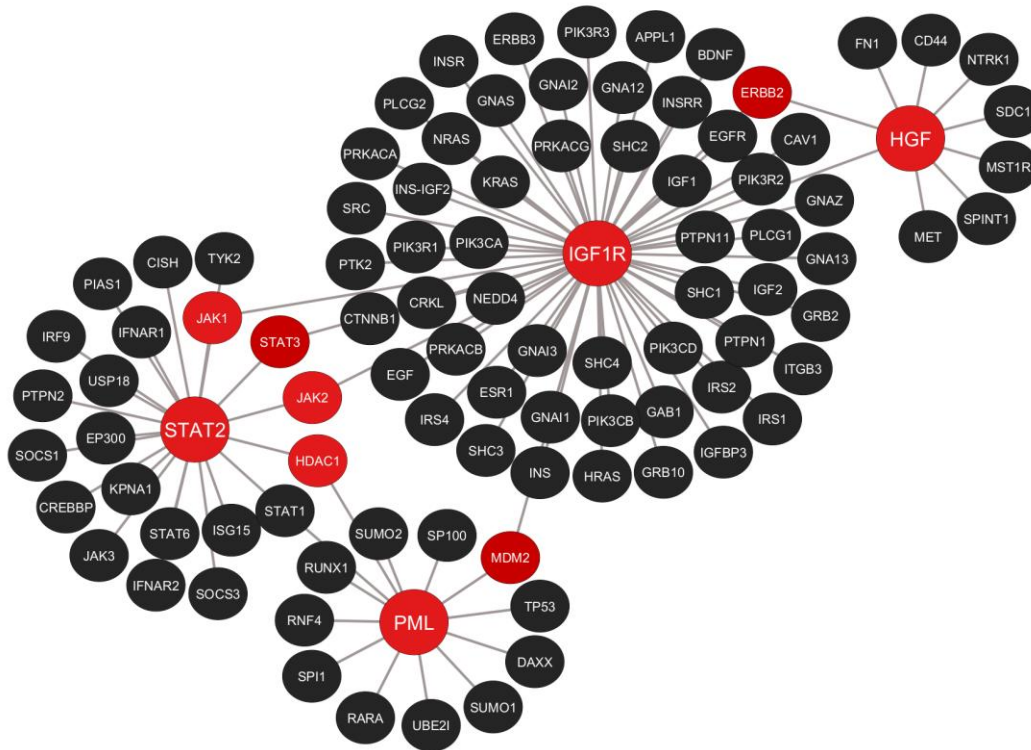


Fig. S11: PPI network of the four shared DEGs that mapped to STRING (IGF1R, STAT2, HGF and PML) and their first-order interactors (97 nodes, 101 edges). Red nodes mark the ten hub genes and black nodes the remaining interactors.

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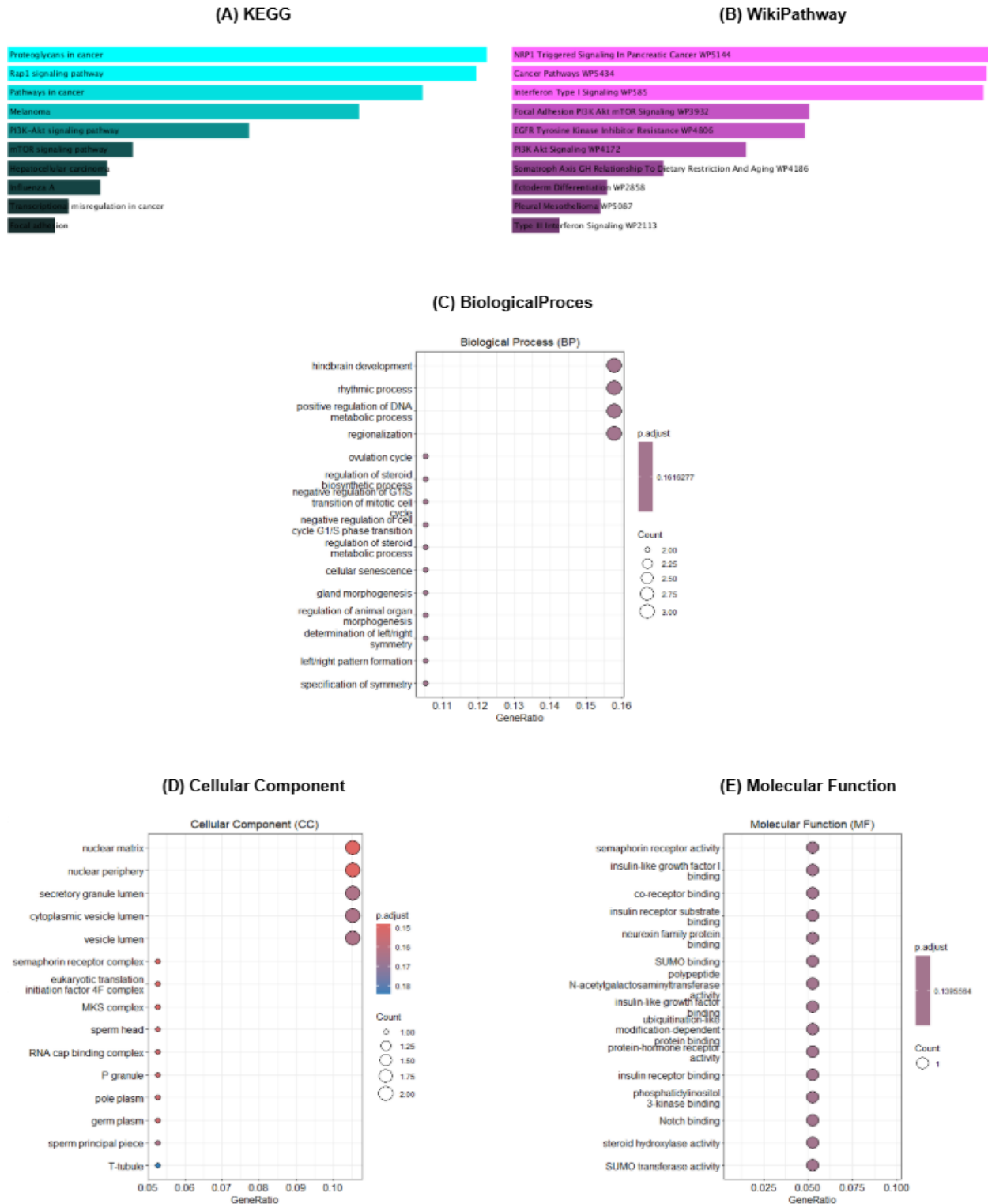
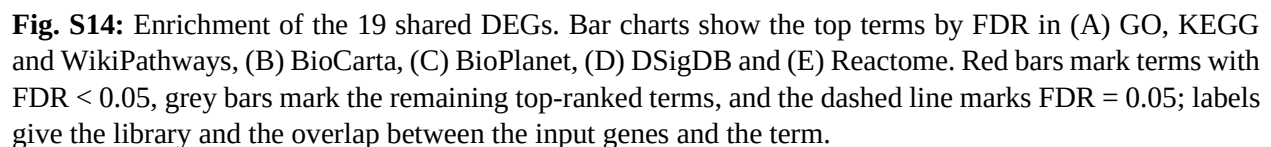


Fig. S12: Top-ranked KEGG (A) and WikiPathways (B) pathways and GO biological process (C), cellular component (D), and molecular function (E) terms for the 19 shared DEGs, ranked by nominal p-value. These plots give an overview of the pathways to which the shared genes map; FDR-adjusted statistics are

provided in Supplementary Table S7.



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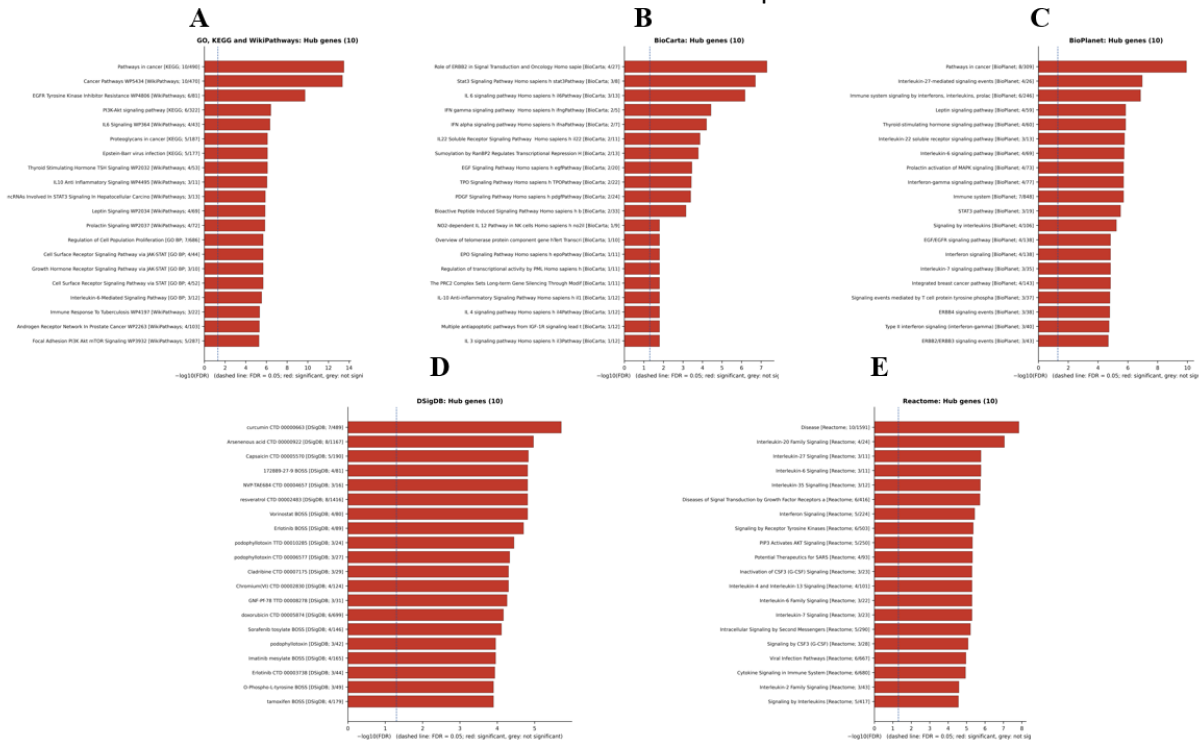
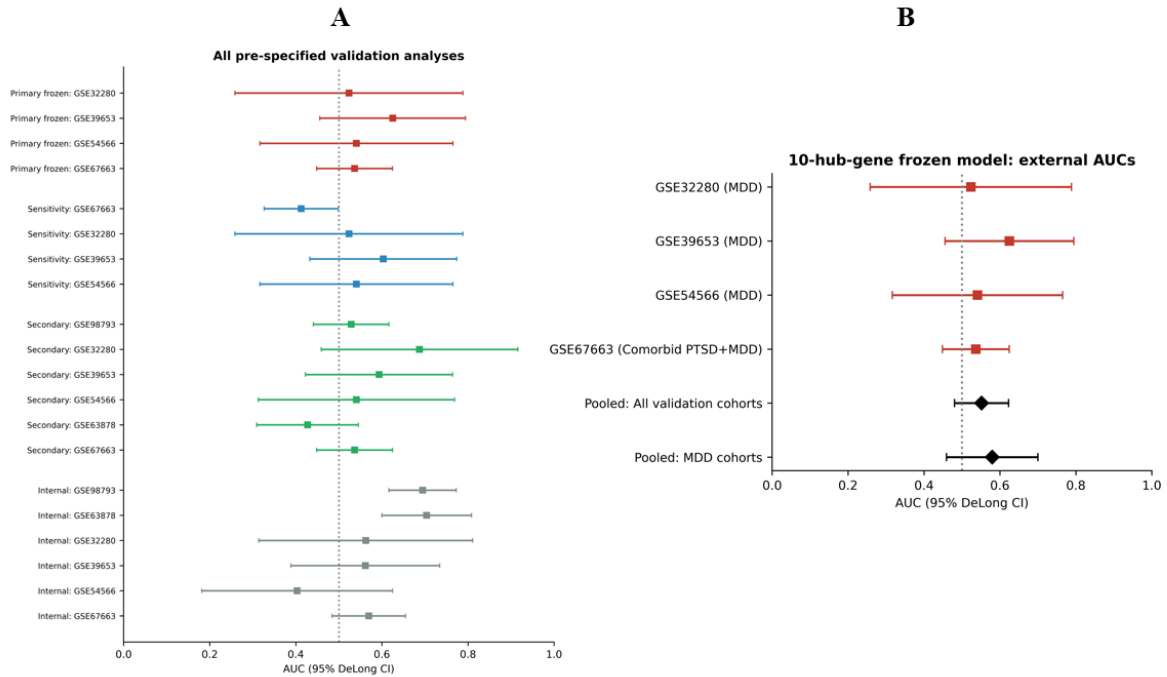


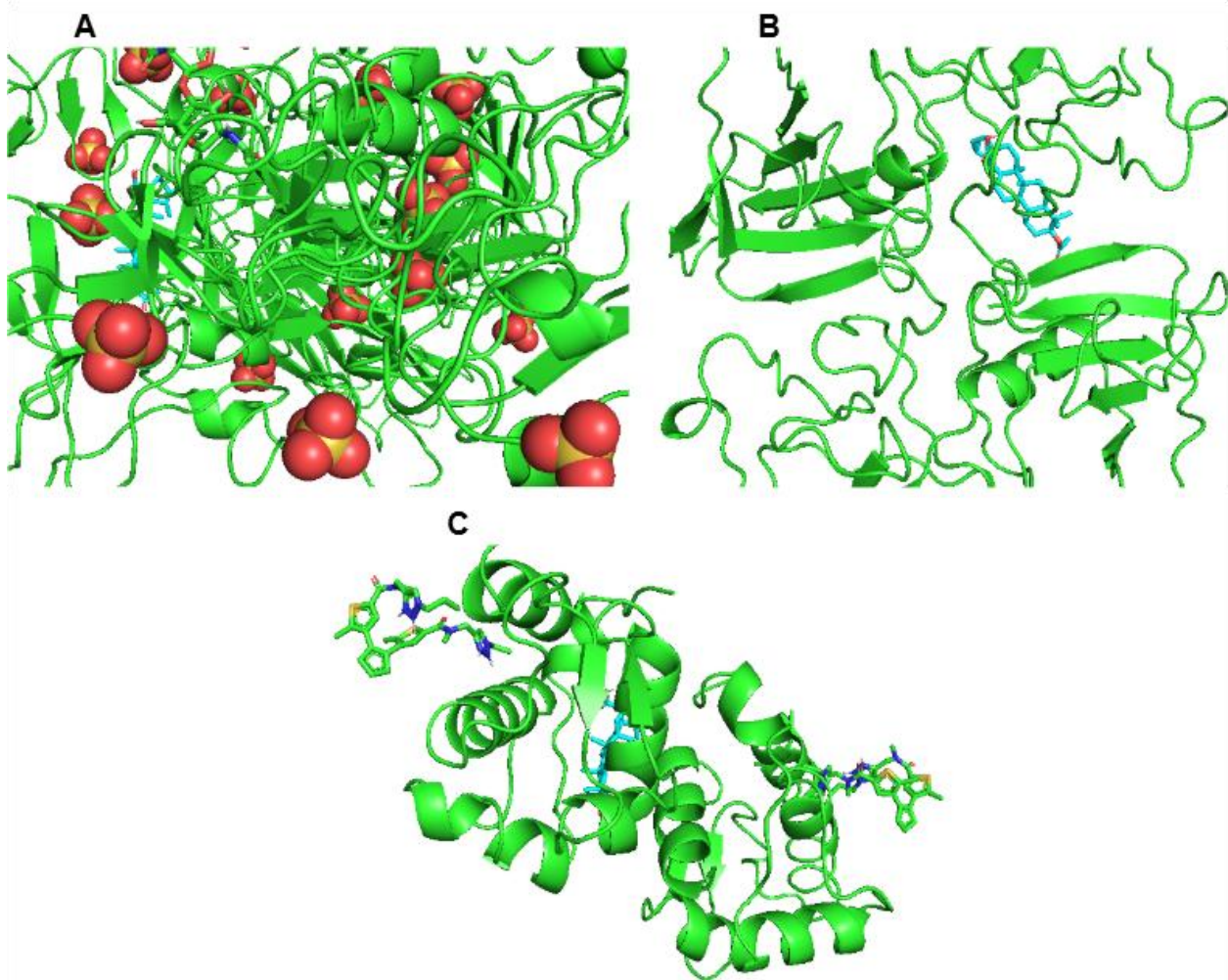
Fig. S15: Enrichment of the 10 hub genes. Bar charts show the top terms by FDR in (A) GO, KEGG and WikiPathways, (B) BioCarta, (C) BioPlanet, (D) DSigDB and (E) Reactome. Red bars mark terms with FDR < 0.05, grey bars mark the remaining top-ranked terms, and the dashed line marks FDR = 0.05; labels give the library and the overlap between the input genes and the term.



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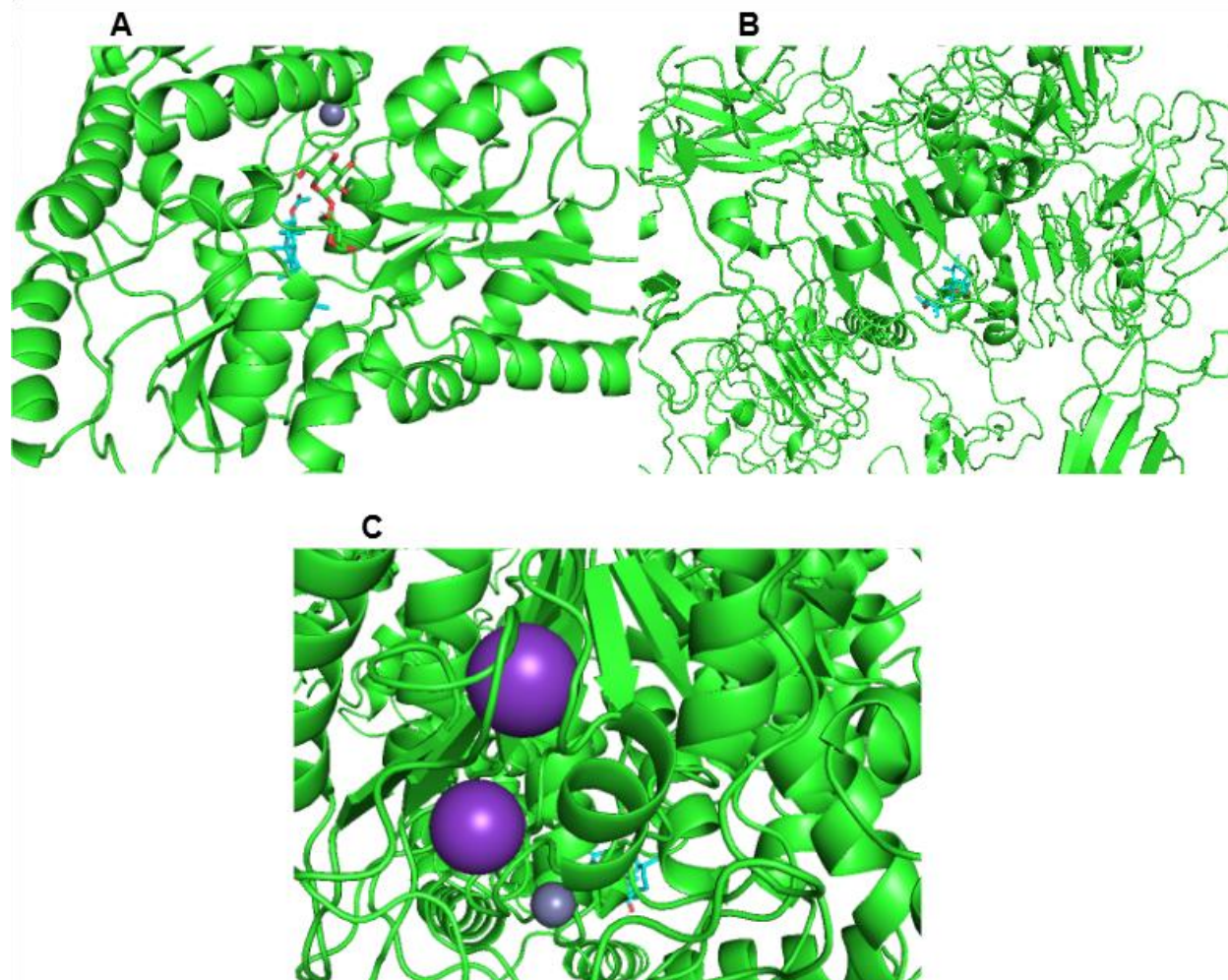
Fig. S16: External validation of the 10-hub-gene frozen model. (A) AUCs (95% DeLong CI) across all pre-specified validation analyses (primary, sensitivity, secondary, internal); most CIs cross 0.5 (chance). (B) External AUCs in GSE32280, GSE39653, GSE54566 (MDD) and GSE67663 (comorbid PTSD+MDD), with pooled estimates for all cohorts and MDD cohorts only.



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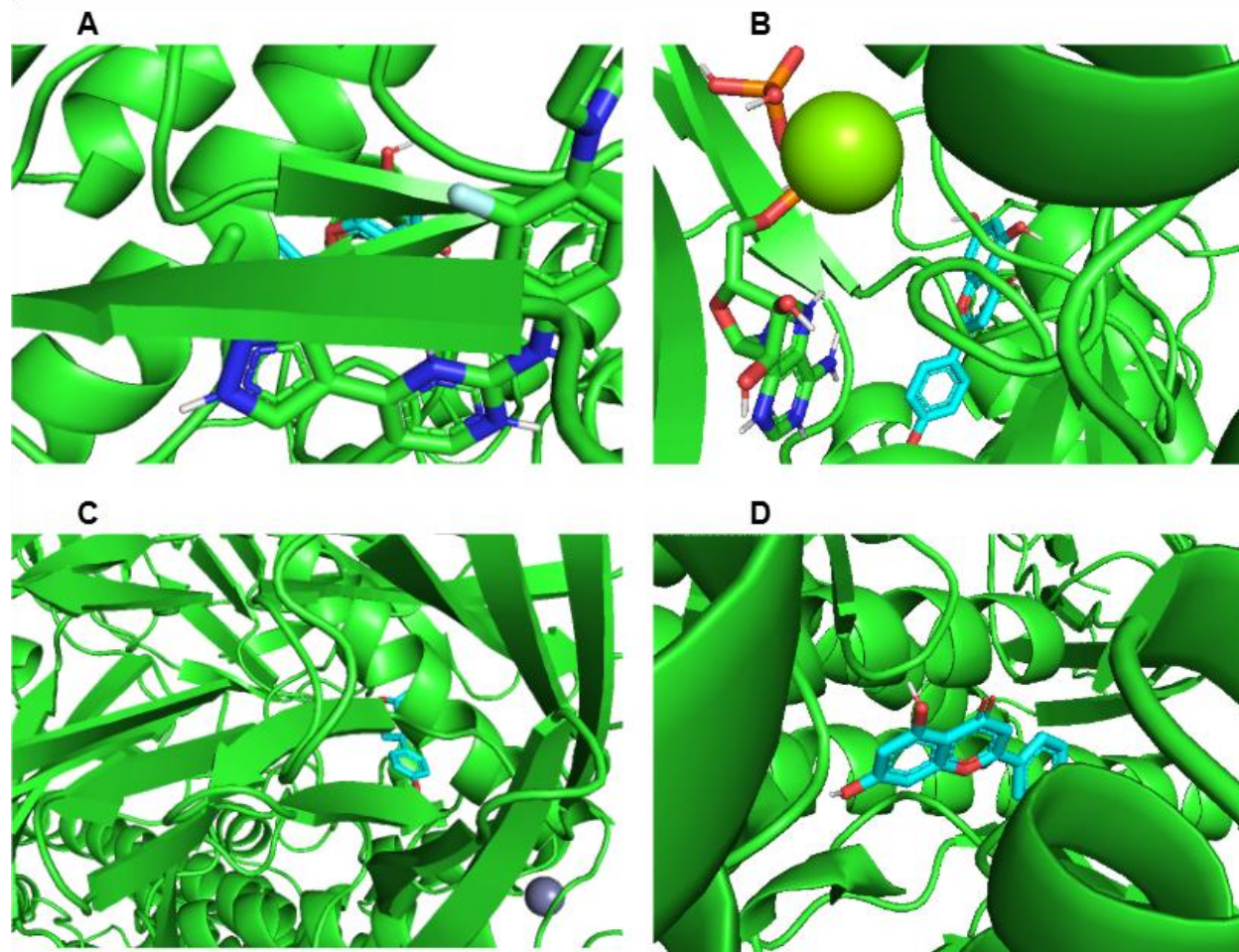
Fig. S17: Predicted docking poses of 3-acetylursolic acid with (A) ERBB2, (B) HGF, and (C) MDM2.



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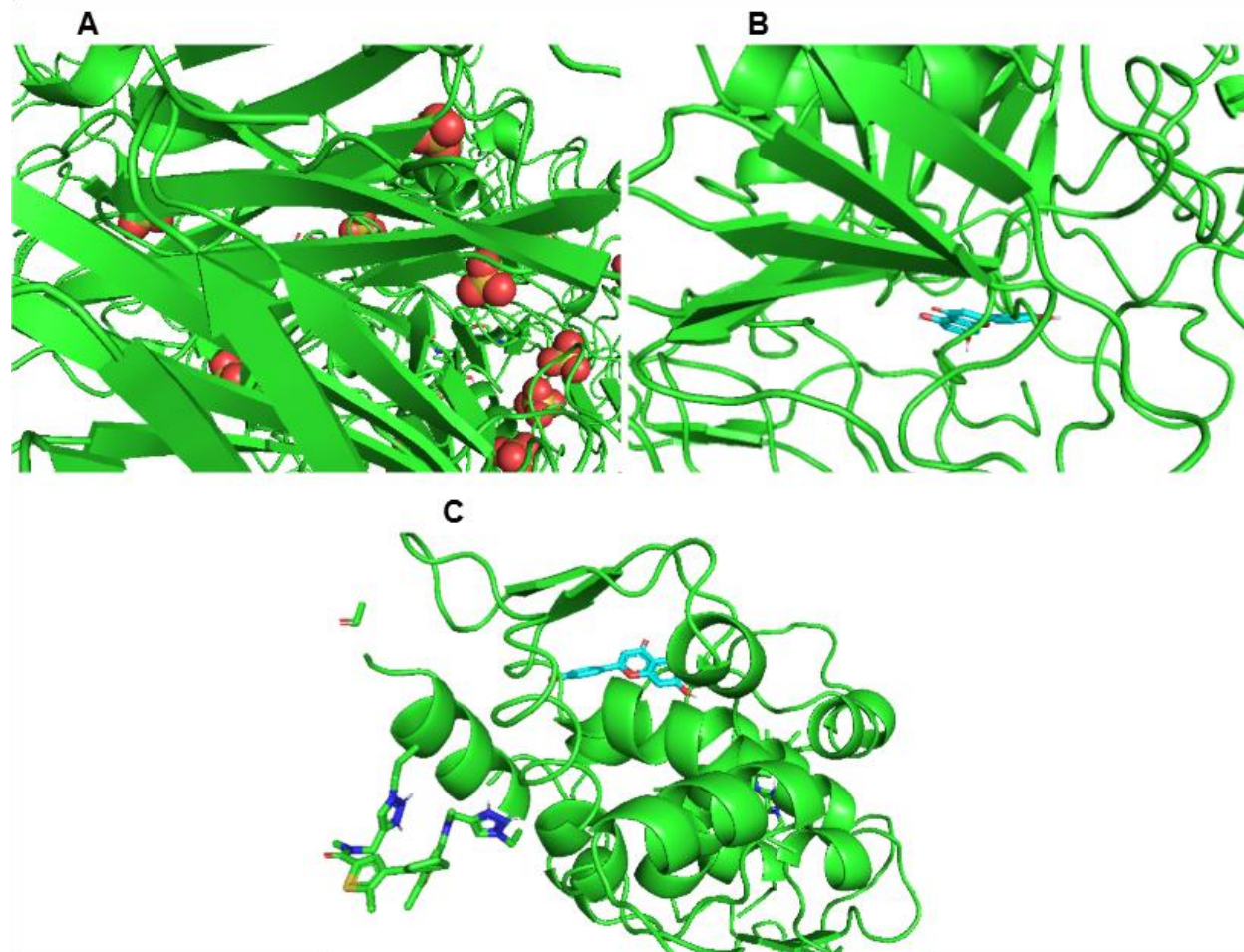
Fig. S18: Predicted docking poses of 3-acetylursolic acid with (A) PML, (B) IGF1R, and (C) HDAC1.



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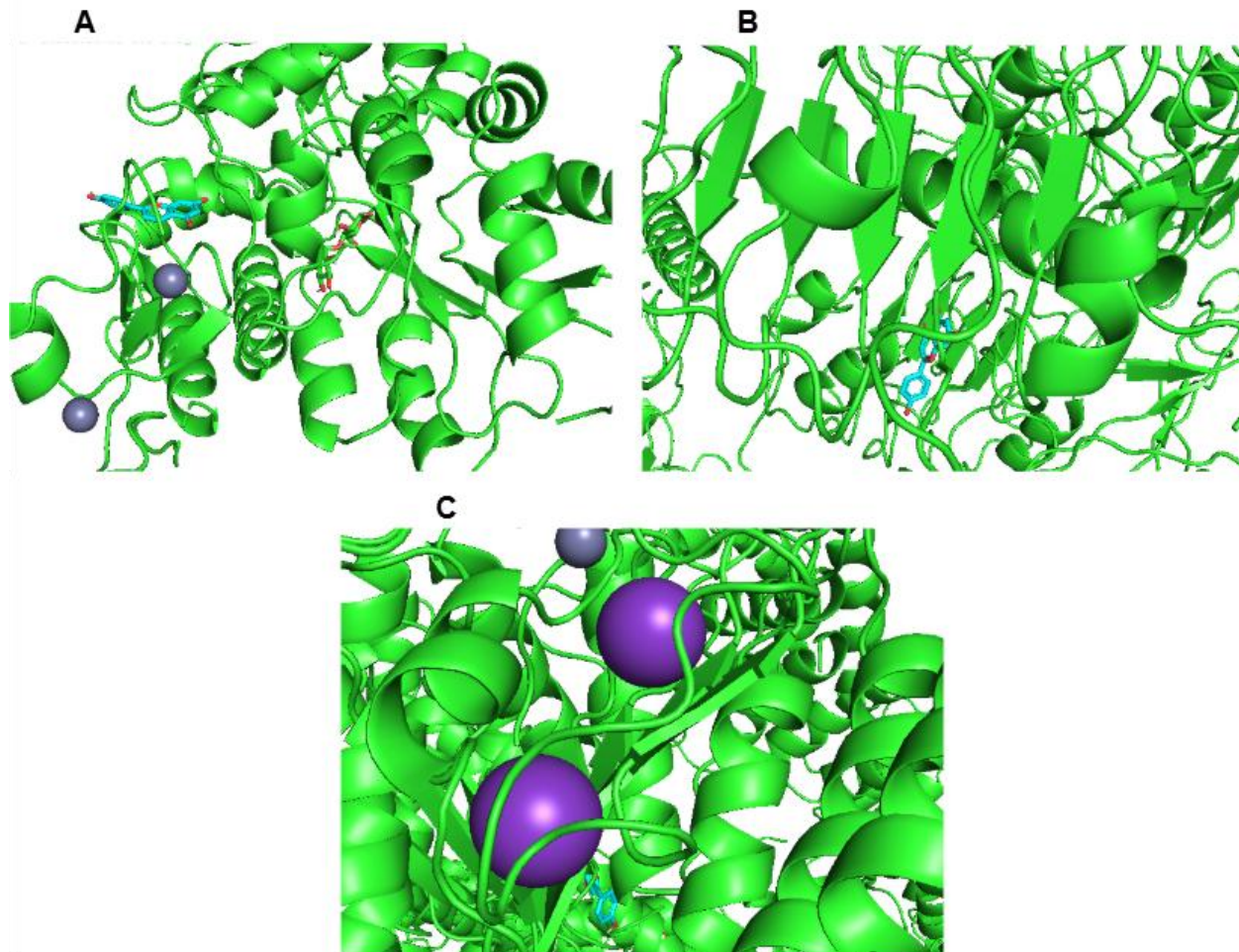
Fig. S19: Predicted docking poses of apigenin with (A) JAK2, (B) JAK1, (C) STAT2, and (D) STAT3.



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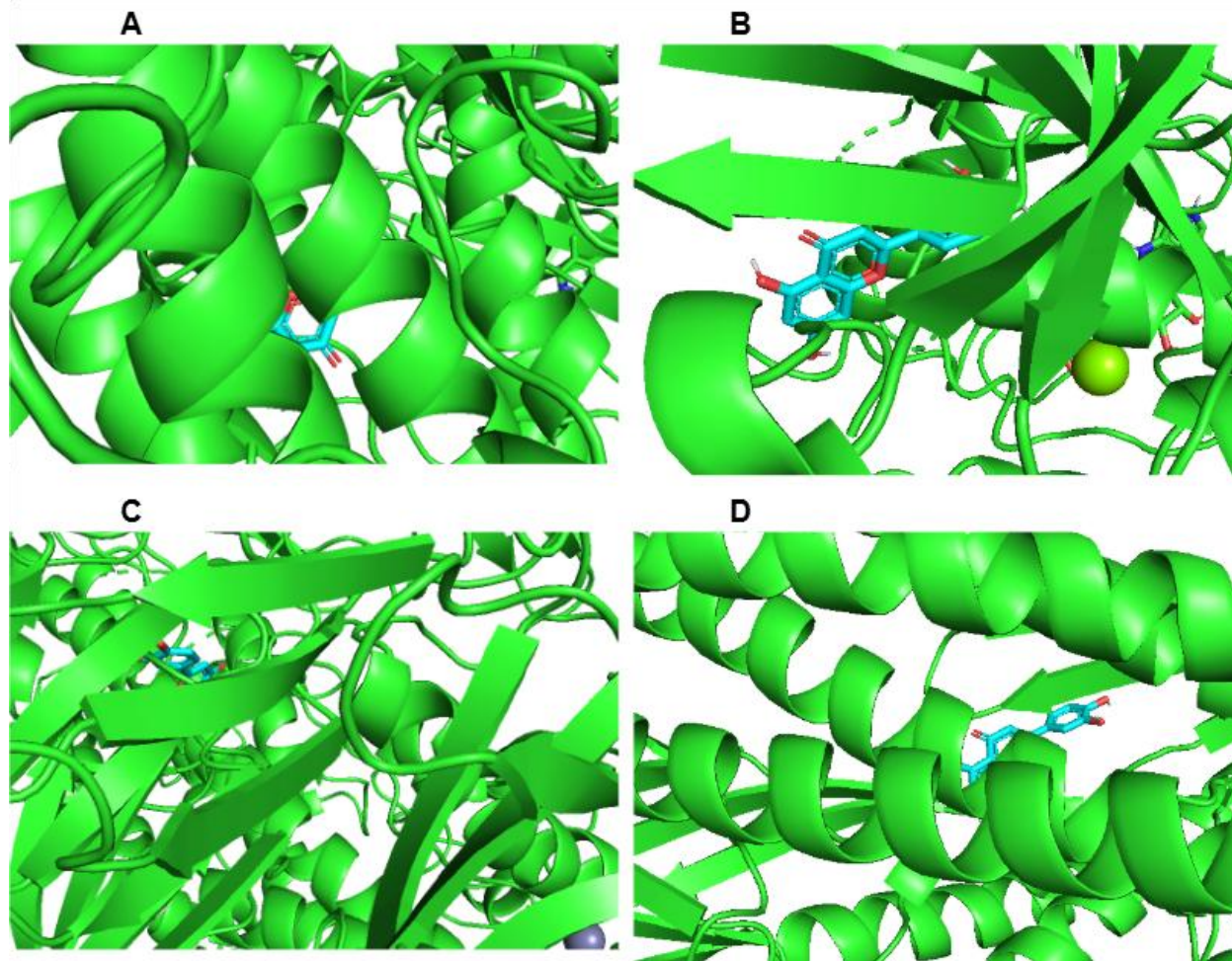
Fig. S20: Predicted docking poses of apigenin with (A) ERBB2, (B) HGF, and (C) MDM2.



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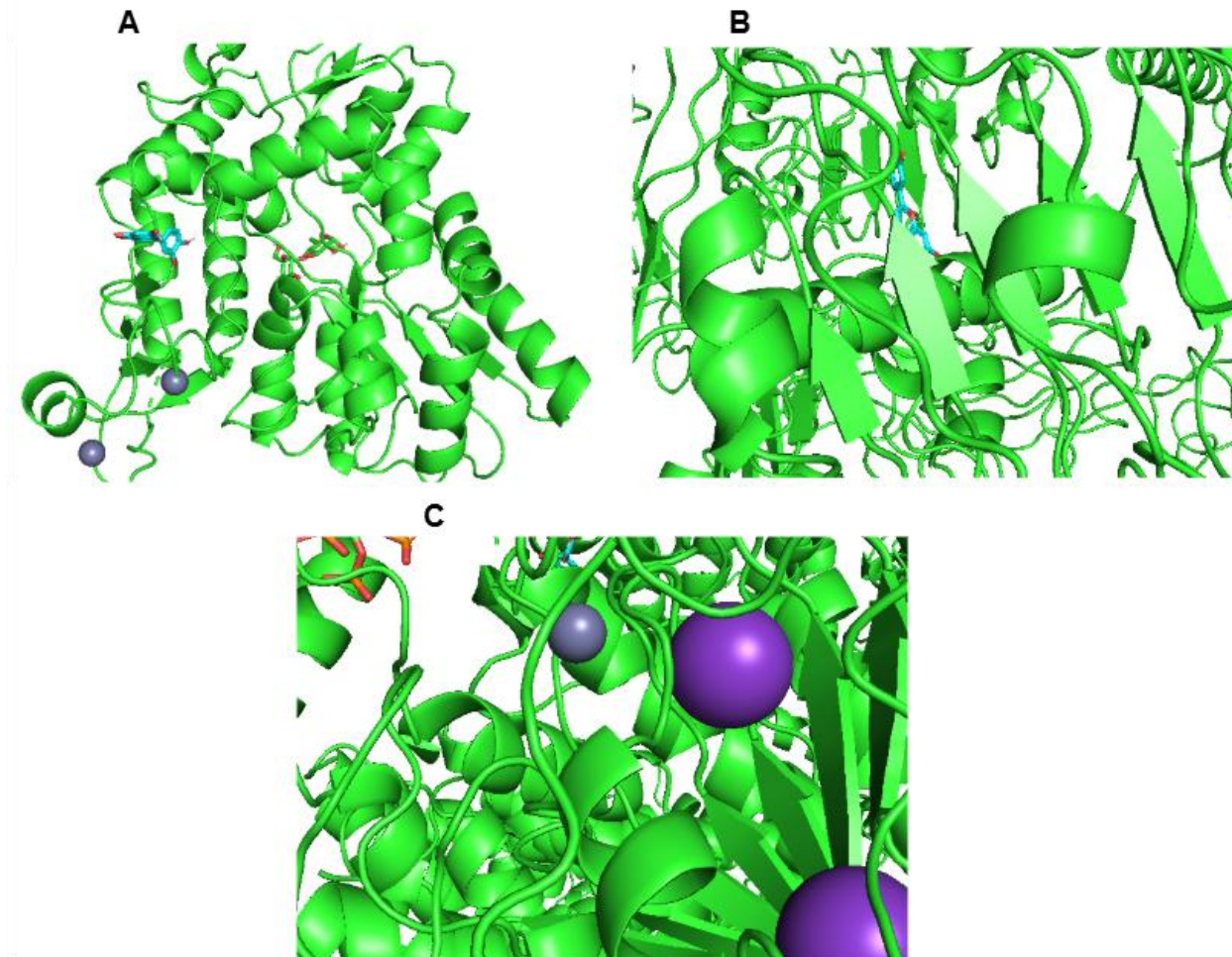
Fig. S21: Predicted docking poses of apigenin with (A) PML, (B) IGF1R and (C) HDAC1.



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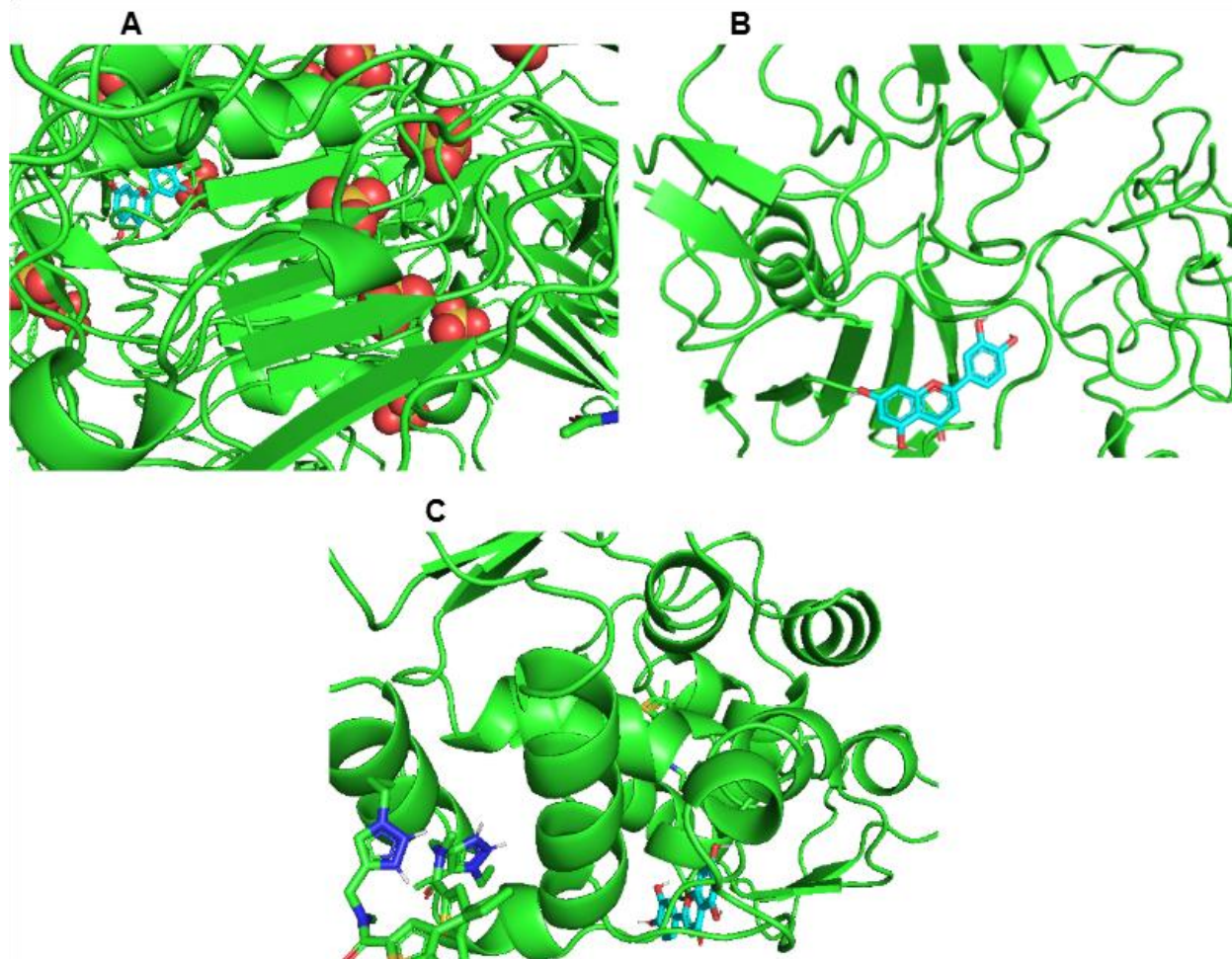
Fig. S22: Predicted docking poses of luteolin with (A) JAK2, (B) JAK1, (C) STAT2, and (D) STAT3.



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Fig. S23: Predicted docking poses of luteolin with (A) ERBB2, (B) HGF, and (C) MDM2.



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Fig. S24: Predicted docking poses of luteolin with (A) PML, (B) IGF1R, and (C) HDAC1.

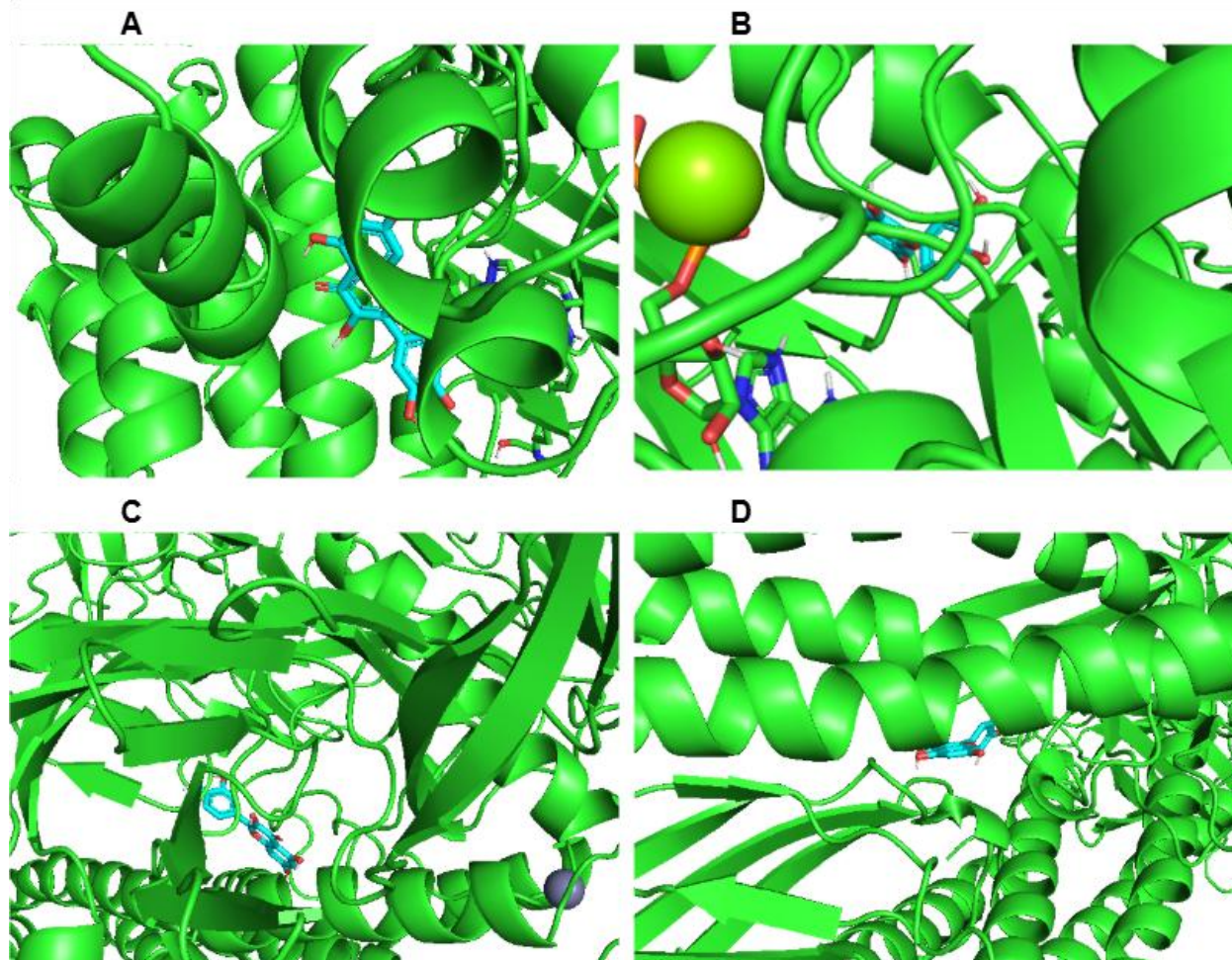


Fig. S25: Predicted docking poses of quercetin with (A) JAK2, (B) JAK1, (C) STAT2, and (D) STAT3.

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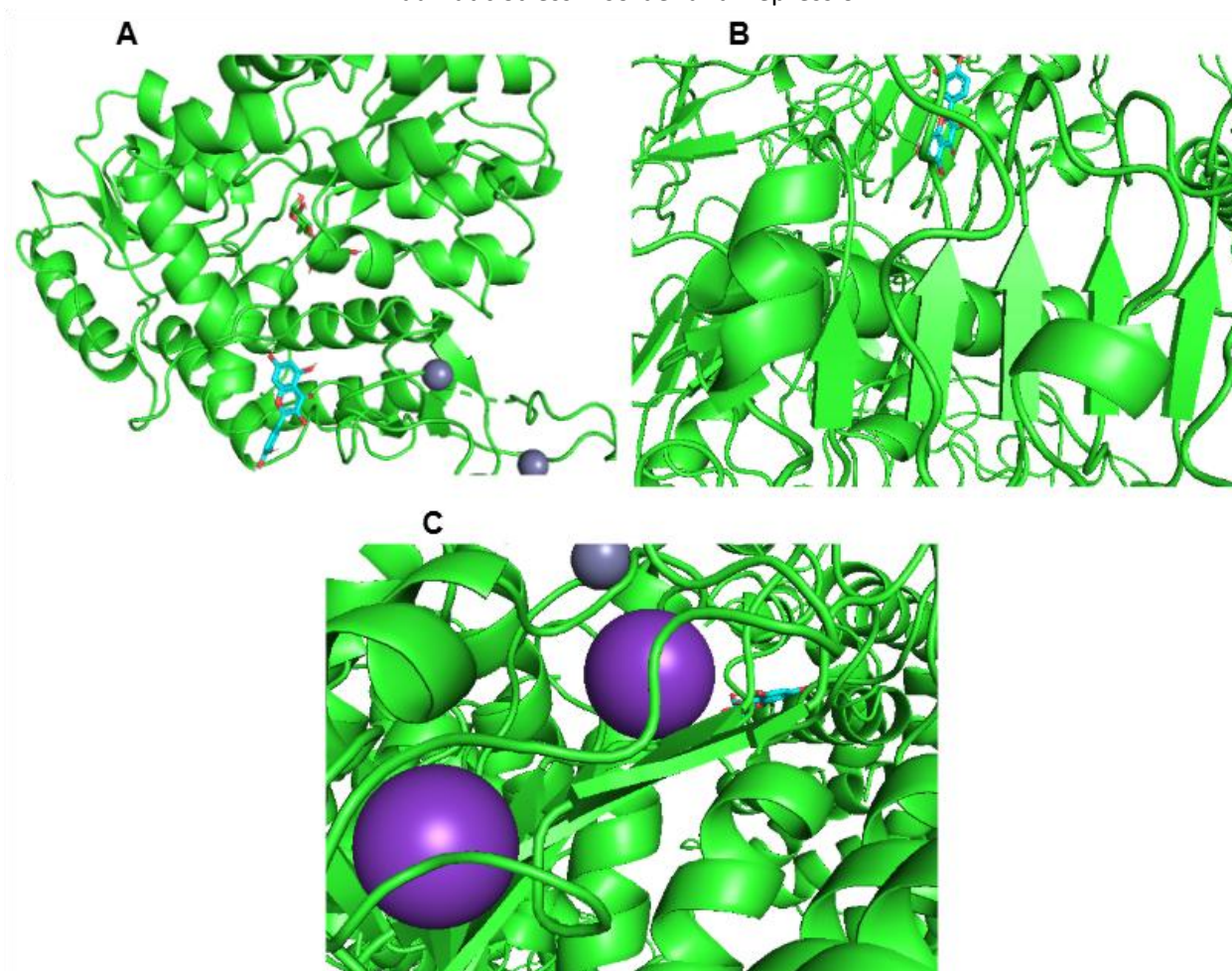


Fig. S26: Predicted docking poses of quercetin with (A) ERBB2, (B) HGF, and (C) MDM2.

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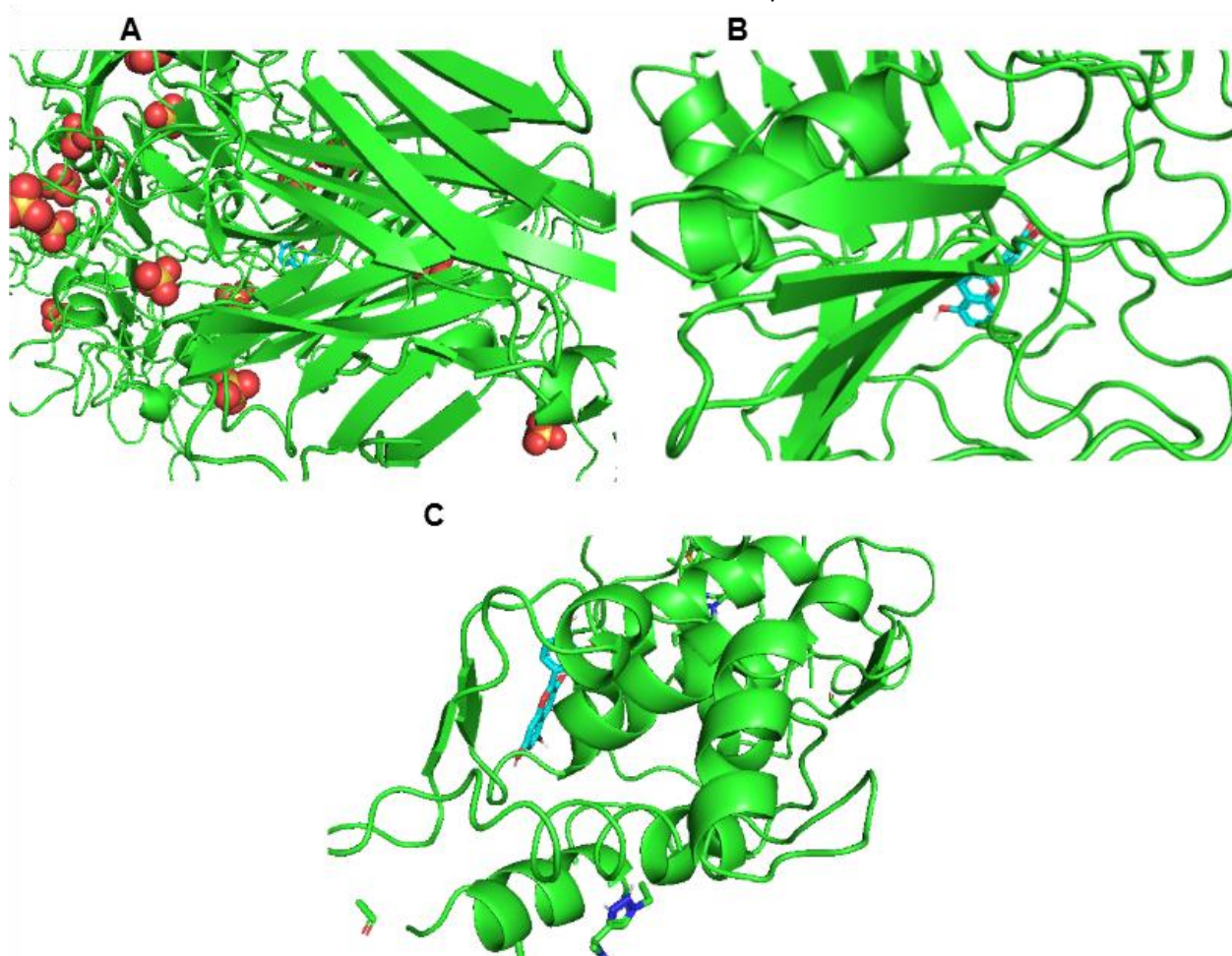


Fig. S27: Predicted docking poses of quercetin with (A) PML, (B) IGF1R, and (C) HDAC1.

GitHub Link: <https://github.com/MostaryKhatun/stress-depression-biomarker-discovery>